

Direct coupling circuit

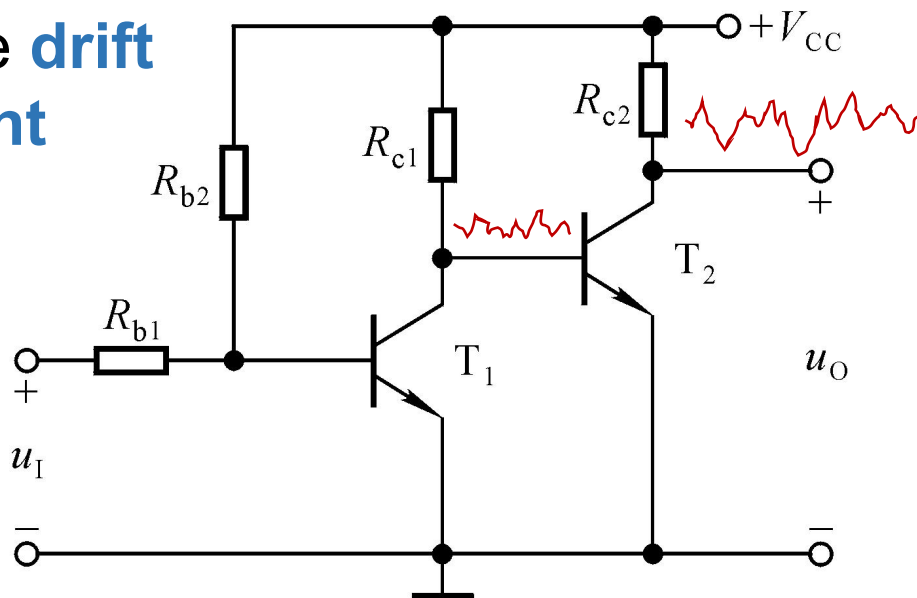
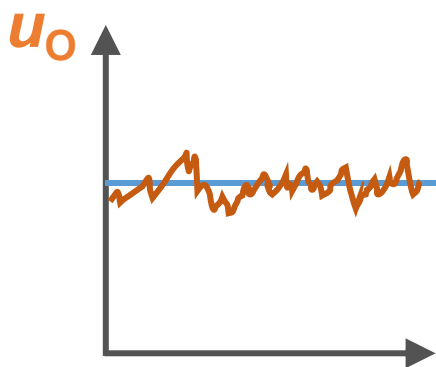
No need of capacitor and inductor, suitable for **large-scale integrated circuits** 适合大规模集成电路

Drawback of direct coupling circuit:

Zero-drift error 零点漂移

When input signal $u_i=0$, the output signal u_o is not stable, but shows a slow and random variation

Zero-drift error is due to the **drift of quiescent working point**



Sources that contribute to the zero-drift error:

- a) Temperature 由温度引起的零漂称为温漂
- b) Aging 由元器件老化引起的零漂称为时漂

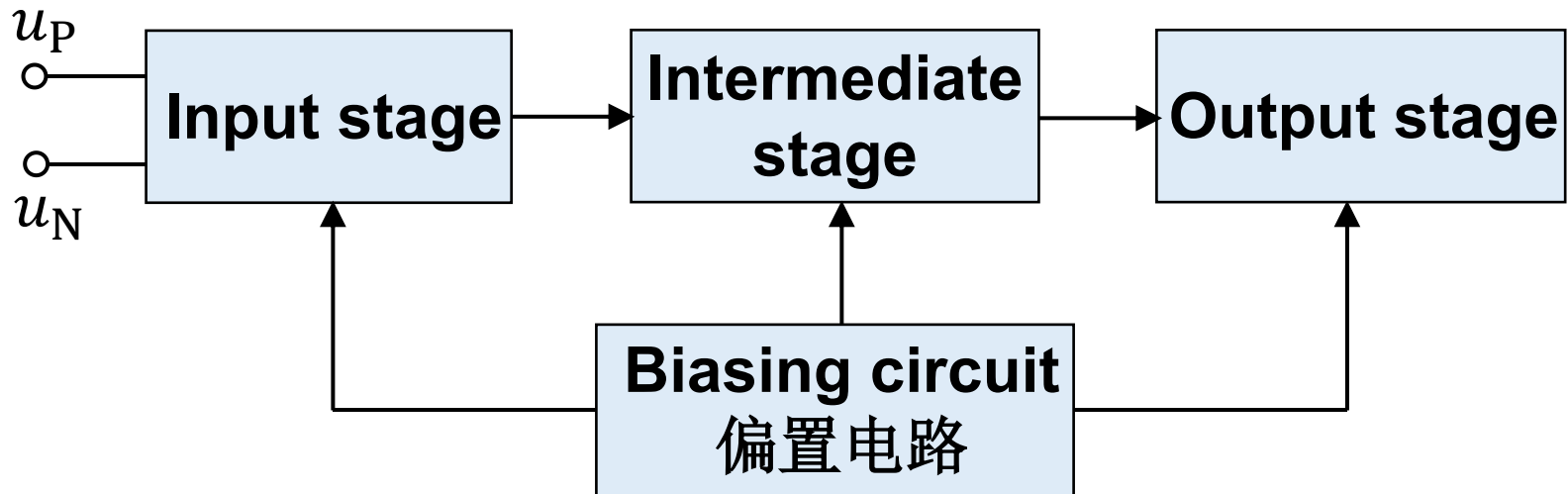
Temperature effect is the major source

Suppress the zero-drift error:

- a) Negative feedback DC circuit 直流负反馈电路
- b) **Differential amplifier circuit** 采用差分放大电路

4.2 Differential amplifier 差分放大电路

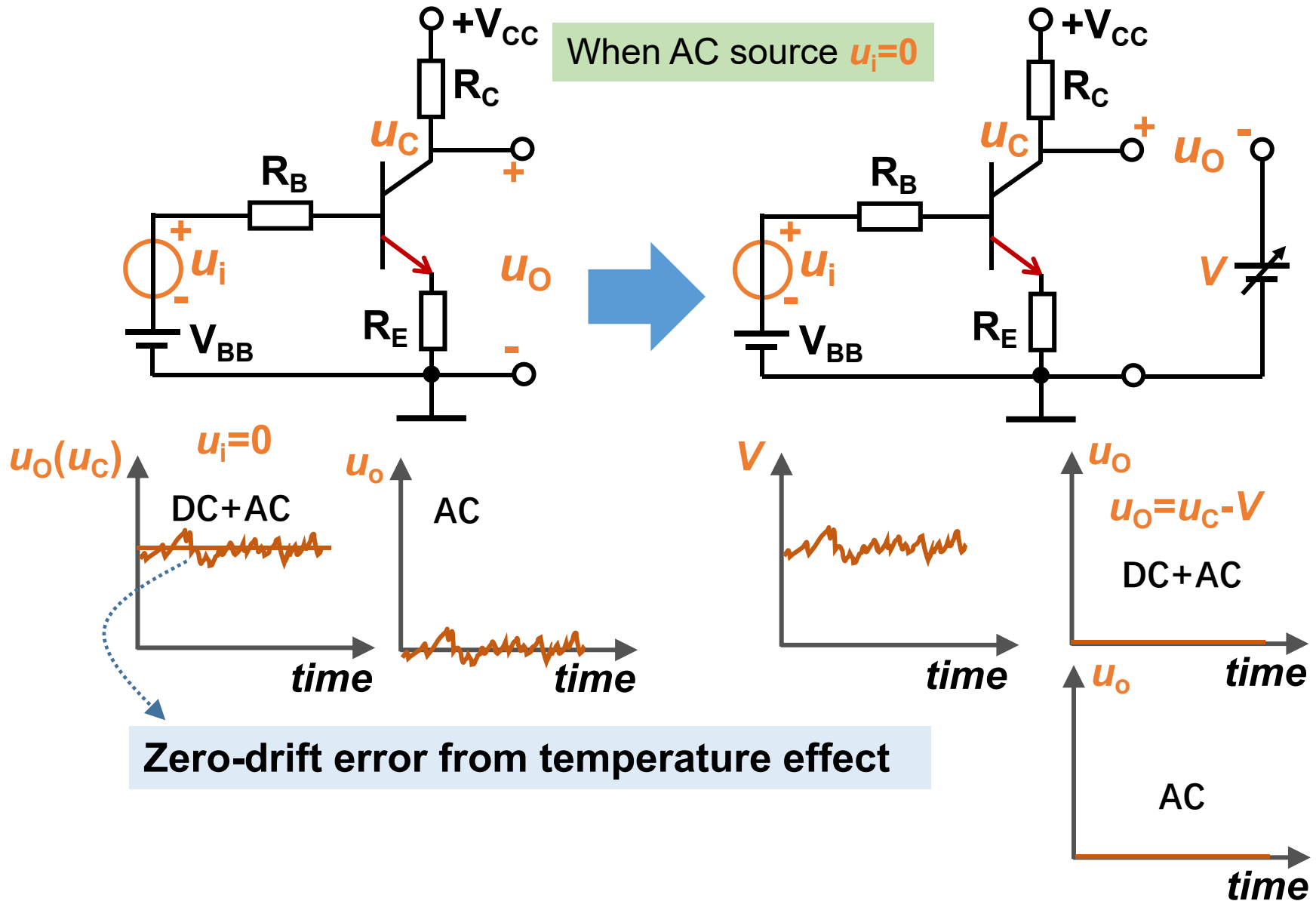
The structure of the integrated operational circuit (IOC)



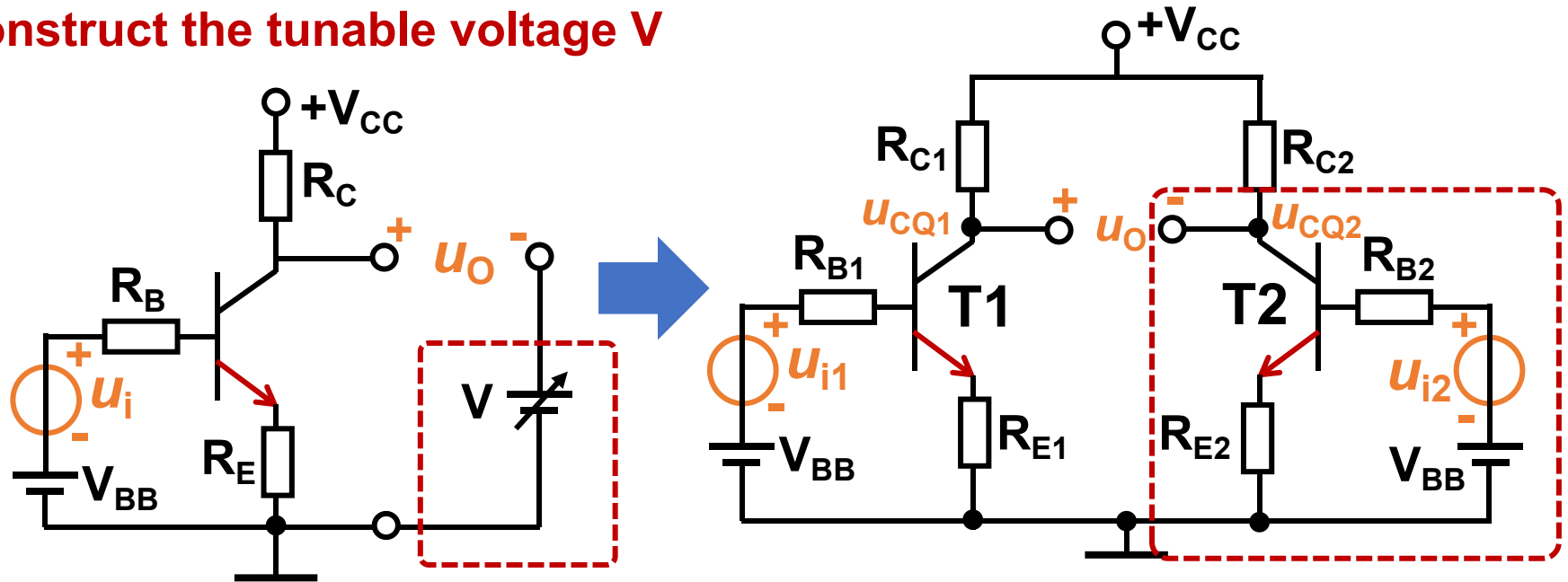
Input stage: suppress zero-drift error, large input resistance, high voltage gain

Use **differential amplifier circuit** 差分放大电路

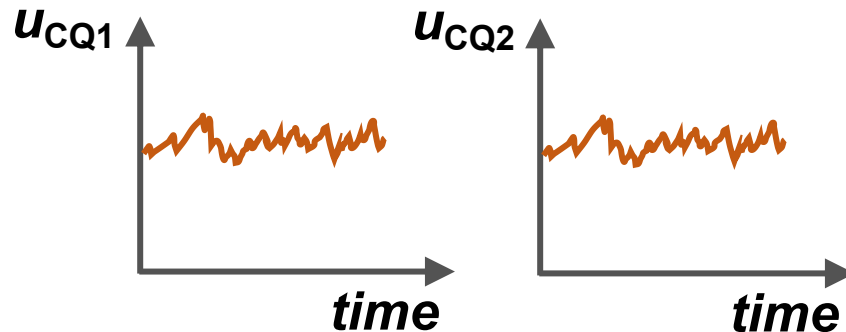
4.2 Differential amplifier 差分放大电路



Construct the tunable voltage V



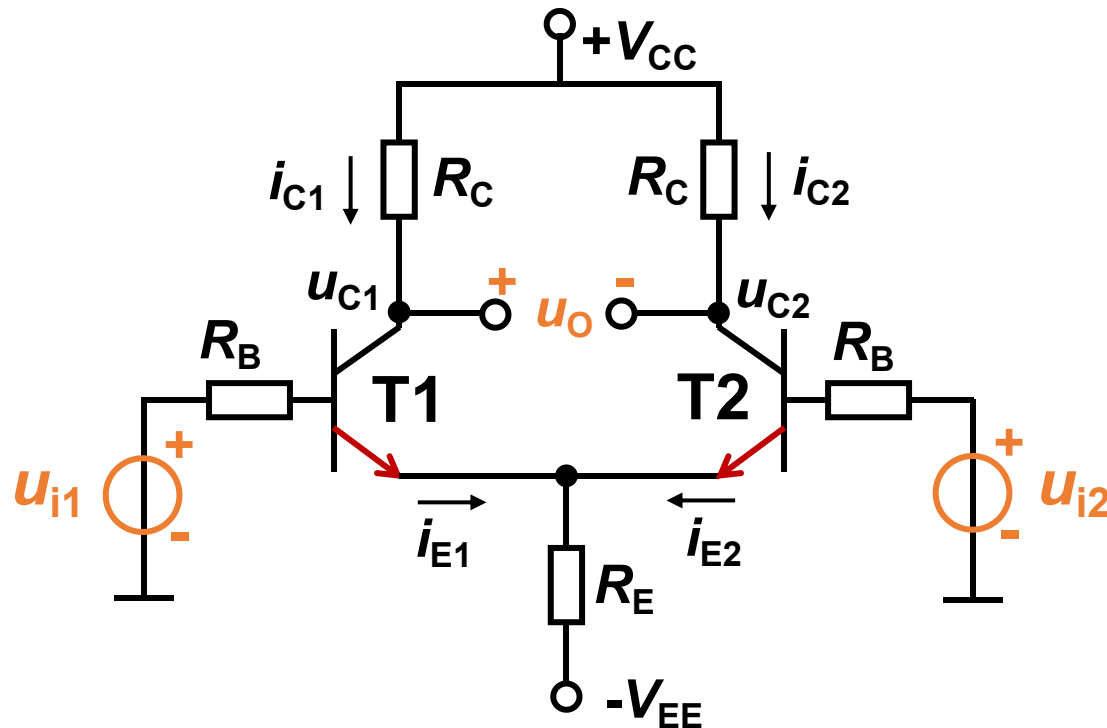
When T_1 and T_2 are identical, $R_{B1}=R_{B2}$, $R_{C1}=R_{C2}$, $R_{E1}=R_{E2}$



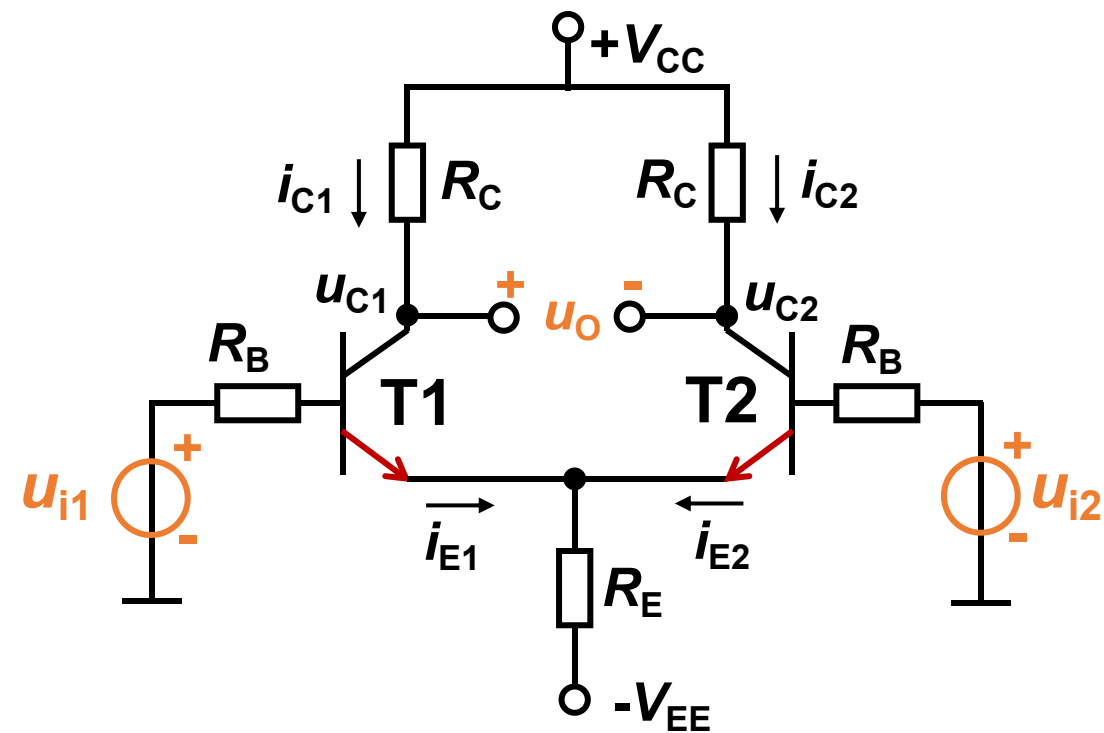
$$u_{CQ1} = u_{CQ2}$$

$$u_o = u_{CQ1} - u_{CQ2} = 0$$

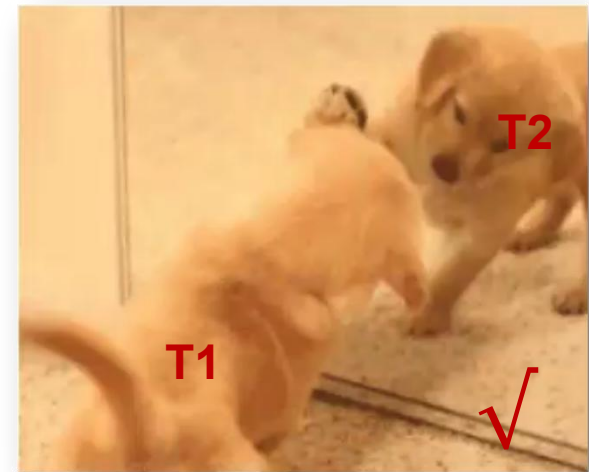
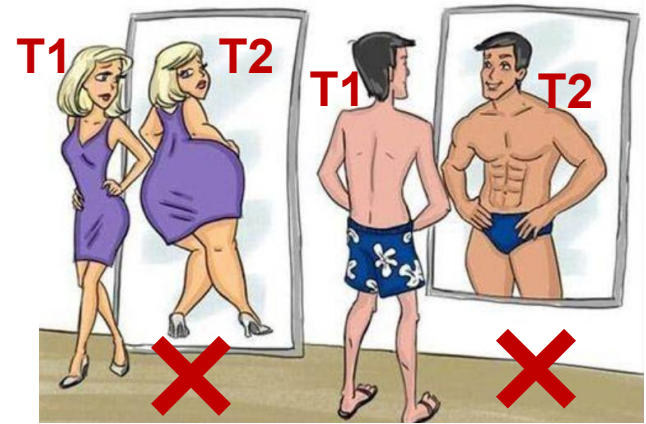
4.2.1 “Long-tailed” differential amplifier 长尾式差分放大电路

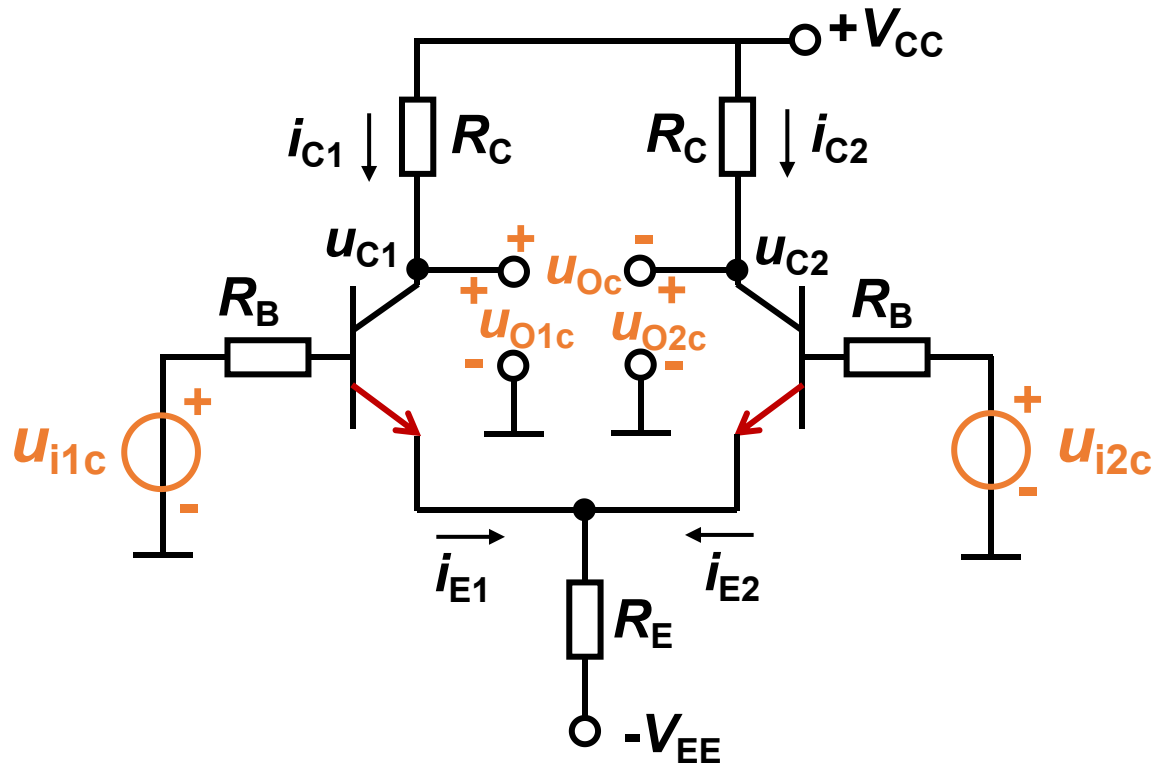


- T1 and T2 are identical
- The circuit is symmetric
- Share R_E
- Two input signal sources: u_{i1} and u_{i2}



□ T_1 and T_2 are identical

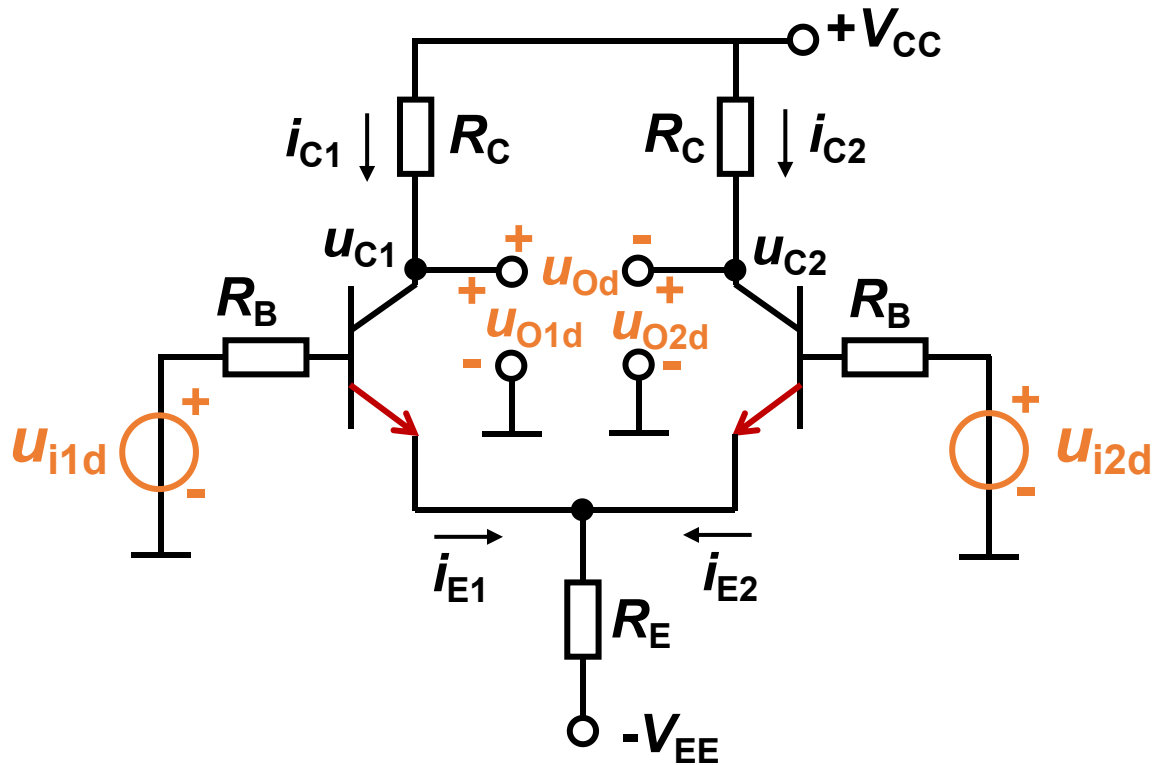




□ Common-mode signal 共模信号 ($u_{i1c}=u_{i2c}$)

When $u_{i1c}=u_{i2c} \Rightarrow u_{O1c}=u_{O2c} \Rightarrow u_{Oc}=u_{O1c}-u_{O2c}=0$

The common-mode signal is strongly suppressed in differential amplifier (no output AC signal)



□ Differential-mode signal 差模信号 ($u_{i1d} = -u_{i2d}$)

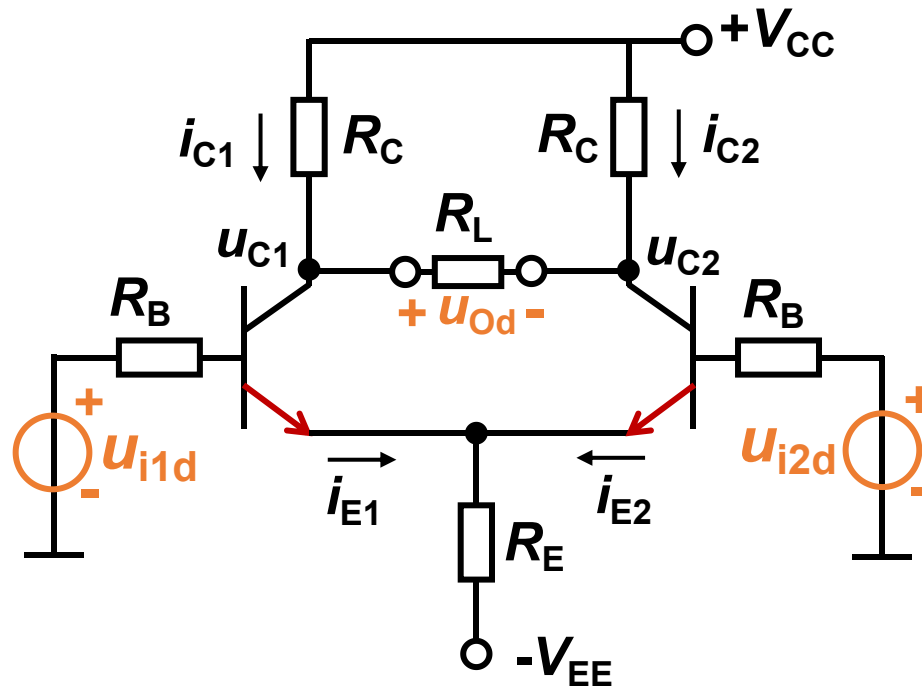
When $u_{i1d} = -u_{i2d} \Rightarrow u_{O1d} = -u_{O2d} \Rightarrow u_{Od} = u_{O1d} - u_{O2d} = 2u_{O1d}$

Question: will R_E decrease the voltage gain?

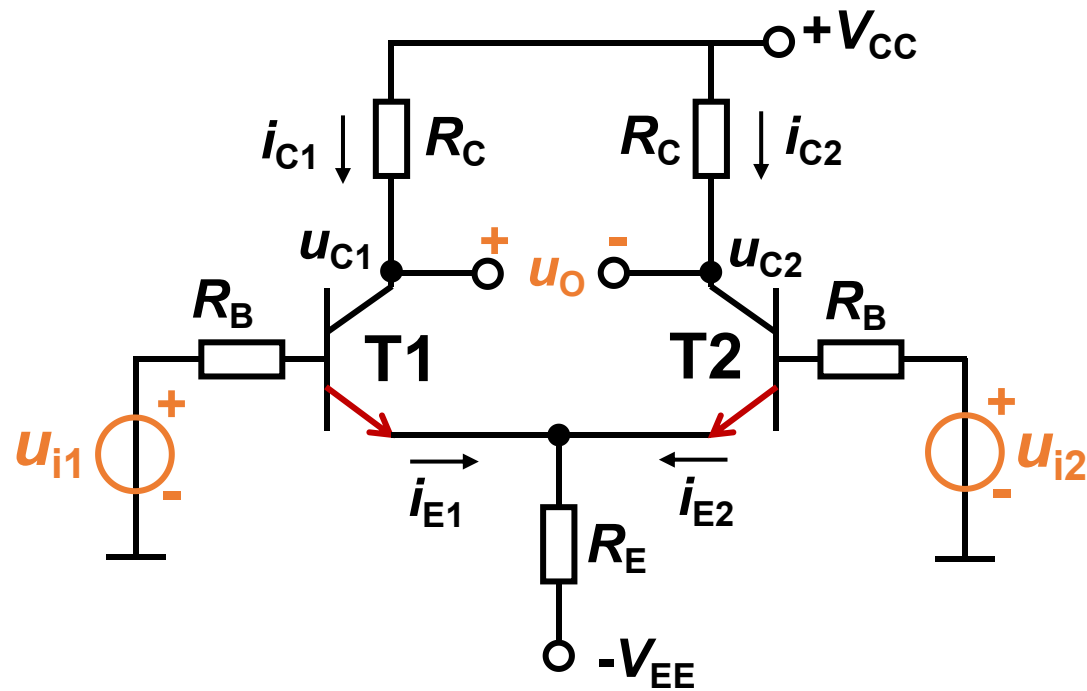
1) Differential-mode signal 差模信号 ($u_{i1d} = -u_{i2d}$)

The input signal: $u_{id} = 2u_{i1d} = -2u_{i2d}$

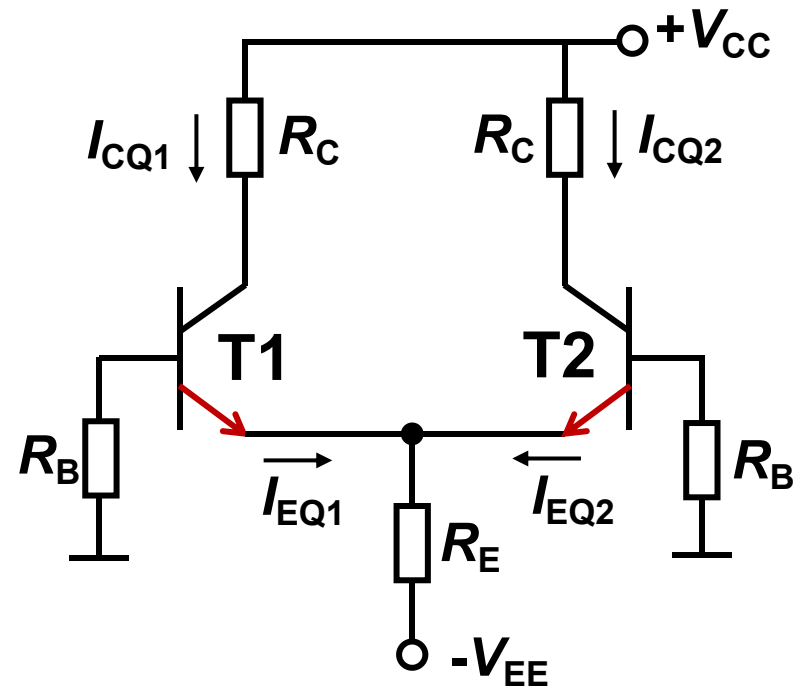
Dual-ended input and output mode 双端输入/输出模式

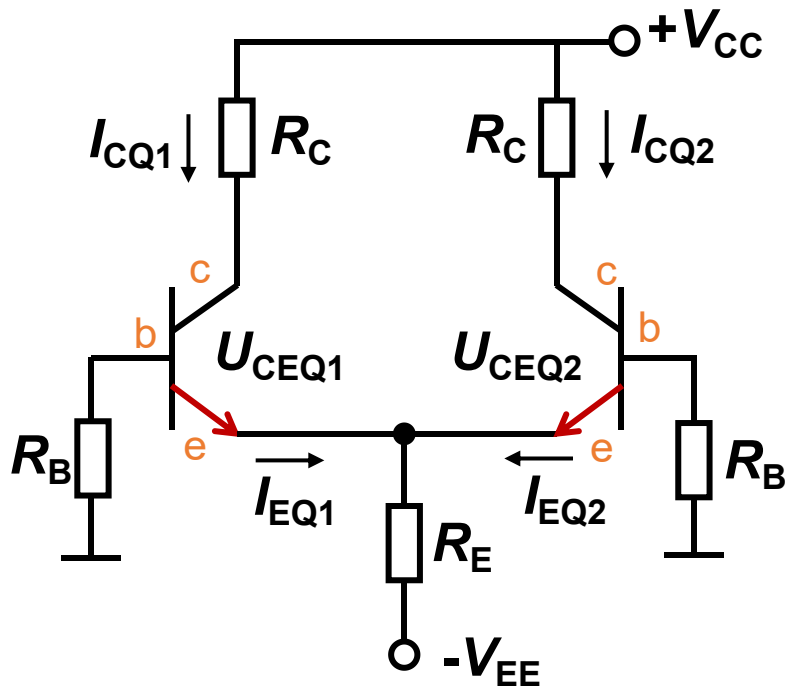


Analyze DC circuit 静态分析



DC circuit





Due to the symmetry

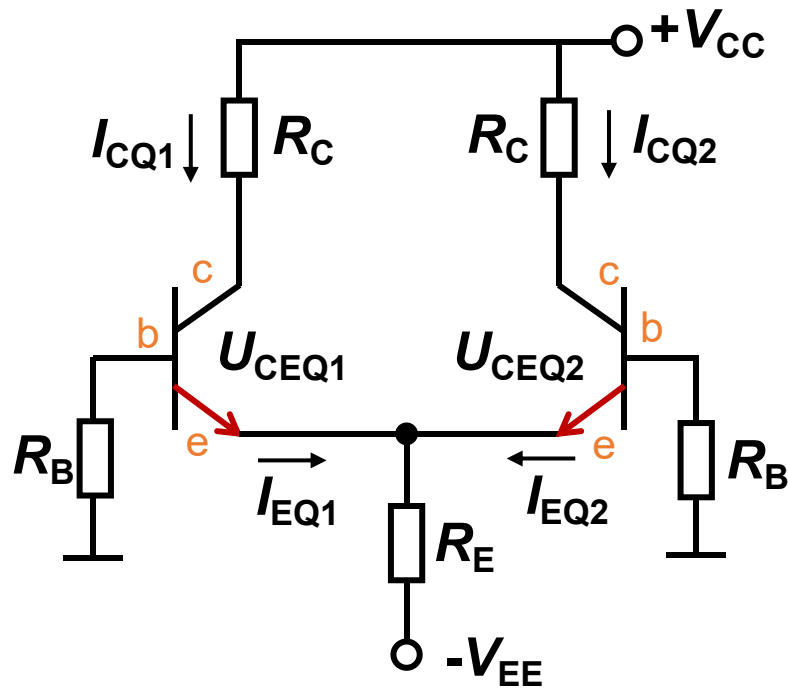
$$I_{CQ1} = I_{CQ2} = I_{CQ}$$

$$I_{EQ1} = I_{EQ2} = I_{EQ}$$

$$U_{CEQ1} = U_{CEQ2} = U_{CEQ}$$

$$U_{BEQ1} = U_{BEQ2} = U_{BEQ}$$

$$\beta_1 = \beta_2 = \beta$$



Analyze the Input circuit:

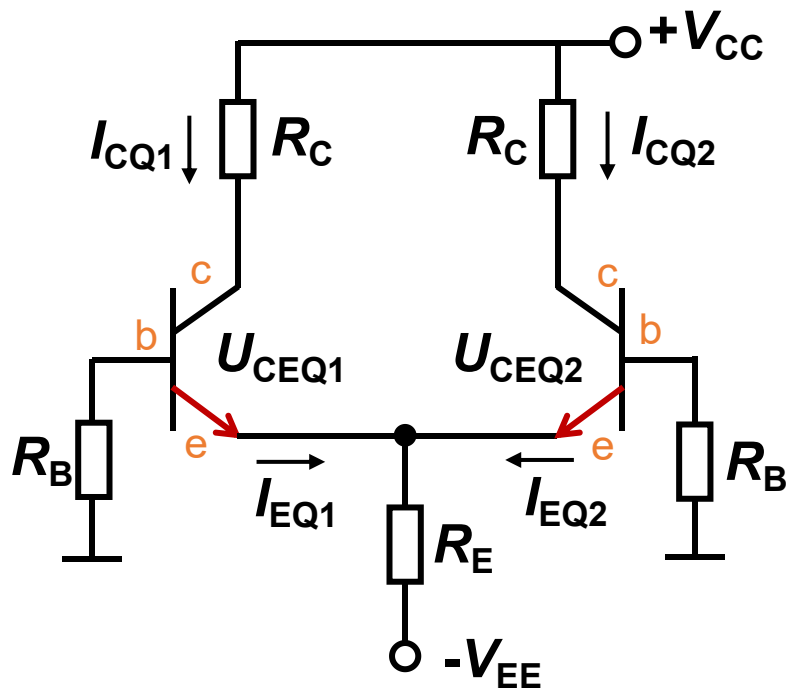
$$V_{EE} = 2I_{EQ}R_E + U_{BEQ} + I_{BQ}R_B$$

$$I_{EQ} = (1 + \bar{\beta})I_{BQ}$$

$$I_{BQ} = \frac{V_{EE} - U_{BEQ}}{2(1 + \bar{\beta})R_E + R_B}$$

$$I_{CQ} = \bar{\beta}I_{BQ} = \bar{\beta} \frac{V_{EE} - U_{BEQ}}{2(1 + \bar{\beta})R_E + R_B}$$

$$I_{EQ} = (1 + \bar{\beta}) \frac{V_{EE} - U_{BEQ}}{2(1 + \bar{\beta})R_E + R_B}$$



Analyze the output circuit:

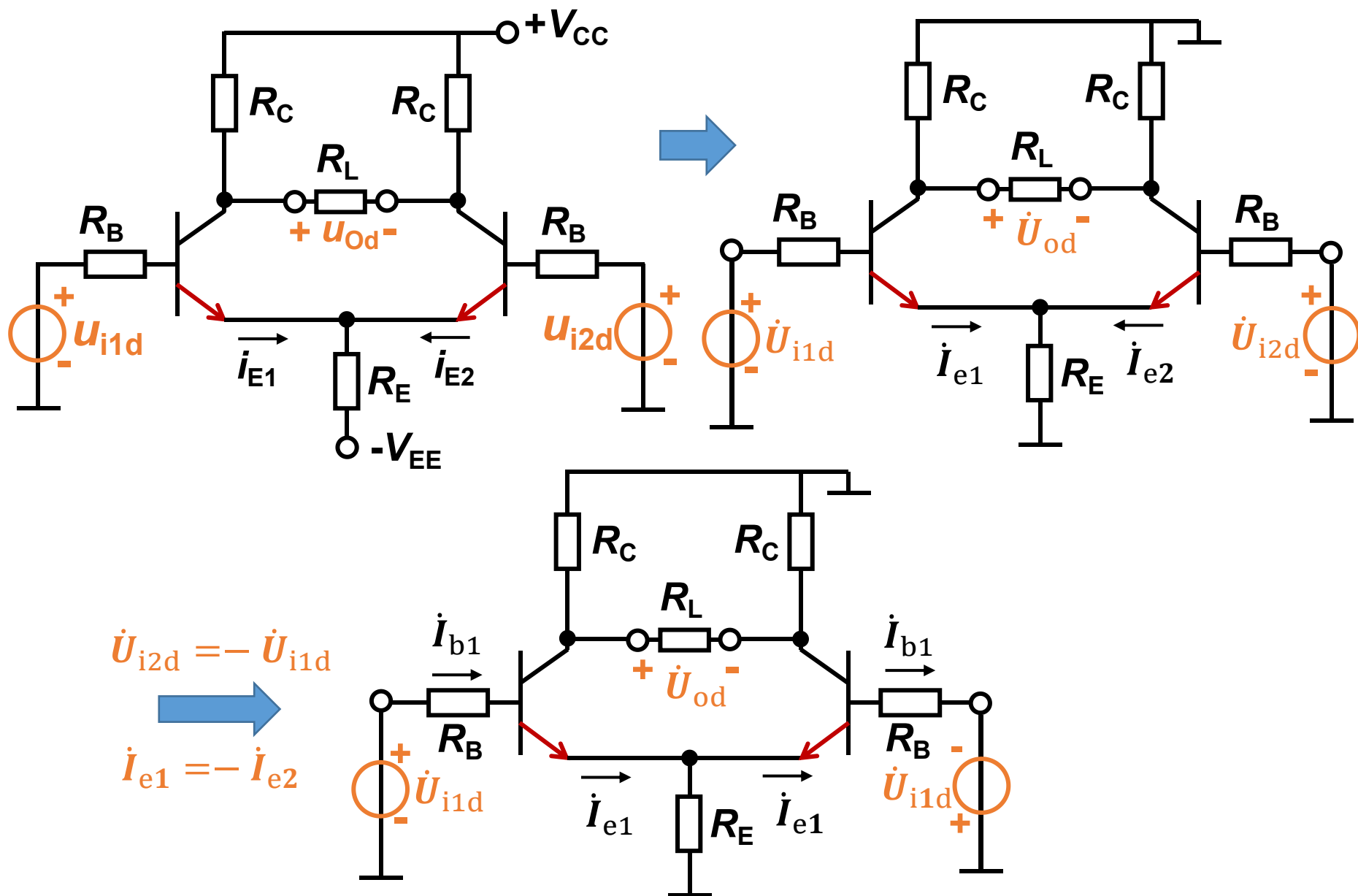
$$V_{CC} + V_{EE} = I_{CQ} R_C + U_{CEQ} + 2I_{EQ} R_E$$

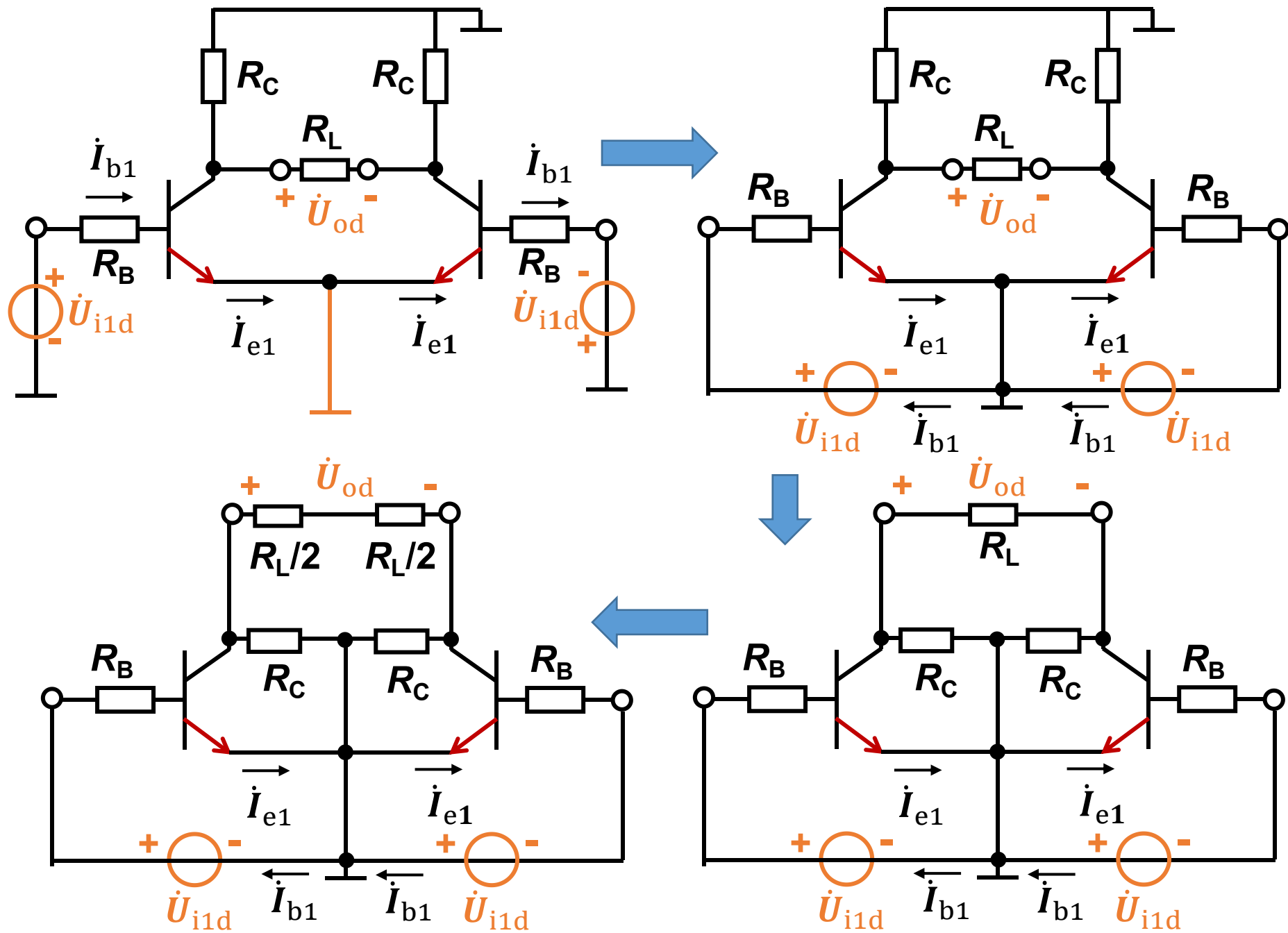


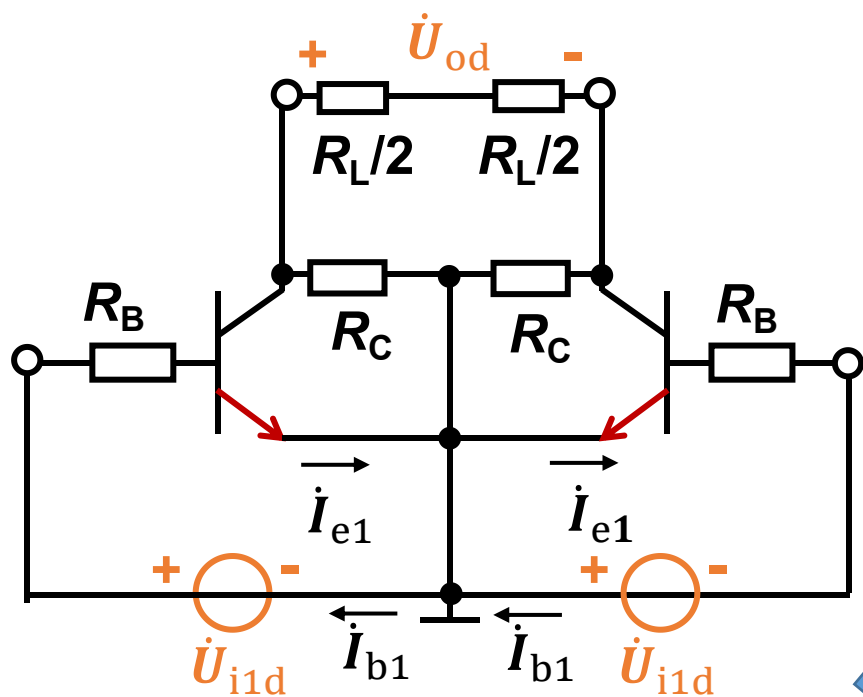
$$U_{CEQ} =$$

$$V_{CC} + V_{EE} - \frac{(V_{EE} - U_{BEQ})(\bar{\beta} R_C + 2\bar{\beta} R_E + 2R_E)}{2(1 + \bar{\beta}) R_E + R_B}$$

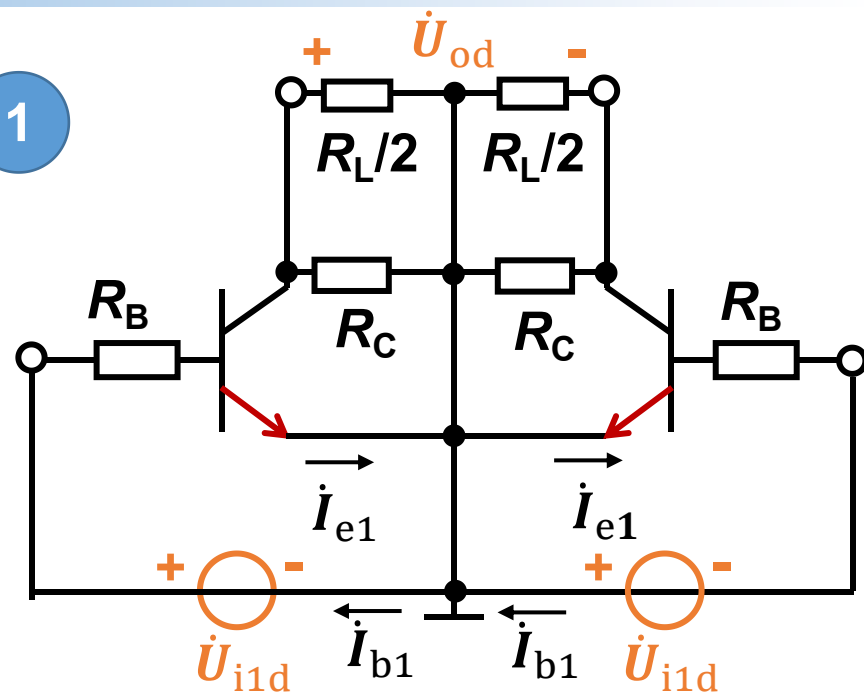
Analyze AC circuits 动态分析



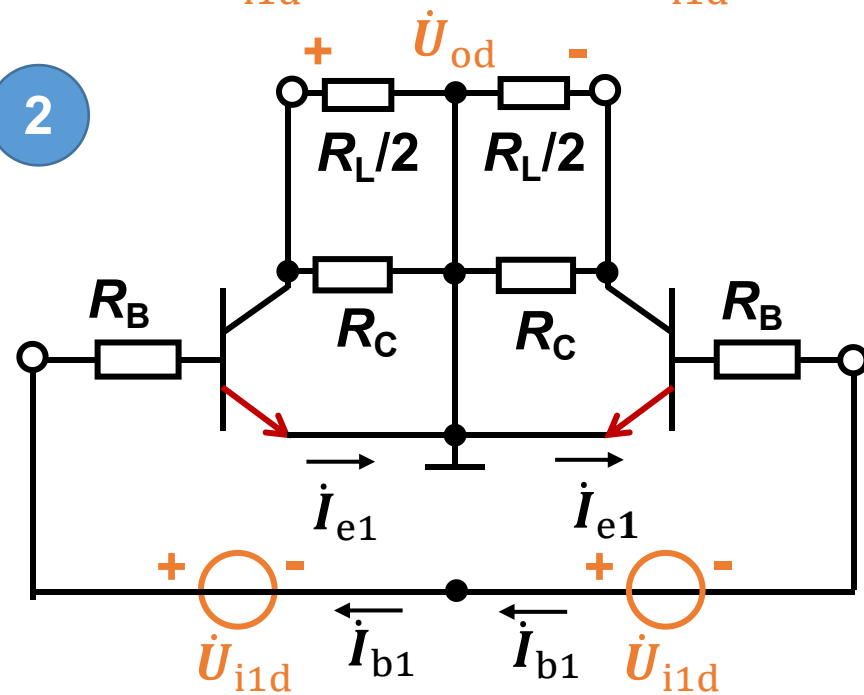


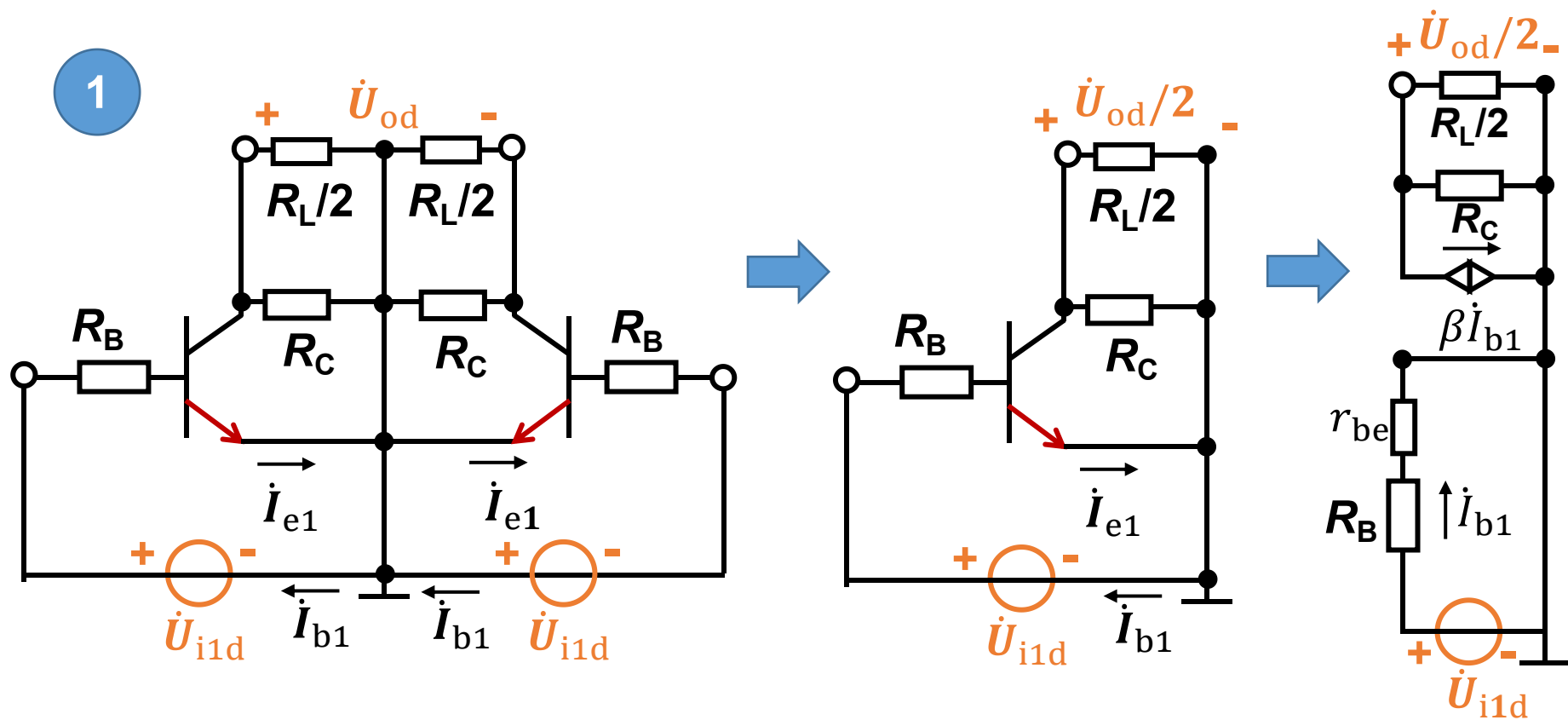


1

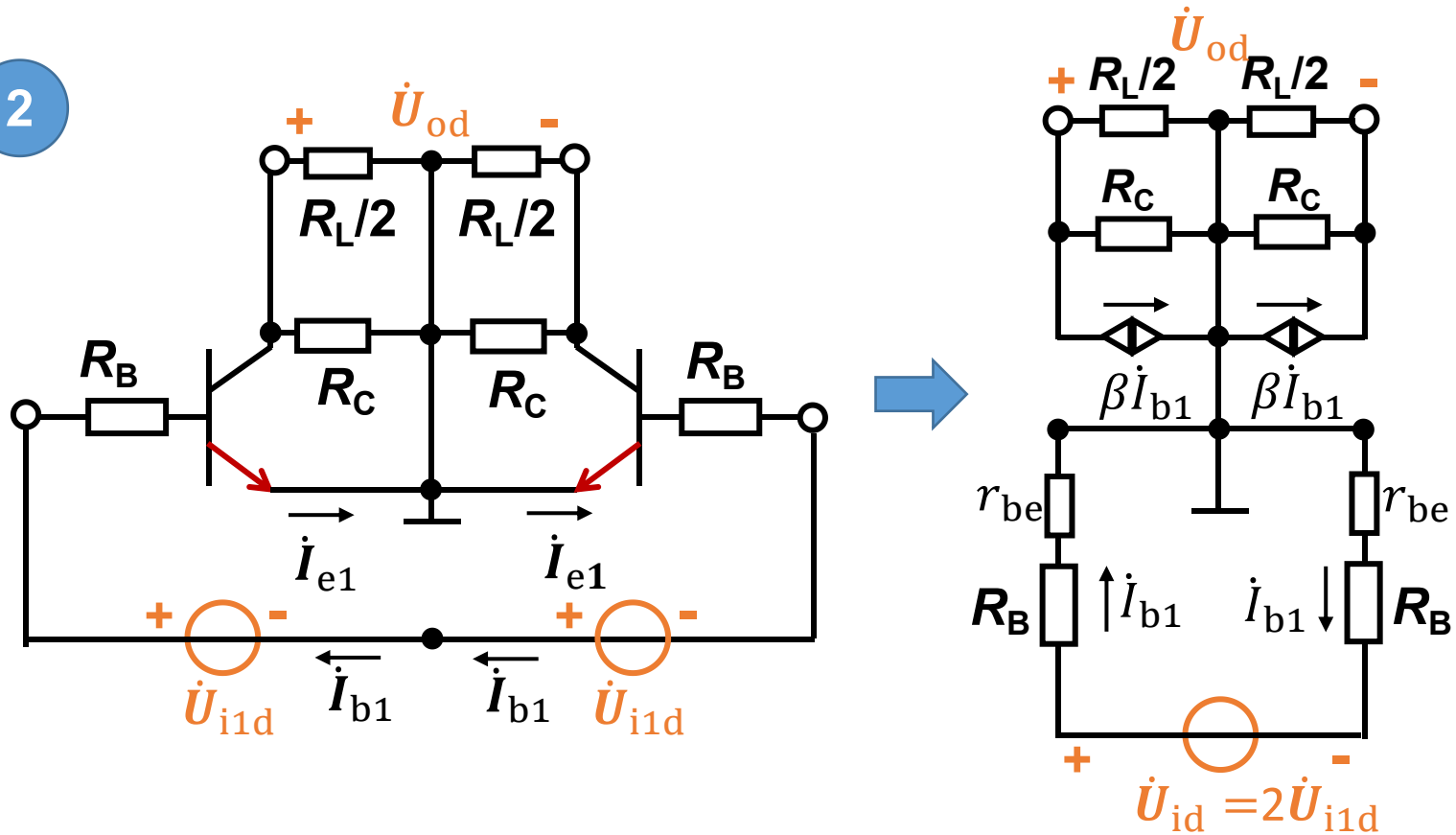


2

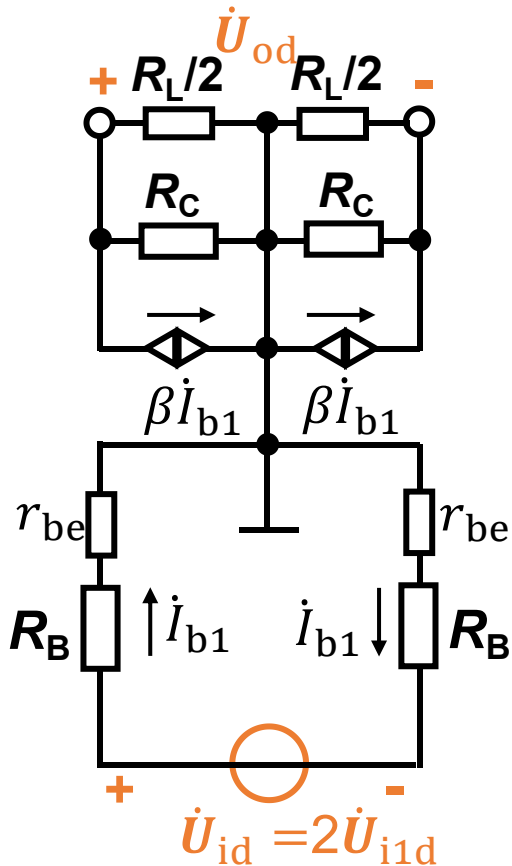




2



That's why the input AC voltage in differential mode is $2\dot{U}_{i1d}$.



Input resistance

$$R_{id} = 2 (r_{be} + R_B)$$

The input voltage

$$\dot{U}_{id} = 2\dot{U}_{i1d} = 2\dot{I}_{b1}(r_{be} + R_B)$$

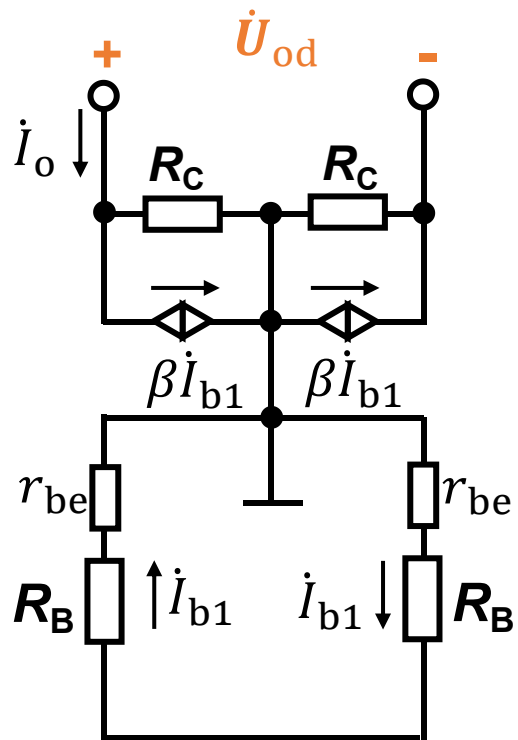
The output voltage

$$\dot{U}_{od} = - 2\beta\dot{I}_{b1}(R_C // R_{L/2})$$

The voltage gain

$$\dot{A}_d = \frac{\dot{U}_{od}}{\dot{U}_{id}} = - \frac{\beta(R_C // R_{L/2})}{r_{be} + R_B}$$

- The voltage gain is similar with the value of common-e amplifier
- Sacrifice one transistor to achieve better stability



Output resistance

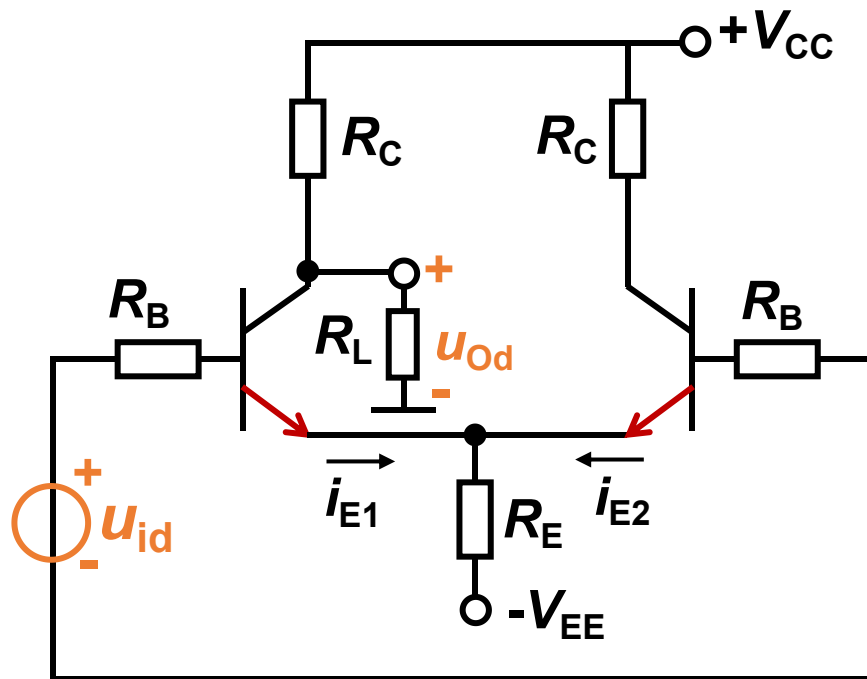
$$i_{b1} = 0$$

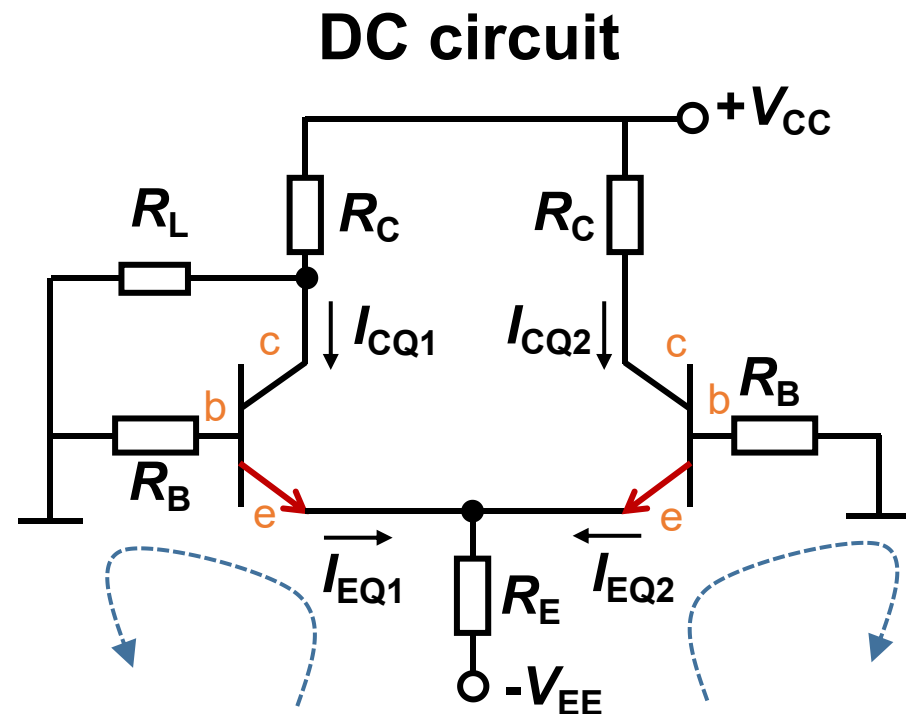
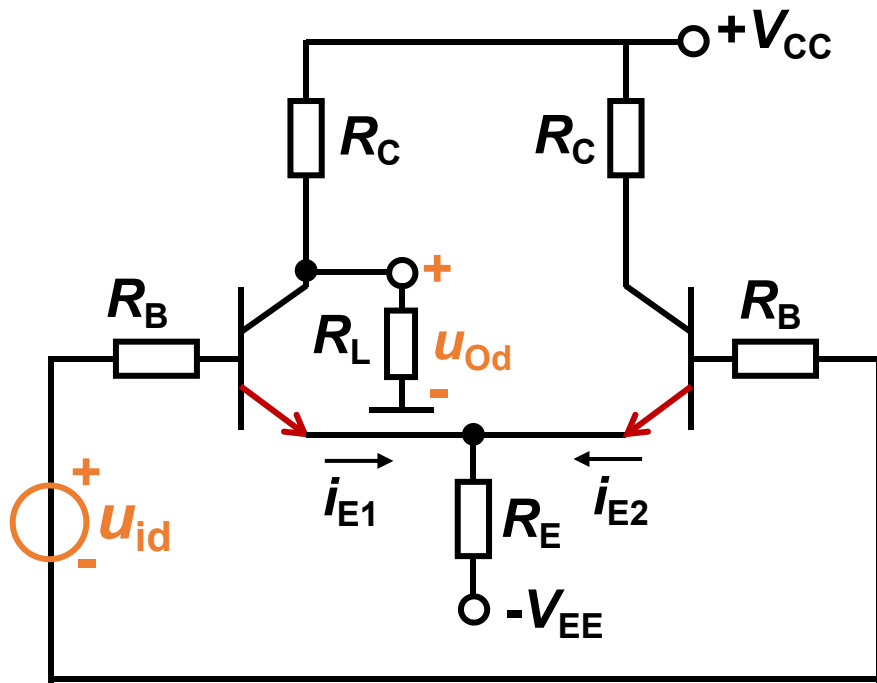
$$R_{od} = 2 R_C$$

2) Differential-mode signal 差模信号

Dual-ended input, Single-ended output

双端输入, 单端输出





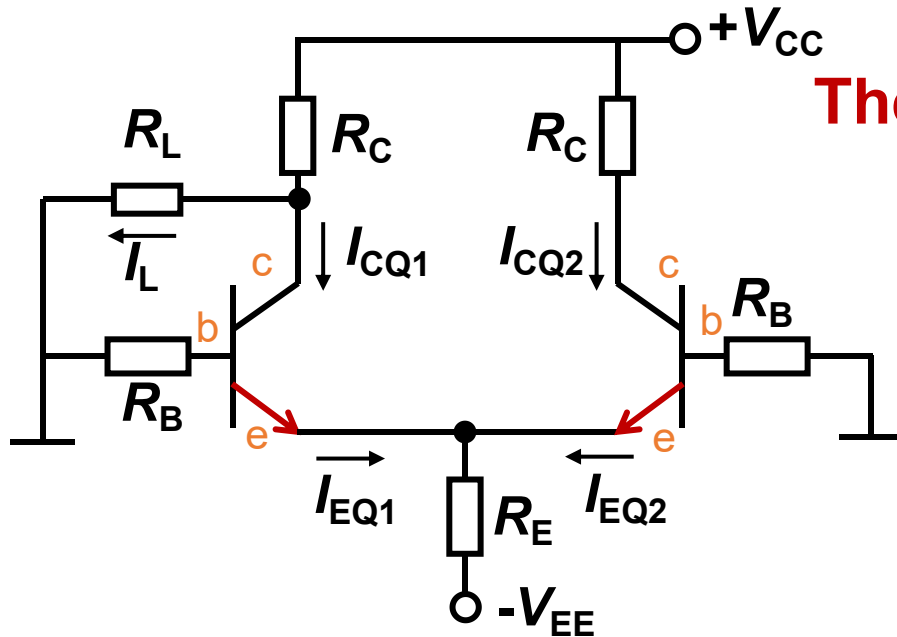
The input circuits $\begin{cases} -V_{EE} + (I_{EQ1} + I_{EQ2})R_E + U_{BEQ} + I_{BQ1}R_B = 0 \\ -V_{EE} + (I_{EQ1} + I_{EQ2})R_E + U_{BEQ} + I_{BQ2}R_B = 0 \end{cases}$

$$I_{BQ1} = I_{BQ2}, I_{CQ1} = I_{CQ2}, I_{EQ1} = I_{EQ2}$$

$$I_{EQ} = (1 + \bar{\beta})I_{BQ} \quad I_{BQ} = \frac{V_{EE} - U_{BEQ}}{2(1 + \bar{\beta})R_E + R_B}$$

The input circuits are still symmetric.

DC circuit

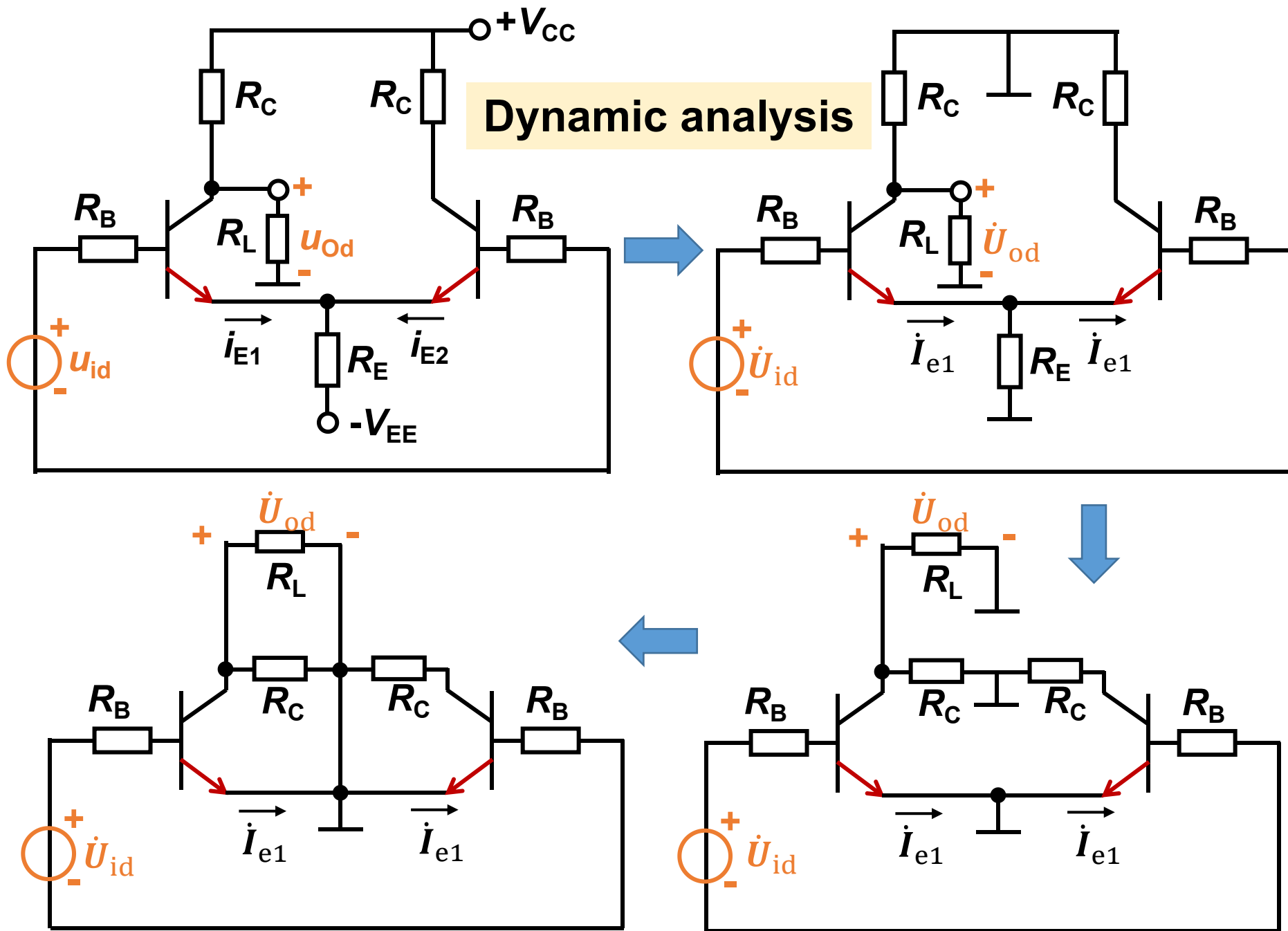


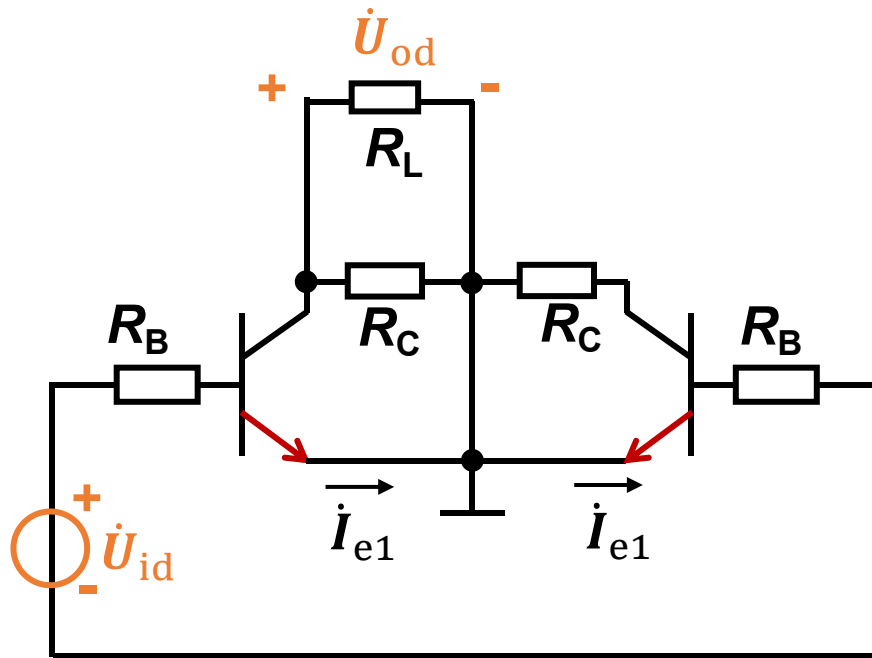
The output circuits are asymmetric

$$\left\{ \begin{array}{l} I_L R_L + (I_{CQ} + I_L) R_C = V_{CC} \\ -V_{EE} + 2I_{EQ} R_E + U_{CEQ1} = I_L R_L \\ V_{CC} + V_{EE} = I_{CQ} R_C + U_{CEQ2} + 2I_{EQ} R_E \end{array} \right.$$

The negative feedback of R_E can reduce the zero-drift error.
The zero-drift error is larger than the dual-ended output mode

Dynamic analysis



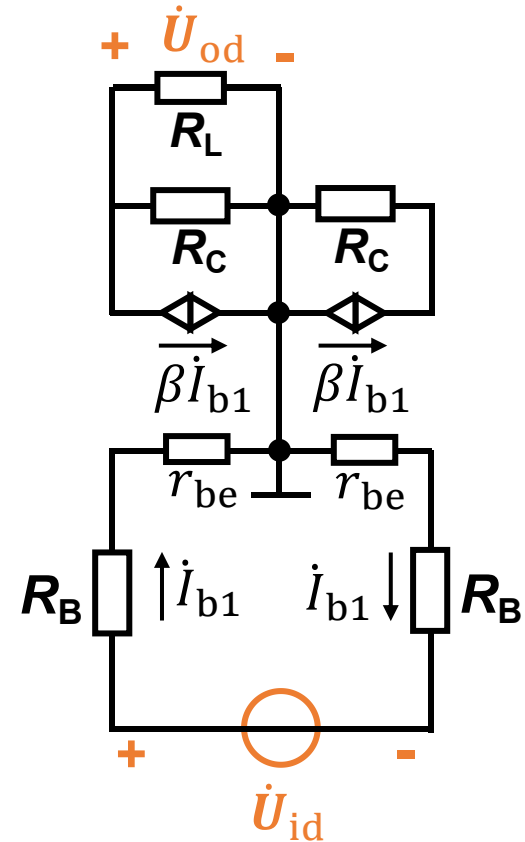


The input voltage

$$\dot{U}_{id} = 2\dot{I}_{b1}(r_{be} + R_B)$$

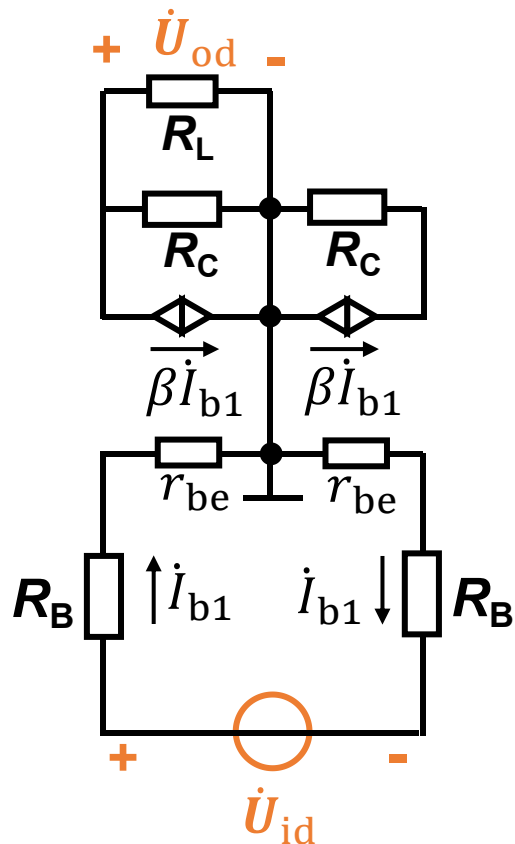
The output voltage

$$\dot{U}_{od} = -\beta\dot{I}_{b1}(R_C // R_L)$$



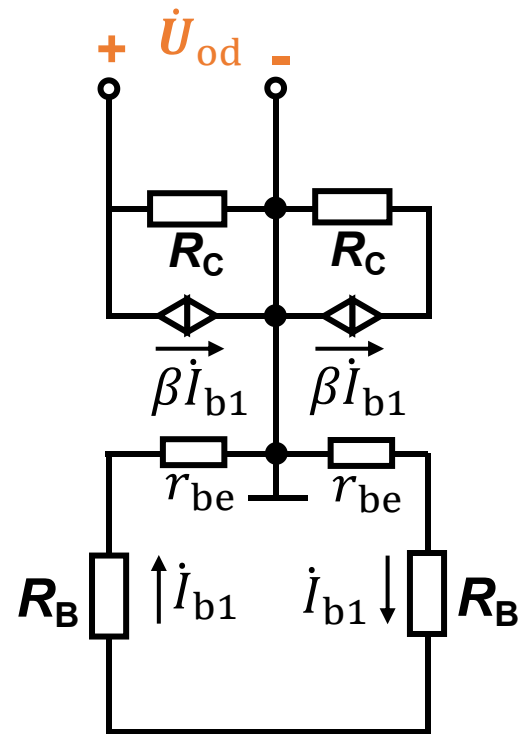
The voltage gain

$$\dot{A}_d = \frac{\dot{U}_{od}}{\dot{U}_{id}} = -\frac{\beta(R_C // R_L)}{2(r_{be} + R_B)}$$



Input resistance

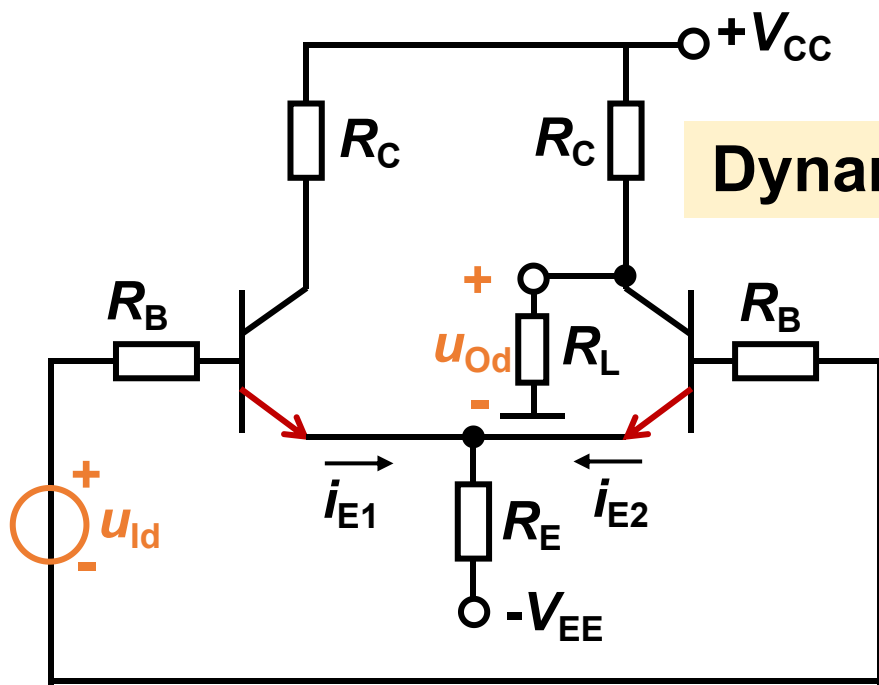
$$R_{id} = 2 (r_{be} + R_B)$$



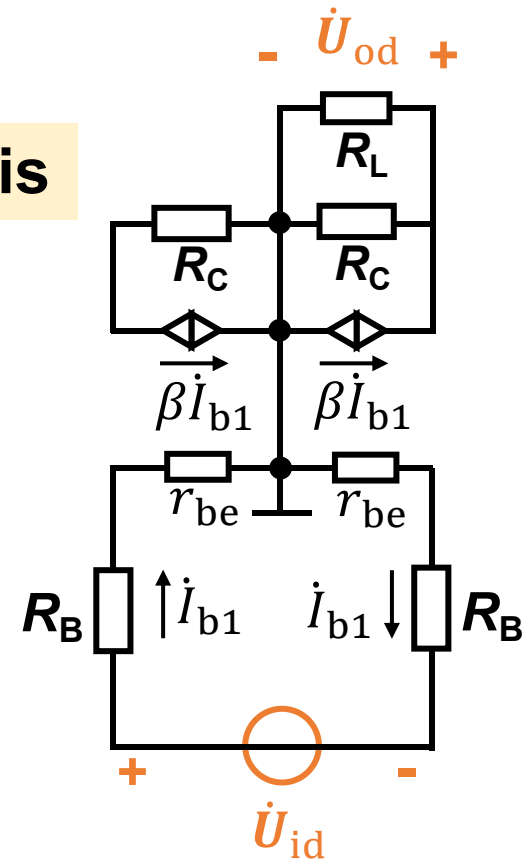
Output resistance

$$i_{b1} = 0$$

$$R_{od} = R_C$$



Dynamic analysis



The input voltage

$$\dot{U}_{id} = 2\dot{I}_{b1}(r_{be} + R_B)$$

The output voltage

$$\dot{U}_{od} = \beta\dot{I}_{b1}(R_C // R_L)$$

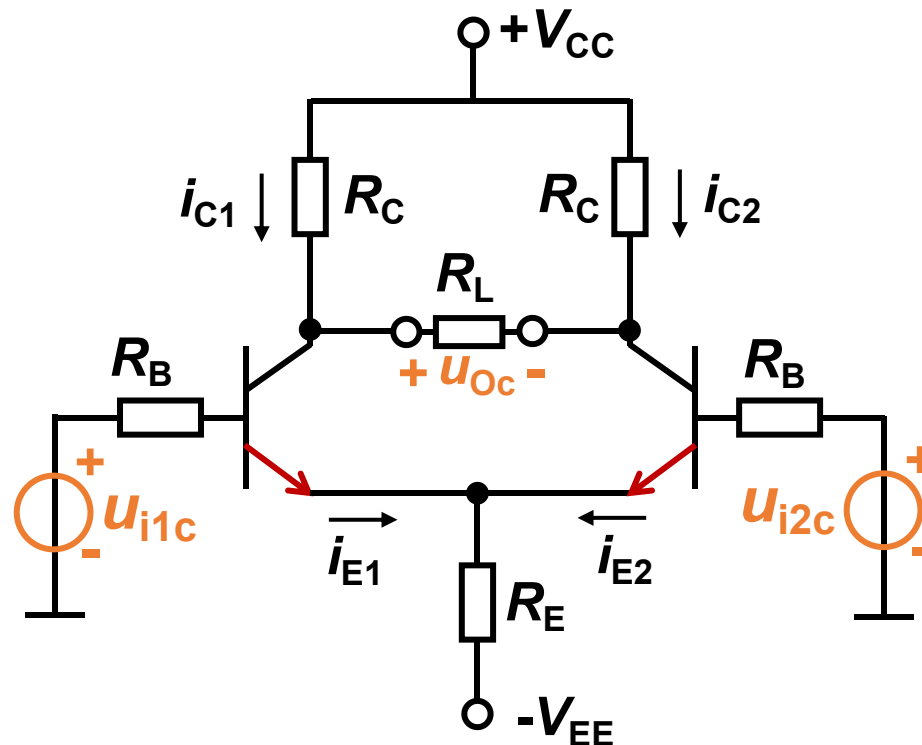
The voltage gain

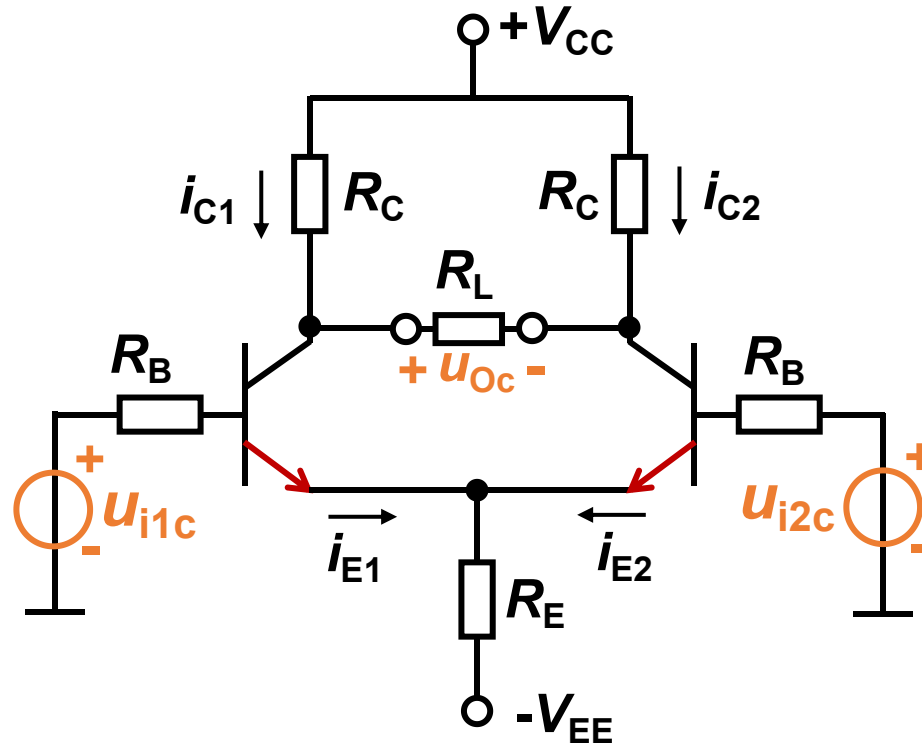
$$\dot{A}_d = \frac{\dot{U}_{od}}{\dot{U}_{id}} = \frac{\beta(R_C // R_L)}{2(r_{be} + R_B)}$$

3) Common-mode signal 共模信号 ($u_{i1c}=u_{i2c}$)

The input signal: $u_{ic}=u_{i1c}=u_{i2c}$

Dual-ended input and output 双端输入/输出





The output voltage

$$\dot{U}_{oc} = 0$$

The voltage gain

$$\dot{A}_c = \frac{\dot{U}_{oc}}{\dot{U}_{ic}} = 0$$

Common-mode rejection ratio 共模抑制比

$$K_{\text{CMR}} = \left| \frac{\dot{A}_d}{\dot{A}_c} \right|$$

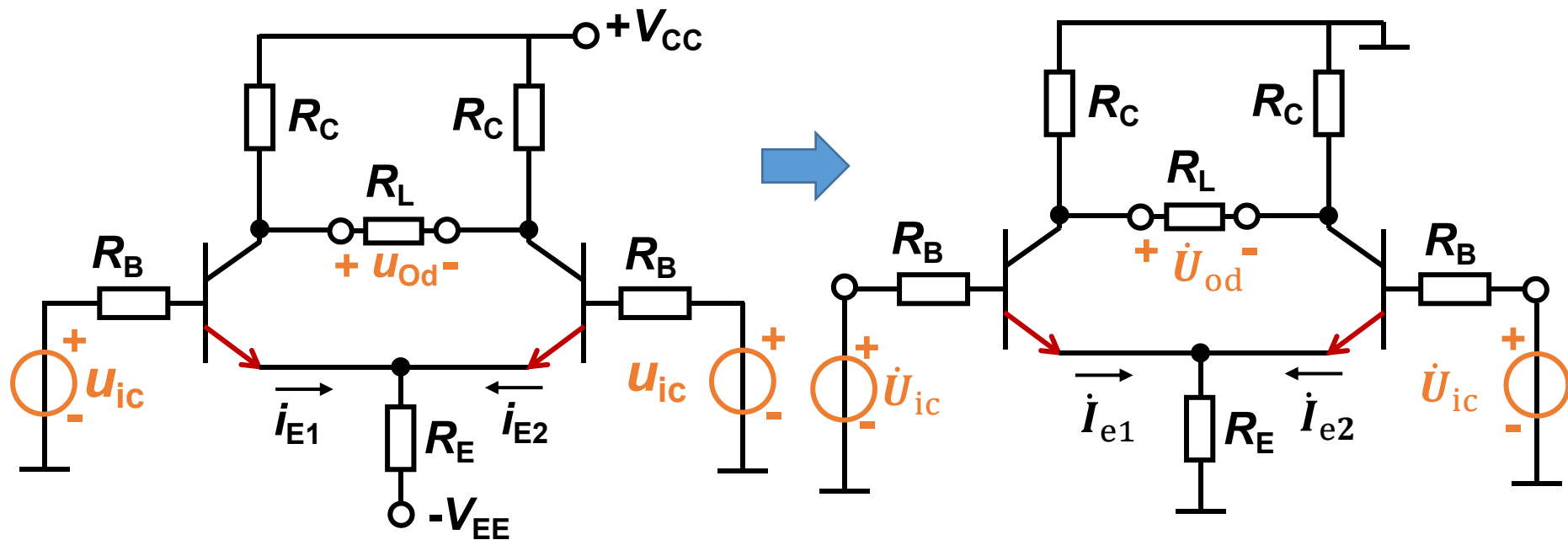
For ideal differential amplifier, in dual-ended output mode

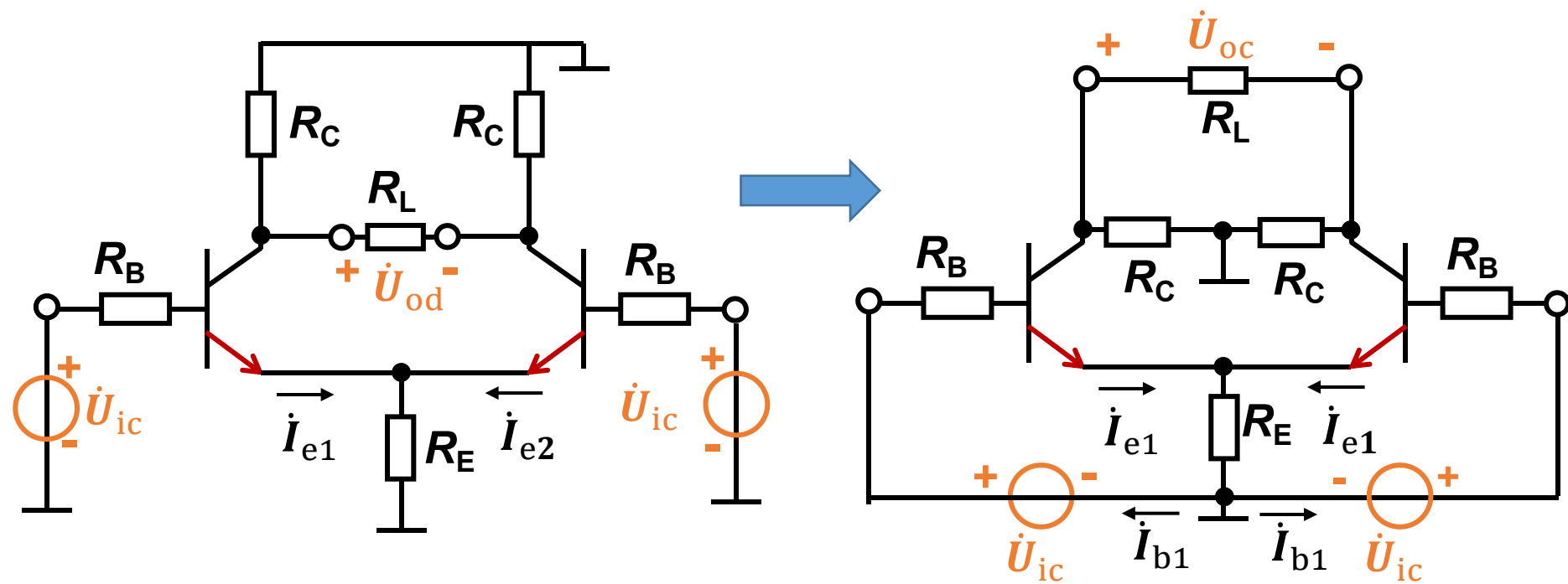
$$K_{\text{CMR}} \rightarrow \infty$$

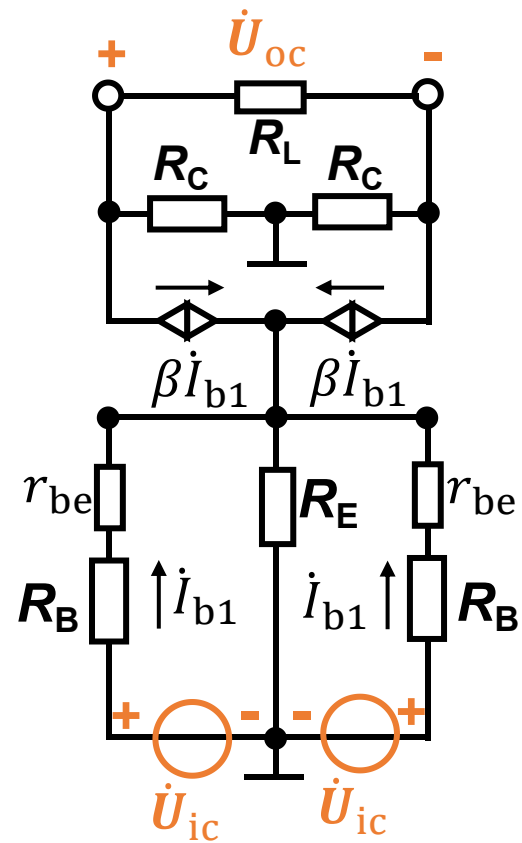
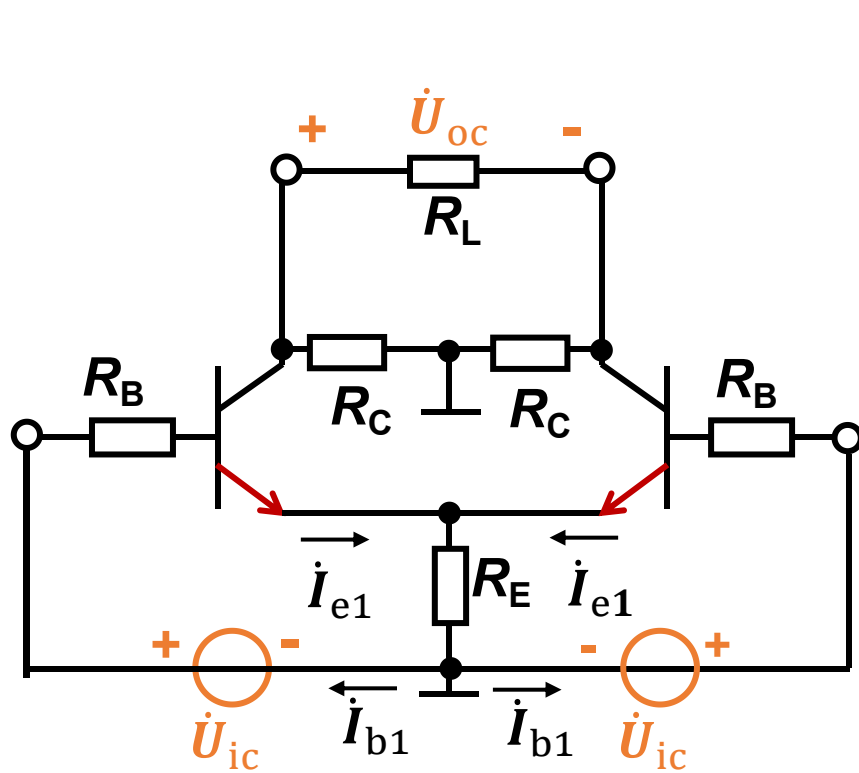
Larger K_{CMR} , better performance

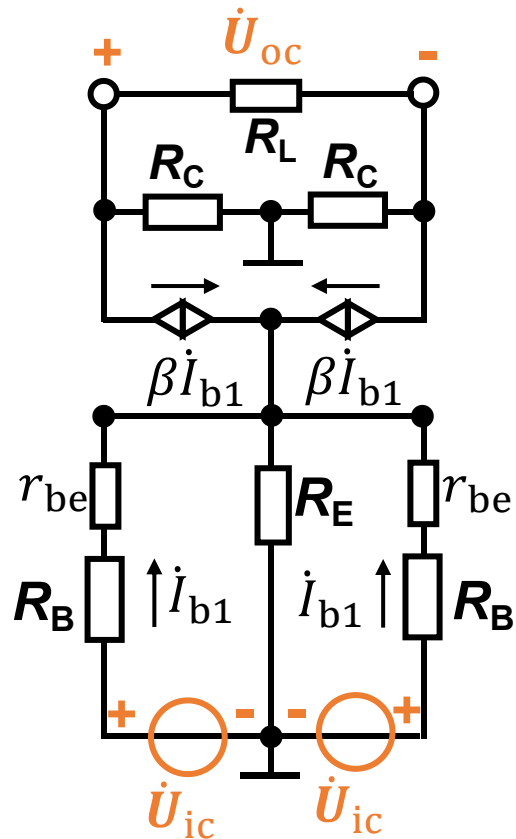
Analyze AC circuits 动态分析

for input and output resistance





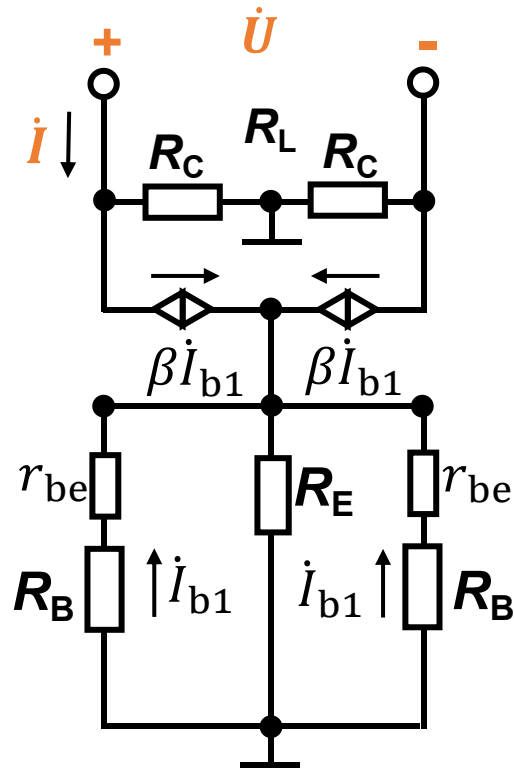




Input resistance

$$R_{ic} = \frac{\text{Input voltage}}{\text{Input current}} = \frac{\dot{U}_{ic}}{\dot{i}_{b1}}$$

$$= r_{be} + R_B + 2(\beta + 1)R_E$$



Output resistance

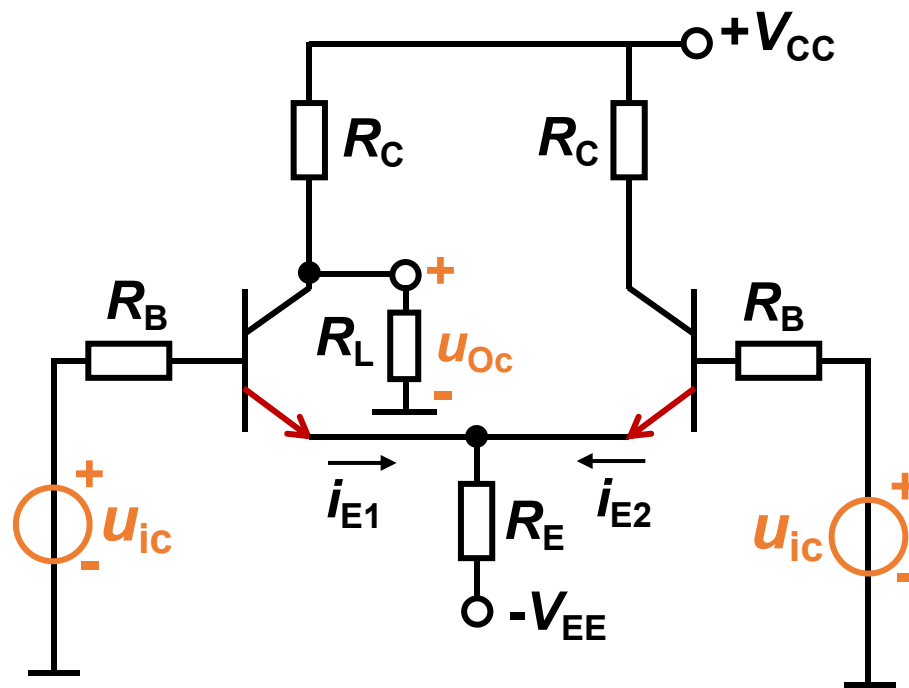
$$\dot{I}_{b1} = 0$$

$$R_{oc} = \frac{\dot{U}}{\dot{I}} = 2R_C$$

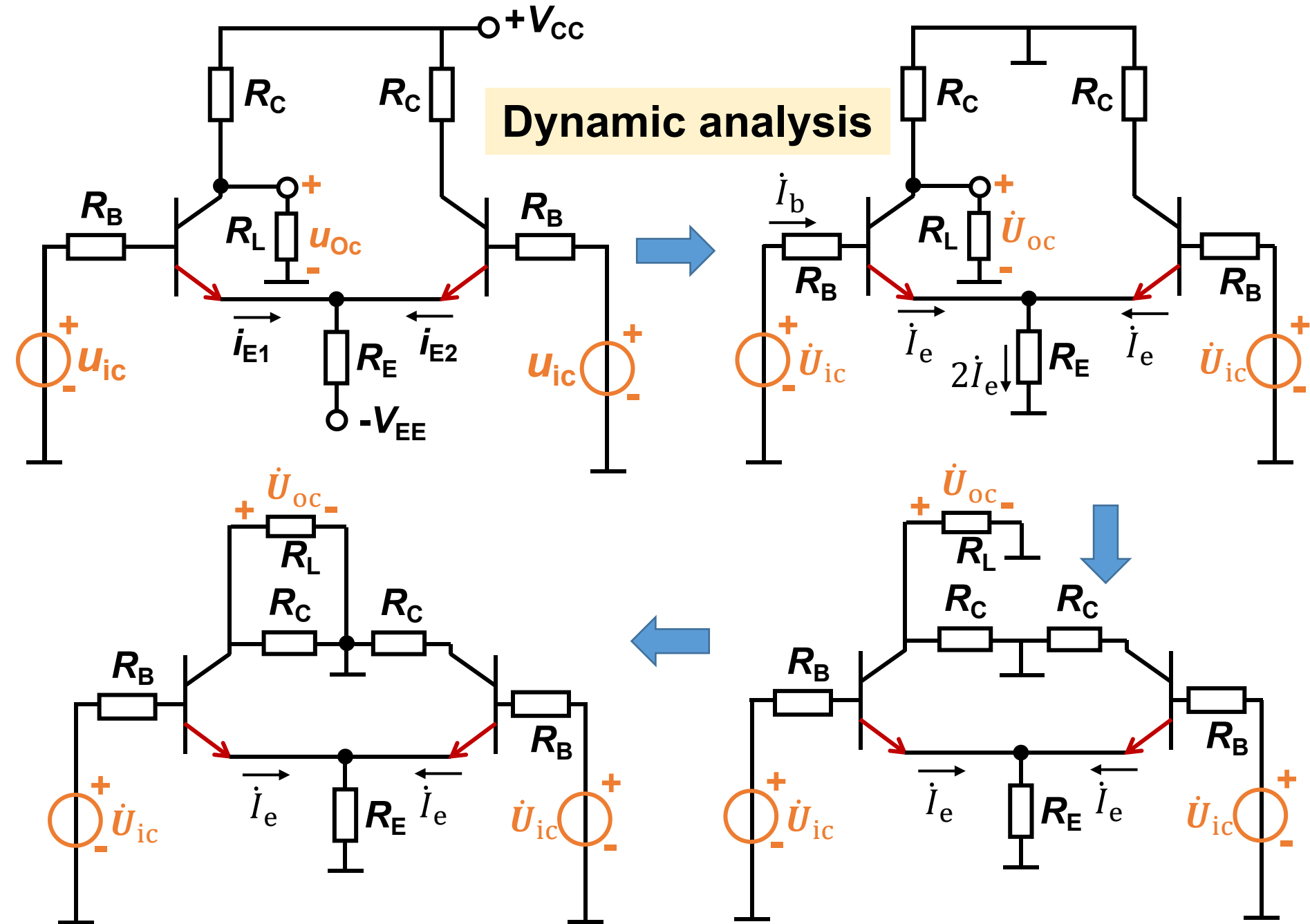
4) Common-mode signal 共模信号 ($u_{i1c}=u_{i2c}$)

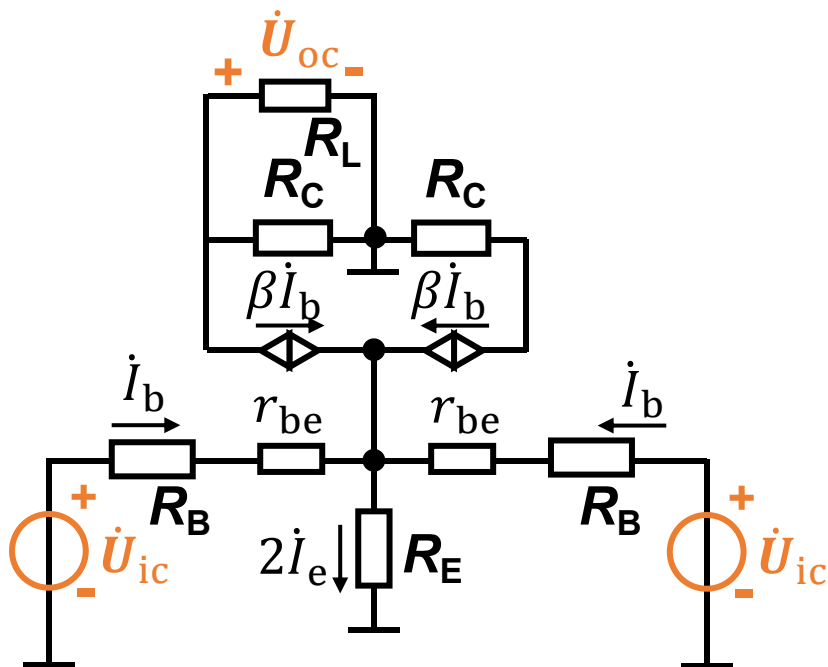
The input signal: $u_{ic}=u_{i1c}=u_{i2c}$

Single-ended output 单端输出



Dynamic analysis





The input voltage

$$\dot{U}_{ic} = \dot{I}_b R_B + \dot{I}_b r_{be} + 2\dot{I}_e R_E$$

The output voltage

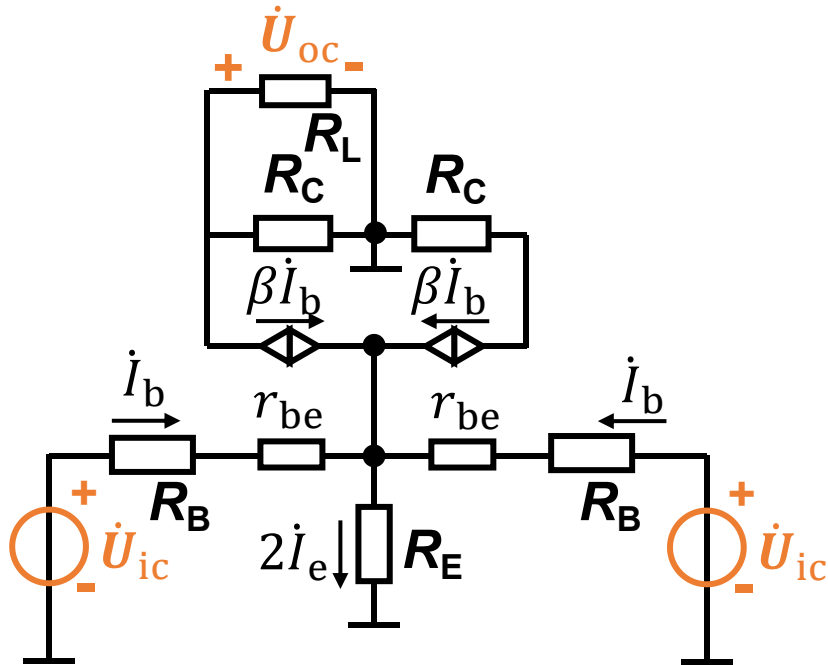
$$\dot{U}_{oc} = -\beta \dot{I}_b (R_C // R_L)$$

The voltage gain

$$\dot{A}_c = \frac{\dot{U}_{oc}}{\dot{U}_{ic}} = -\frac{\beta(R_C // R_L)}{2(1 + \beta)R_E + R_B + r_{be}}$$

Common-mode rejection ratio 共模抑制比

$$K_{CMR} = \left| \frac{\dot{A}_d}{\dot{A}_c} \right| = \frac{2(1 + \beta)R_E + R_B + r_{be}}{2(R_B + r_{be})}$$



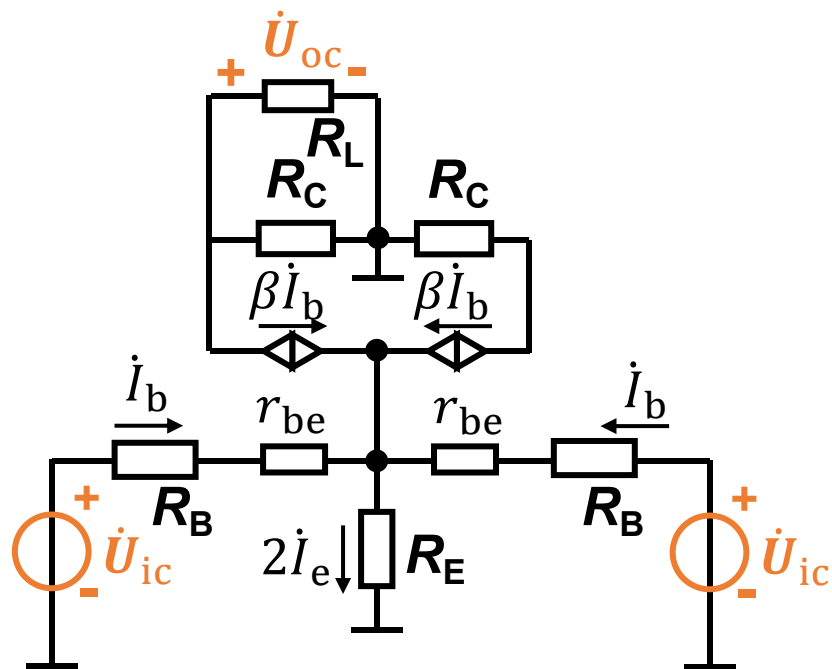
The voltage gain

$$\dot{A}_c = \frac{\dot{U}_{oc}}{\dot{U}_{ic}} = - \frac{\beta(R_C // R_L)}{2(1 + \beta)R_E + R_B + r_{be}}$$

Common-mode rejection ratio

$$K_{CMR} = \frac{2(1 + \beta)R_E + R_B + r_{be}}{2(R_B + r_{be})}$$

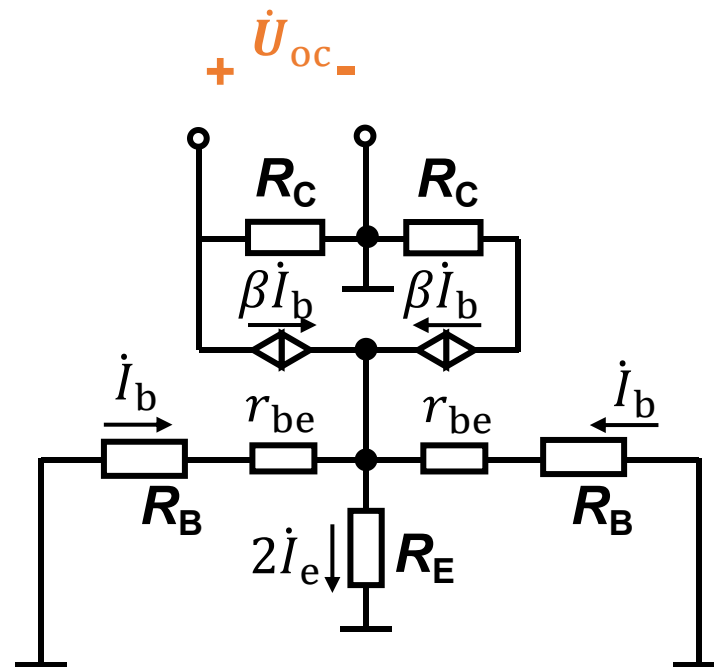
Larger R_E , larger K_{CMR} , better performance!



The input resistance

$$\dot{U}_{ic} = \dot{I}_b R_B + \dot{I}_b r_{be} + 2\dot{I}_e R_E$$

$$R_{ic} = R_B + r_{be} + 2(1 + \beta)R_E$$



The output resistance

$$R_{oc} = R_C$$

Summary

Differential-mode signal 差模信号

□ Dual-ended input & output

$$\dot{A}_d = - \frac{\beta(R_C // R_L/2)}{r_{be} + R_B}$$

Input resistance Output resistance

$$R_{id} = 2(r_{be} + R_B)$$

$$R_{od} = 2 R_C$$

□ Dual-ended input & single-ended output

$$\dot{A}_d = - \frac{\beta(R_C // R_L)}{2(r_{be} + R_B)}$$

Input resistance Output resistance

$$R_{id} = 2(r_{be} + R_B)$$

$$R_{od} = R_C$$

Common-mode signal 共模信号

□ Dual-ended input & output

$$\dot{A}_c = \frac{\dot{U}_{oc}}{\dot{U}_{ic}} = 0$$

Input resistance Output resistance

$$R_{ic} = R_B + r_{be} + 2(1 + \beta)R_E \qquad R_{od} = 2R_C$$

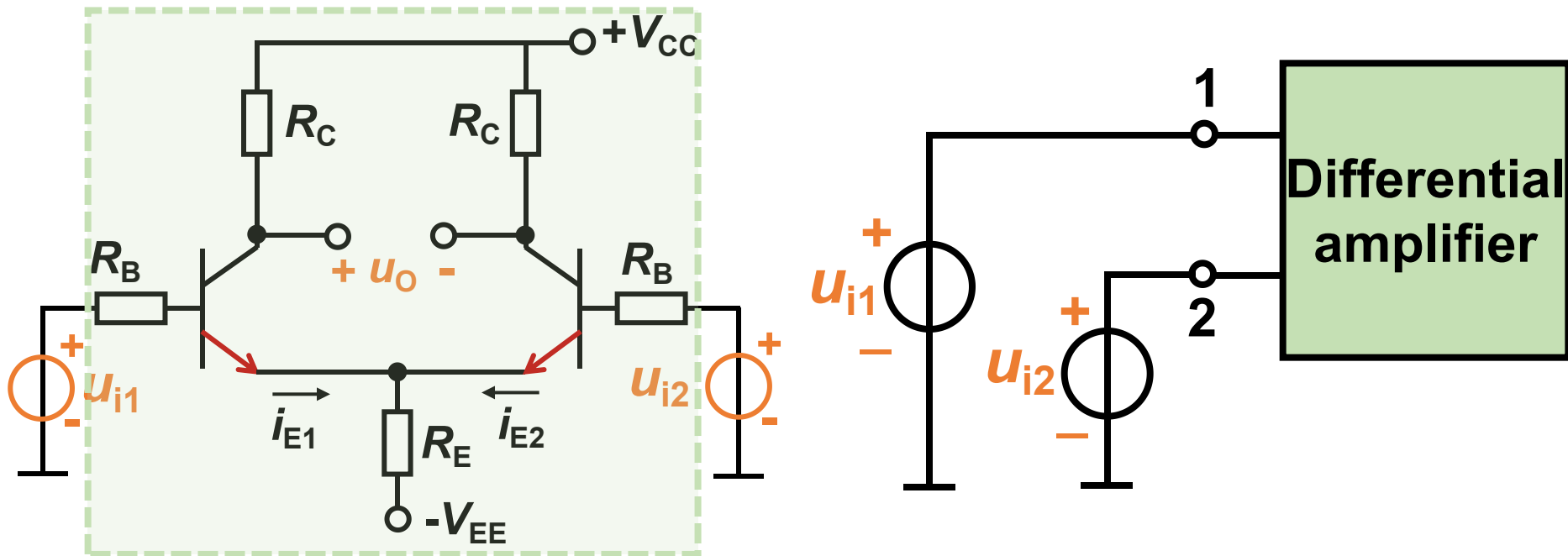
□ Dual-ended input & single-ended output

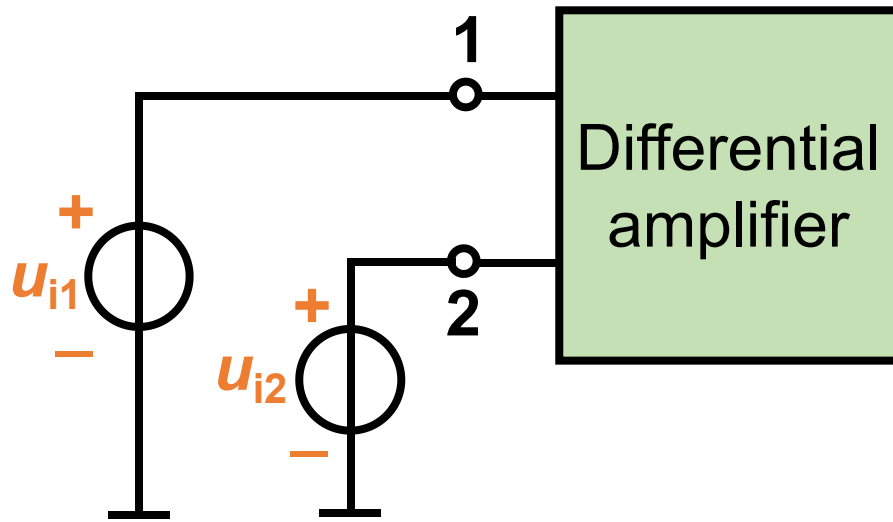
$$\dot{A}_c = - \frac{\beta(R_C // R_L)}{2(1 + \beta)R_E + R_B + r_{be}}$$

Input resistance Output resistance

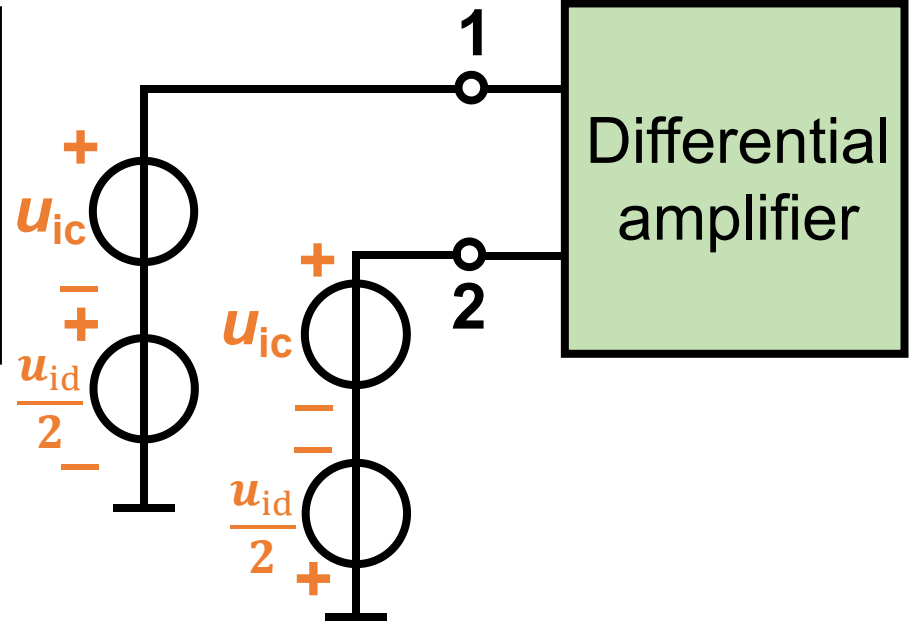
$$R_{ic} = R_B + r_{be} + 2(1 + \beta)R_E \qquad R_{oc} = R_C$$

5) General case: u_{i1} and u_{i2} are arbitrary

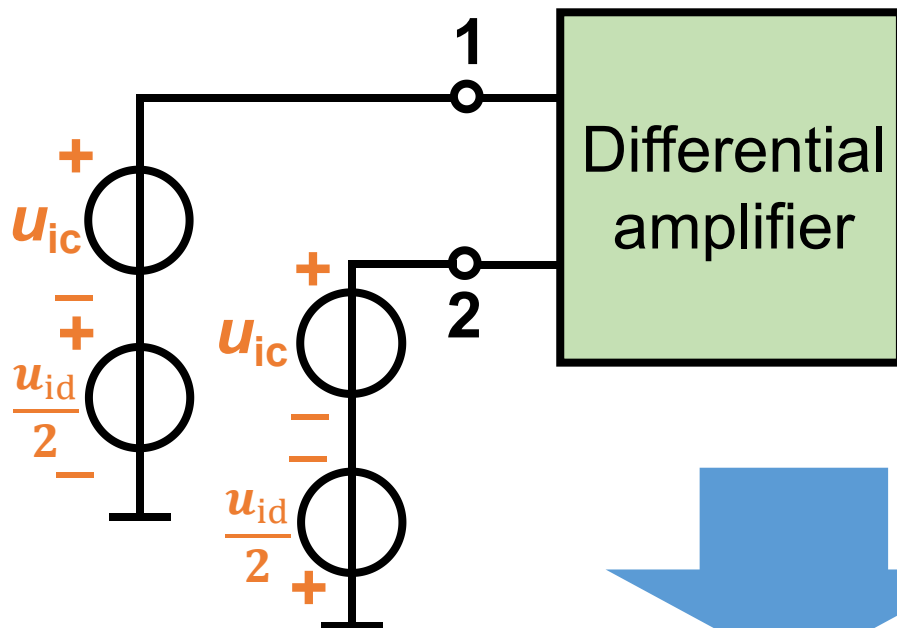




$$\left\{ \begin{array}{l} u_{id} = u_{i1} - u_{i2} \\ u_{ic} = \frac{u_{i1} + u_{i2}}{2} \end{array} \right.$$



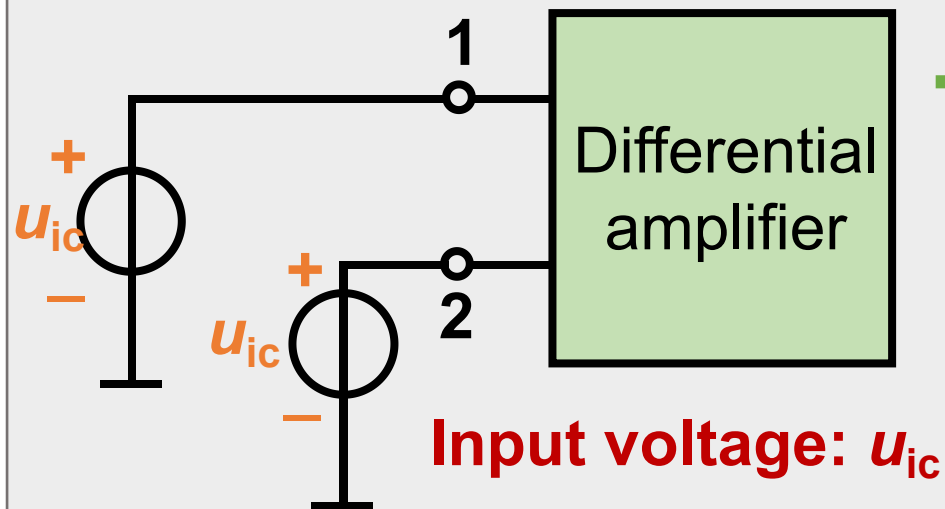
$$\left\{ \begin{array}{l} u_{i1} = u_{ic} + \frac{u_{id}}{2} \\ u_{i2} = u_{ic} - \frac{u_{id}}{2} \end{array} \right.$$



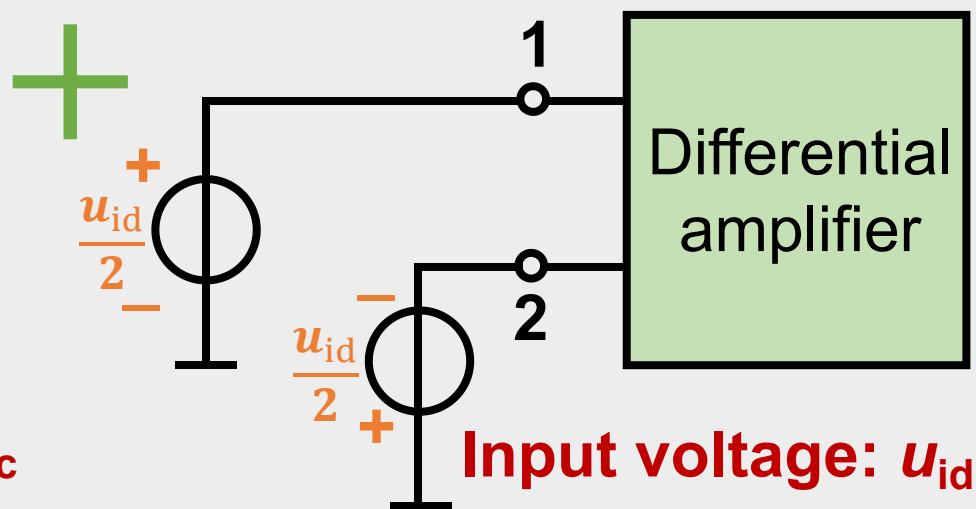
Arbitrary signal can be treated as a superposition of differential-mode and common-mode signals

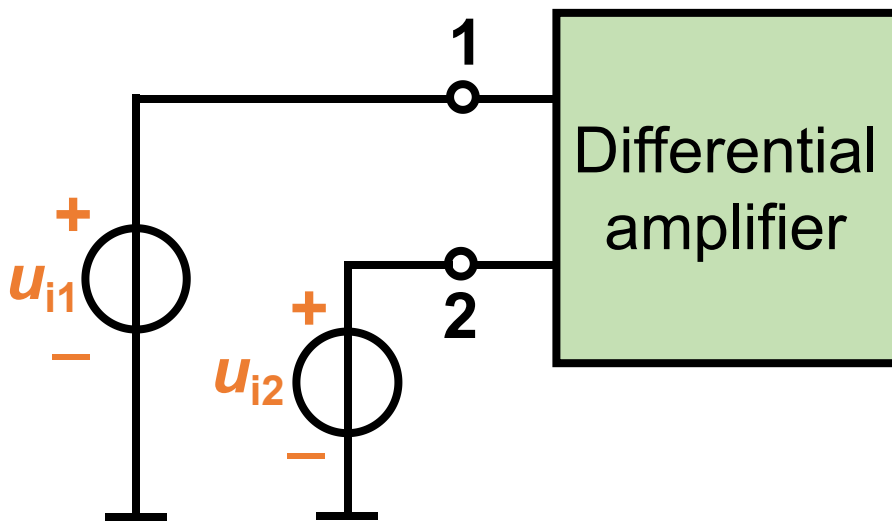
对于任意输入信号，可以写成一个差模信号与一个共模信号的线性叠加

Common-mode

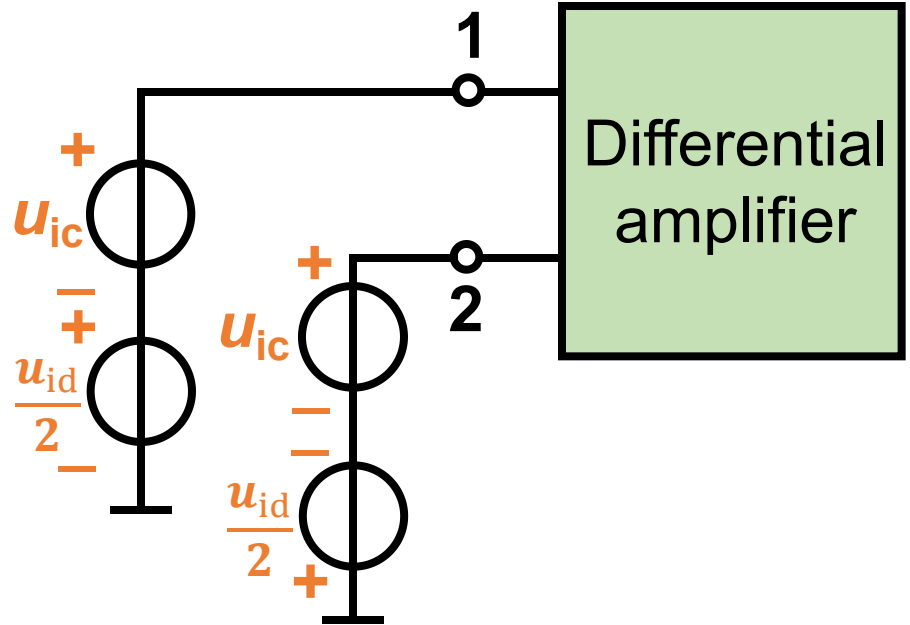


Differential-mode





$$\begin{cases} u_{id} = u_{i1} - u_{i2} \\ u_{ic} = \frac{u_{i1} + u_{i2}}{2} \end{cases}$$



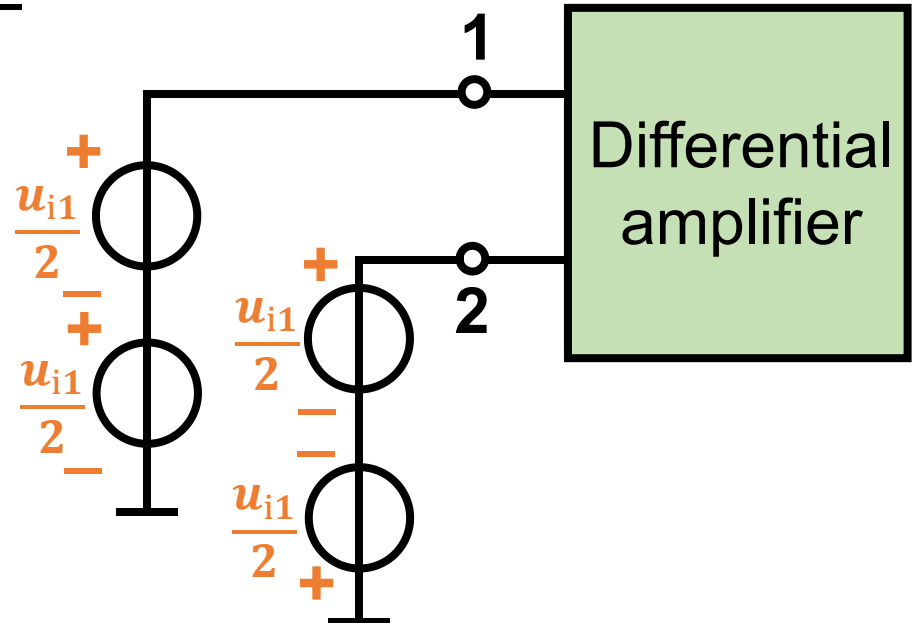
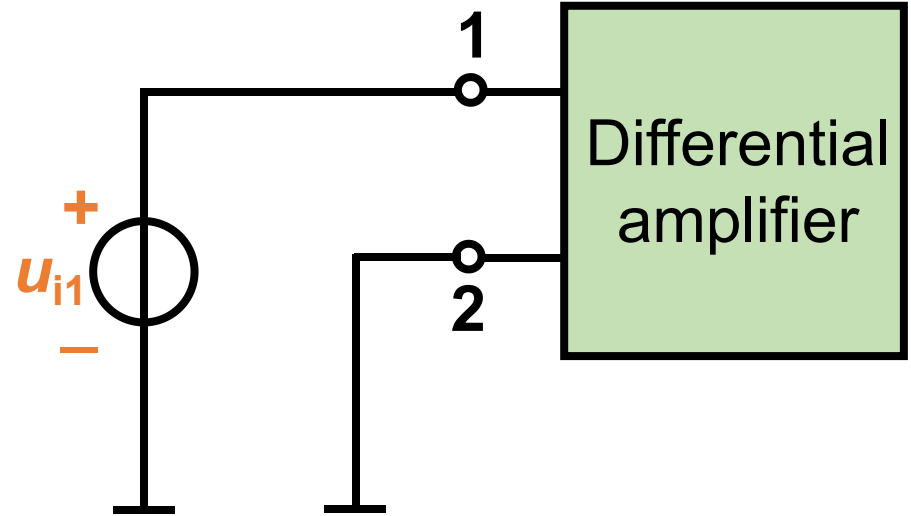
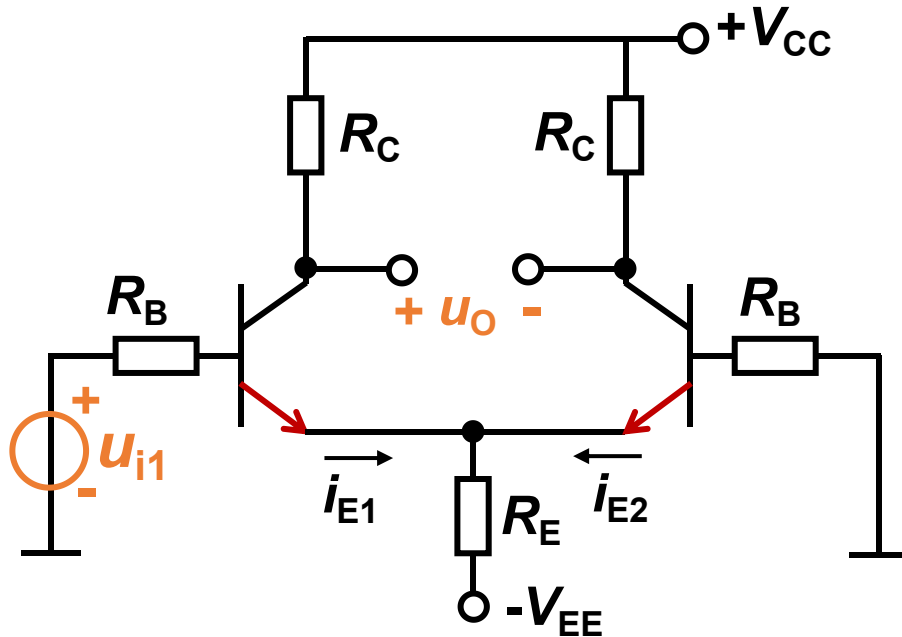
$$\begin{cases} u_{i1} = u_{ic} + \frac{u_{id}}{2} \\ u_{i2} = u_{ic} - \frac{u_{id}}{2} \end{cases}$$

The output voltage

$$u_o = \dot{A}_d u_{id} + \dot{A}_c u_{ic}$$

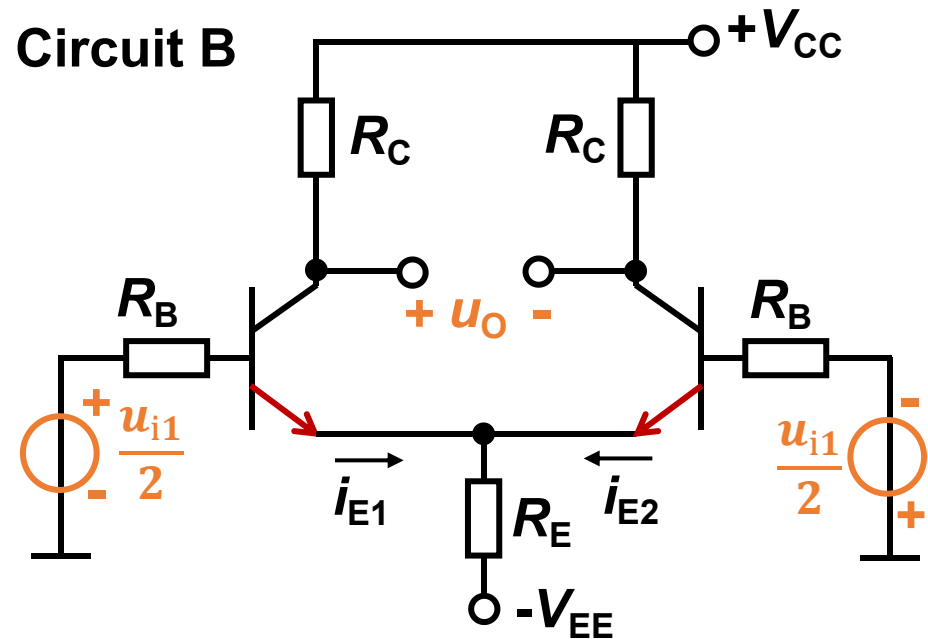
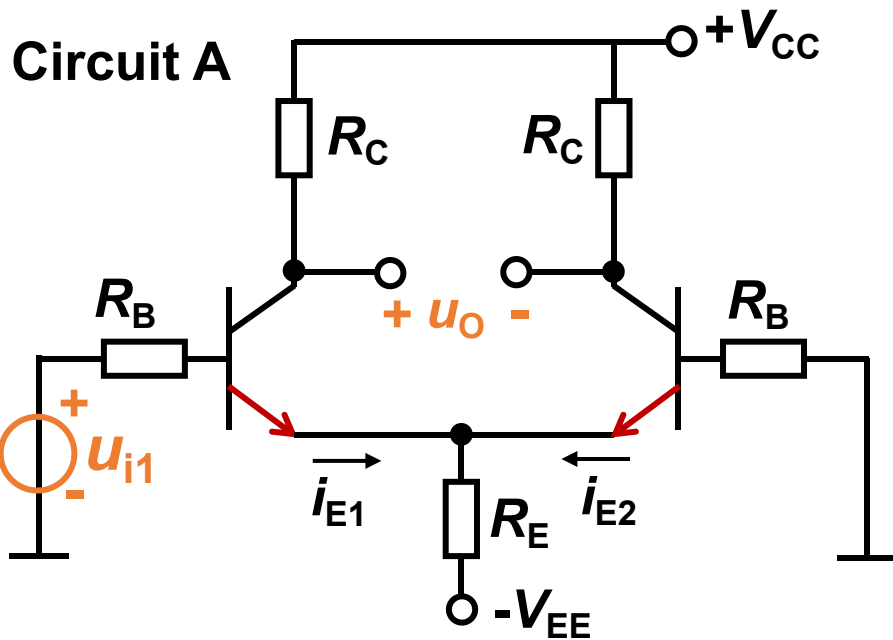
$$u_o = \dot{A}_d (u_{i1} - u_{i2}) + \dot{A}_c \left(\frac{u_{i1} + u_{i2}}{2} \right)$$

Single-ended input: $u_{i2} = 0$



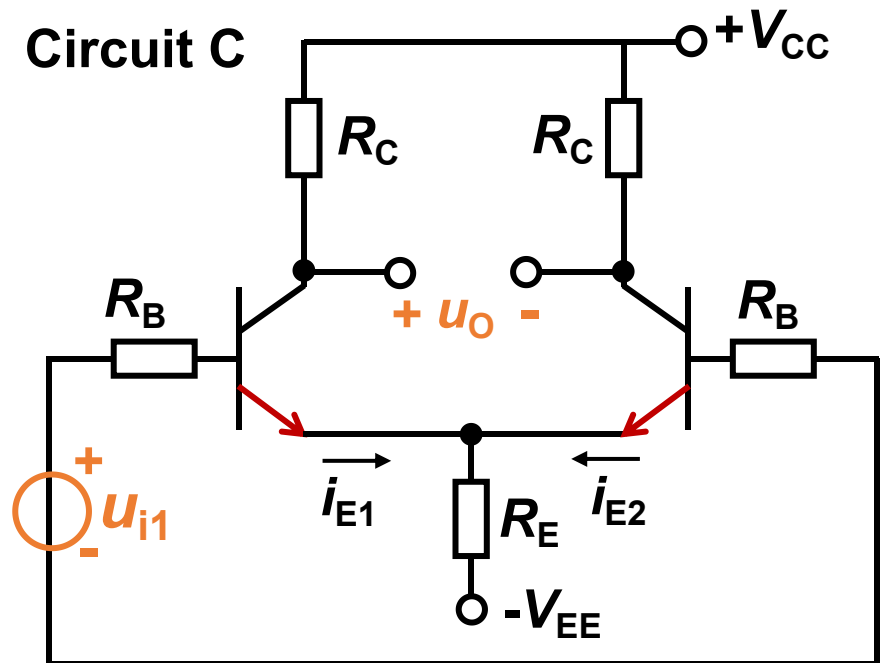
$$\begin{cases} u_{id} = u_{i1} - u_{i2} = u_{i1} \\ u_{ic} = \frac{u_{i1} + u_{i2}}{2} = \frac{u_{i1}}{2} \end{cases}$$

$$u_o = \dot{A}_d u_{i1} + \dot{A}_c \frac{u_{i1}}{2}$$



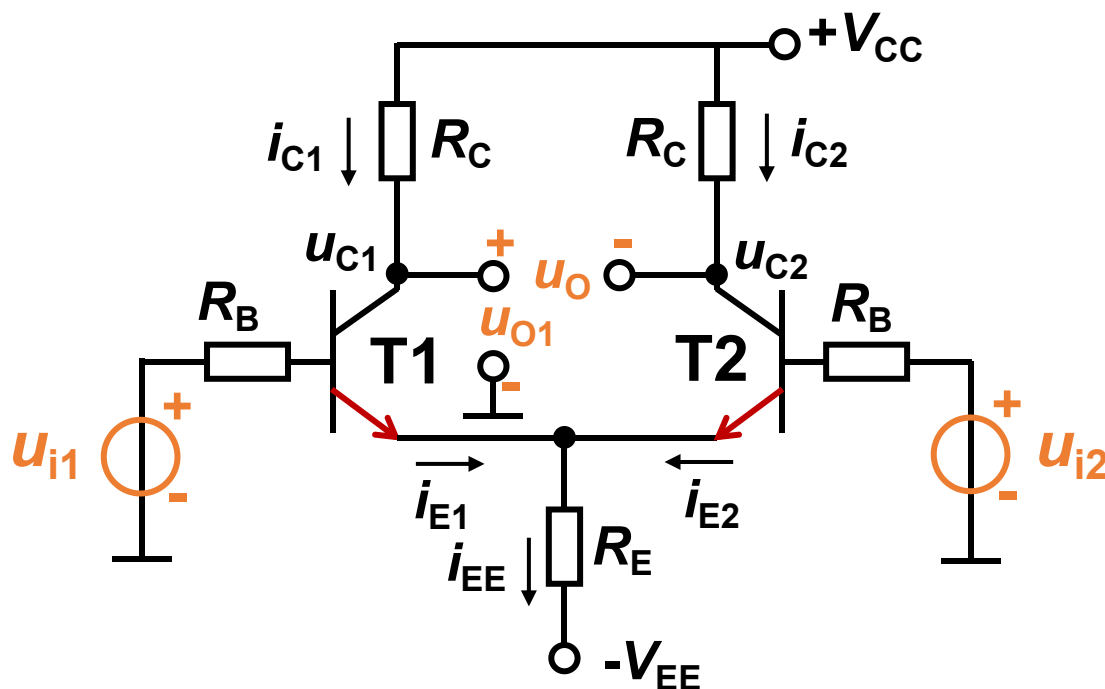
Example:

**What's the difference
between these 3 circuits?**



4.2.2 Improved differential amplifier: with constant-current source

改进型：带恒流源的差分放大电路

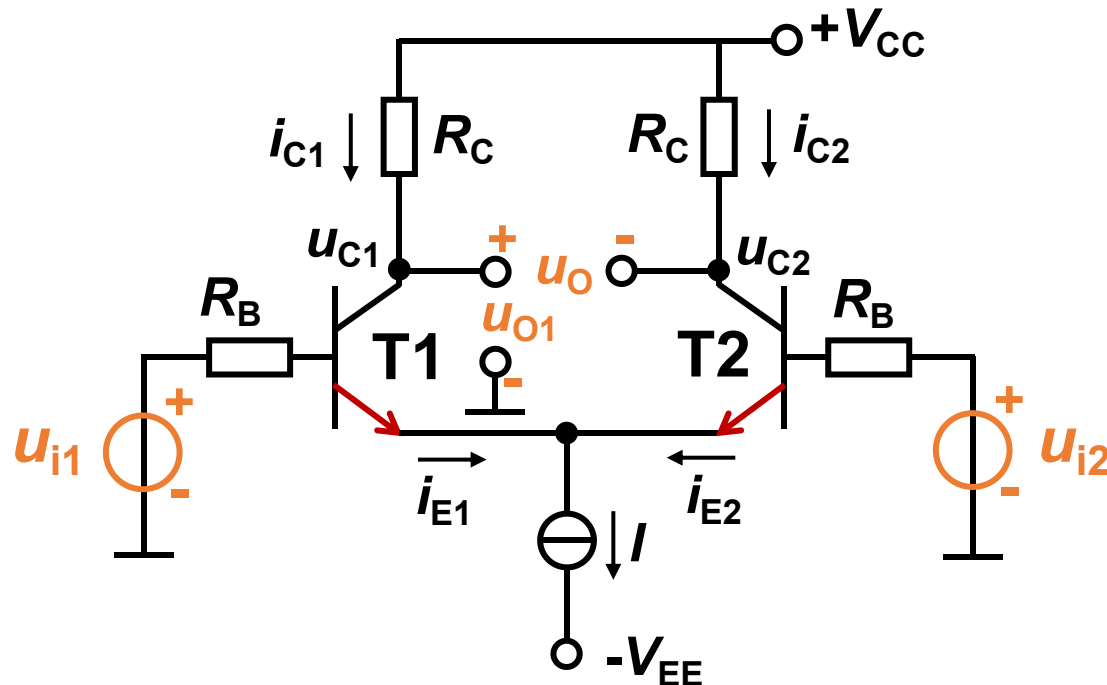


Larger R_E , larger K_{CMR} , better performance!

However, when R_E is too large, V_{EE} is also very large

E.g. $R_E=100\text{ k}\Omega$, $I_{EE}=0.5\text{ mA}$, $V_{EE}=100.7\text{ V}$

Use constant current source to replace R_E



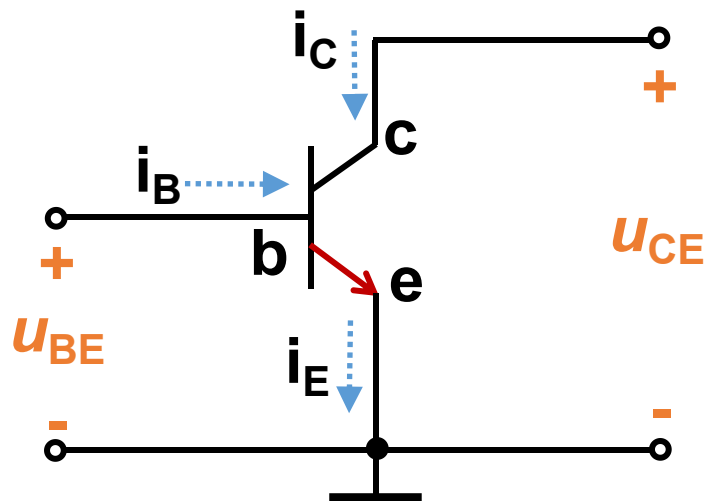
The property of constant current source:

1) Very high AC resistance

➡ Low common-mode voltage gain A_C ➡ High K_{CMR}

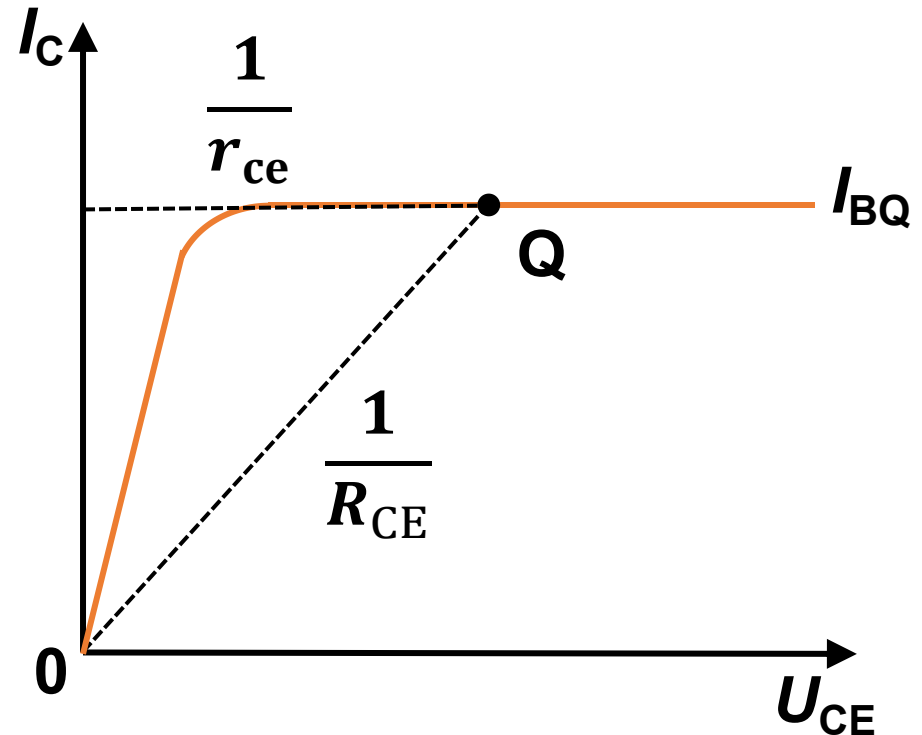
2) Low DC resistance ➡ Low V_{EE}

Constant current source made from transistor



DC resistance: R_{CE}

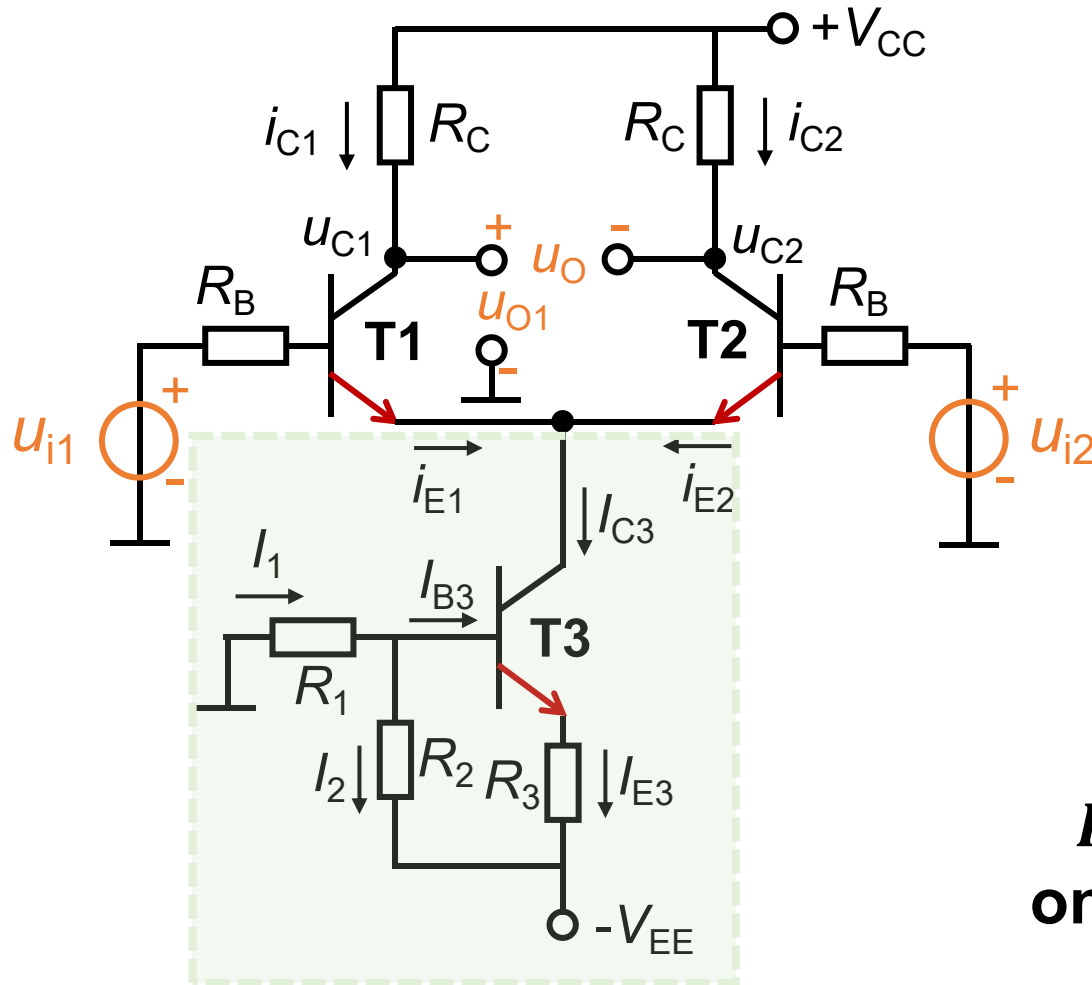
Output curve of transistor



AC resistance: r_{ce}

$$r_{ce} \gg R_{CE}$$

Use constant current source to replace R_E



Constant current source

Choose R_1 , R_2 and R_3 ,
let $I_2 \gg I_{B3}$, $I_2 \approx I_1$

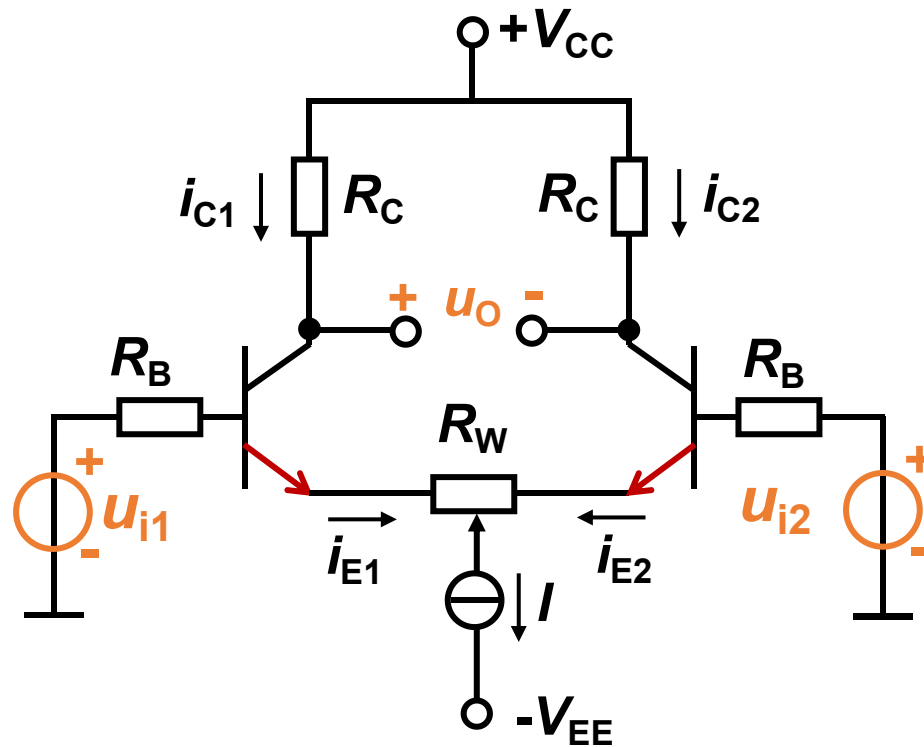
The voltage drop on R_2 :

$$U_{R2} \approx V_{EE} \frac{R_2}{R_1 + R_2}$$

$$I_{C3} \approx I_{E3} = \frac{U_{R2} - U_{BE3}}{R_3}$$

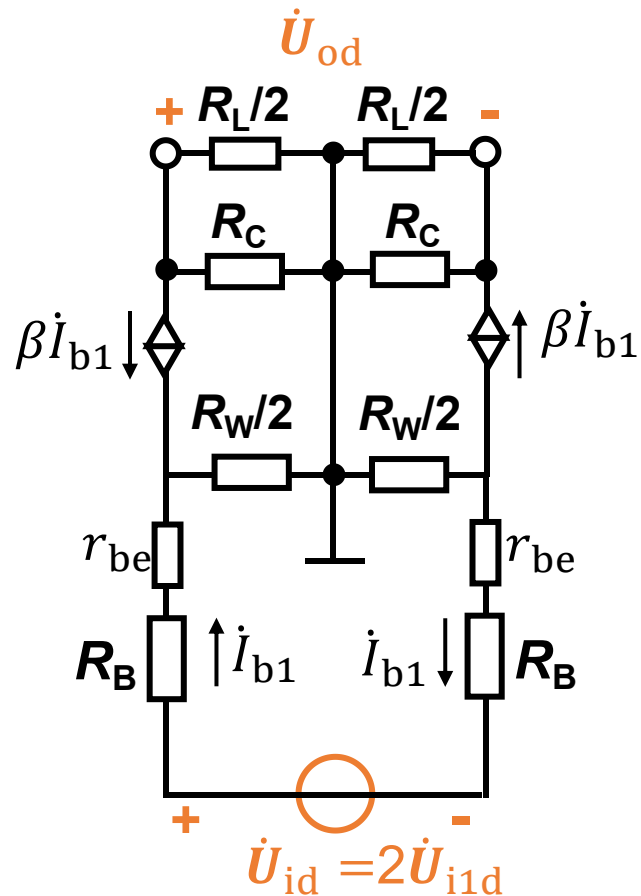
I_{C3} is a constant, which
only depends on the value
of V_{EE} , R_1 , R_2 and R_3 .

For practical applications, it is hard to achieve the absolute symmetric circuit. Use a small value resistor R_W to adjust the symmetry of circuit.

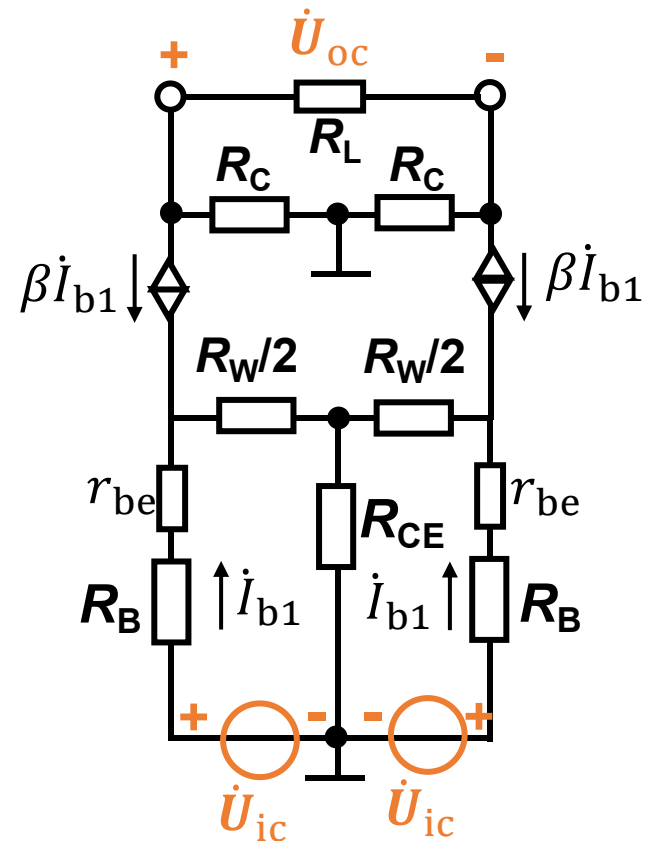


AC equivalent circuit- dual-ended output mode

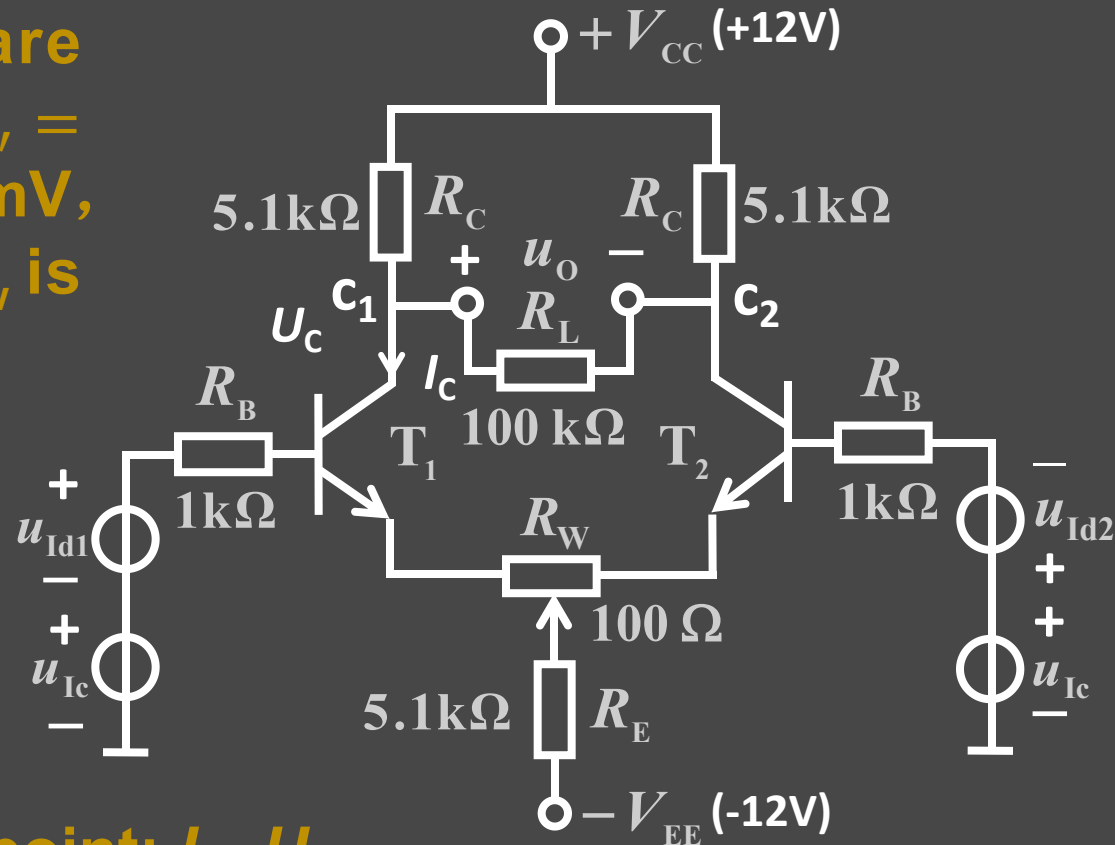
Differential mode



Common mode



[Example] As shown in right figure, T_1 and T_2 are identical, $\beta = 50$, $r_{bb'} = 300 \Omega$, $U_{BE} = 0.7V$, $u_{Id} = 50mV$, $u_{Ic} = 1V$. The slider of R_w is in the middle, ask:



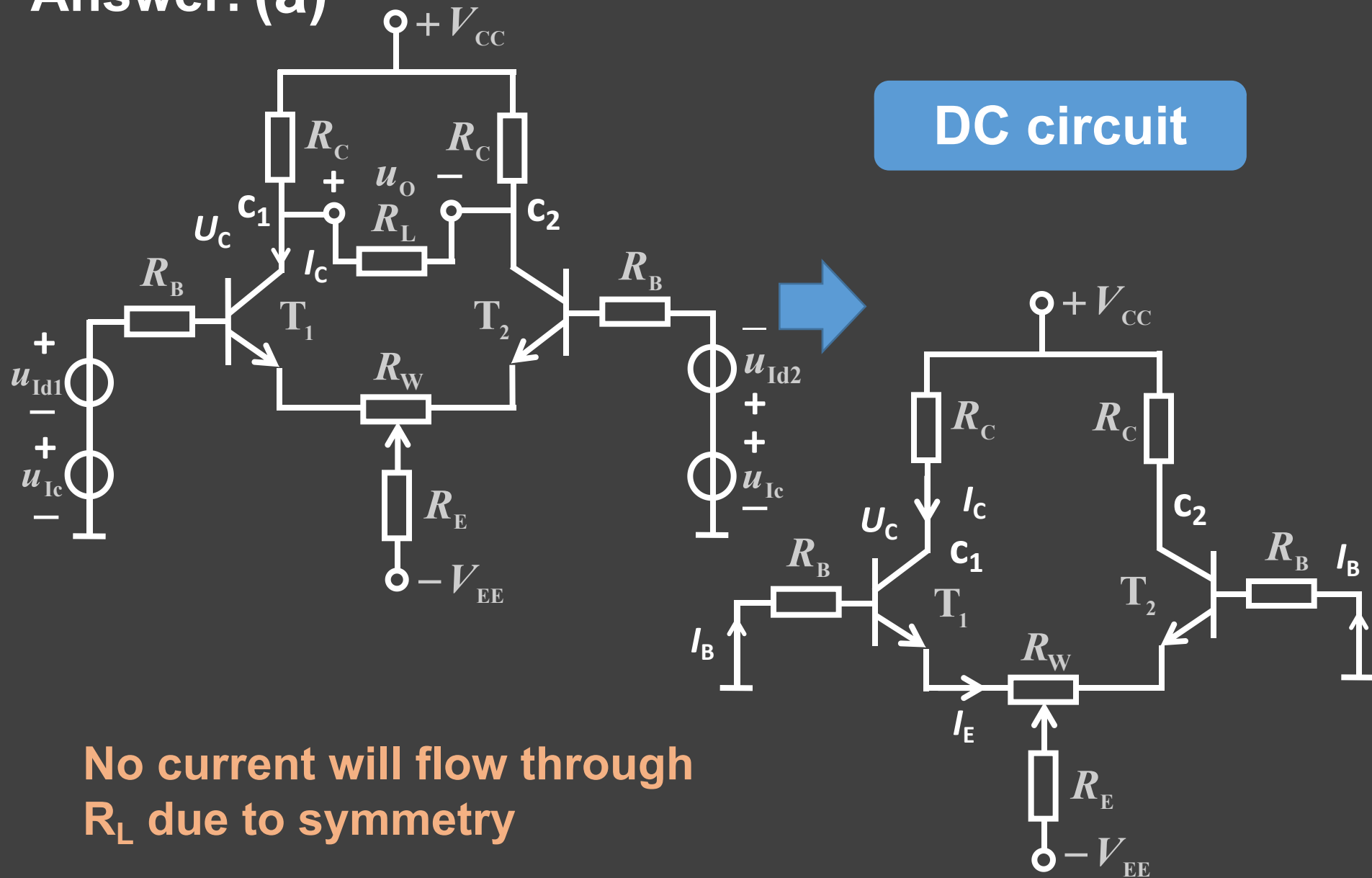
(a) Quiescent working point: I_C , U_C ;

(b) AC output voltage u_o and K_{CMR} ;

(c) If R_L is connected between c_1 and ground, ask: U_C , u_o and K_{CMR} .

(d) R_{id} , R_{ic} and R_o for dual-ended mode.

Answer: (a)



Input circuit:

$$V_{EE} - U_{BE} = I_B R_B + I_E \frac{R_w}{2} + 2I_E R_E$$

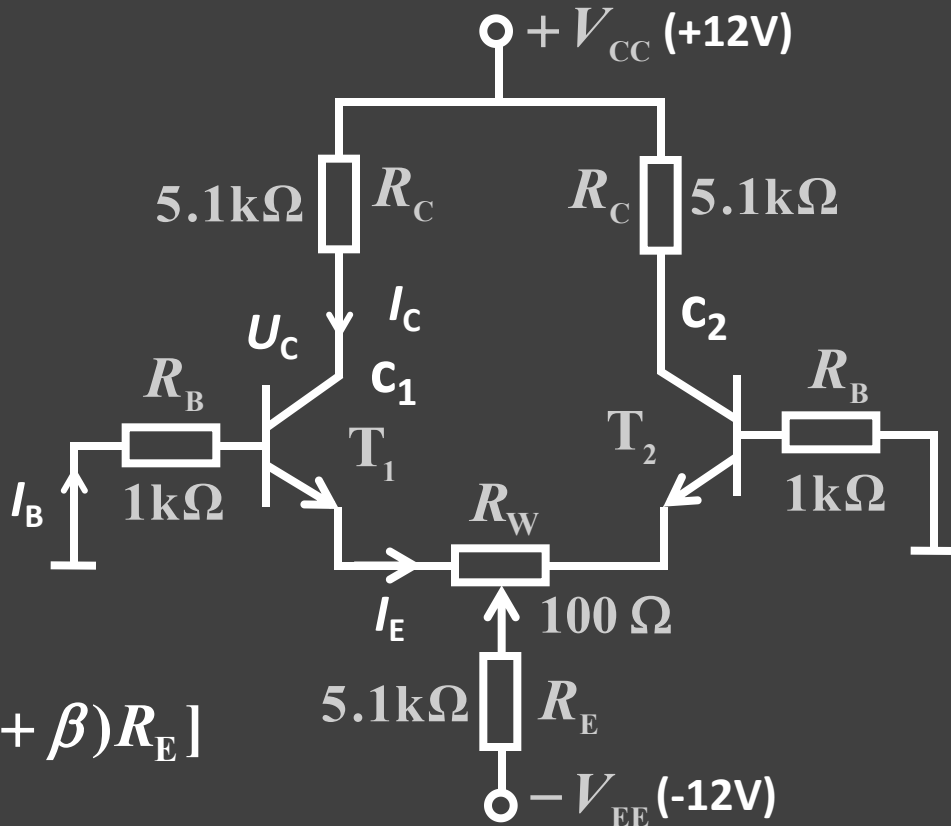
$$I_E = (1 + \beta) I_B$$



$$V_{EE} - U_{BE} = I_B \left[R_B + (1 + \beta) \frac{R_w}{2} + 2(1 + \beta) R_E \right]$$

$$I_{BQ} = \frac{V_{EE} - U_{BEQ}}{2(1 + \beta) R_E + (1 + \beta) \frac{R_w}{2} + R_B} \approx 0.022 \text{ mA}$$

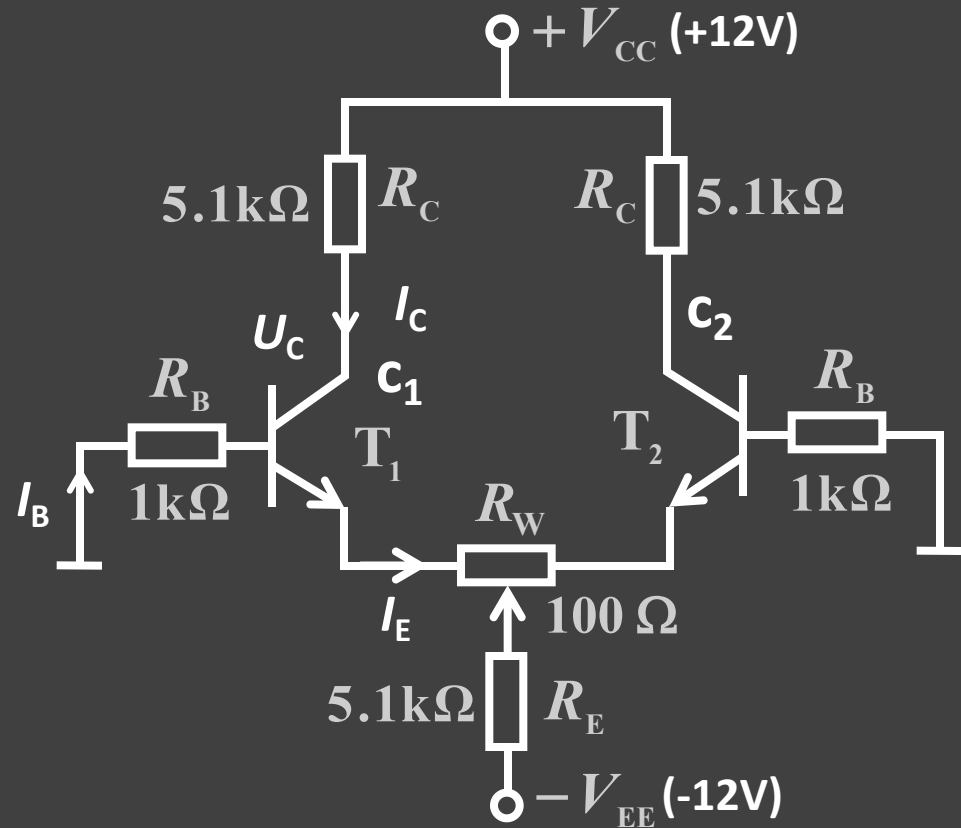
DC circuit



$$I_{CQ} = \beta I_{BQ} \approx 1.1 \text{ mA}$$

$$I_{EQ} = (1 + \beta) I_{BQ} \approx 1.1 \text{ mA}$$

$$\begin{aligned} U_C &= V_{CC} - I_{CQ} R_C \\ &= 12 - 1.1 \times 5.1 \\ &\approx 6.4 \text{ V} \end{aligned}$$



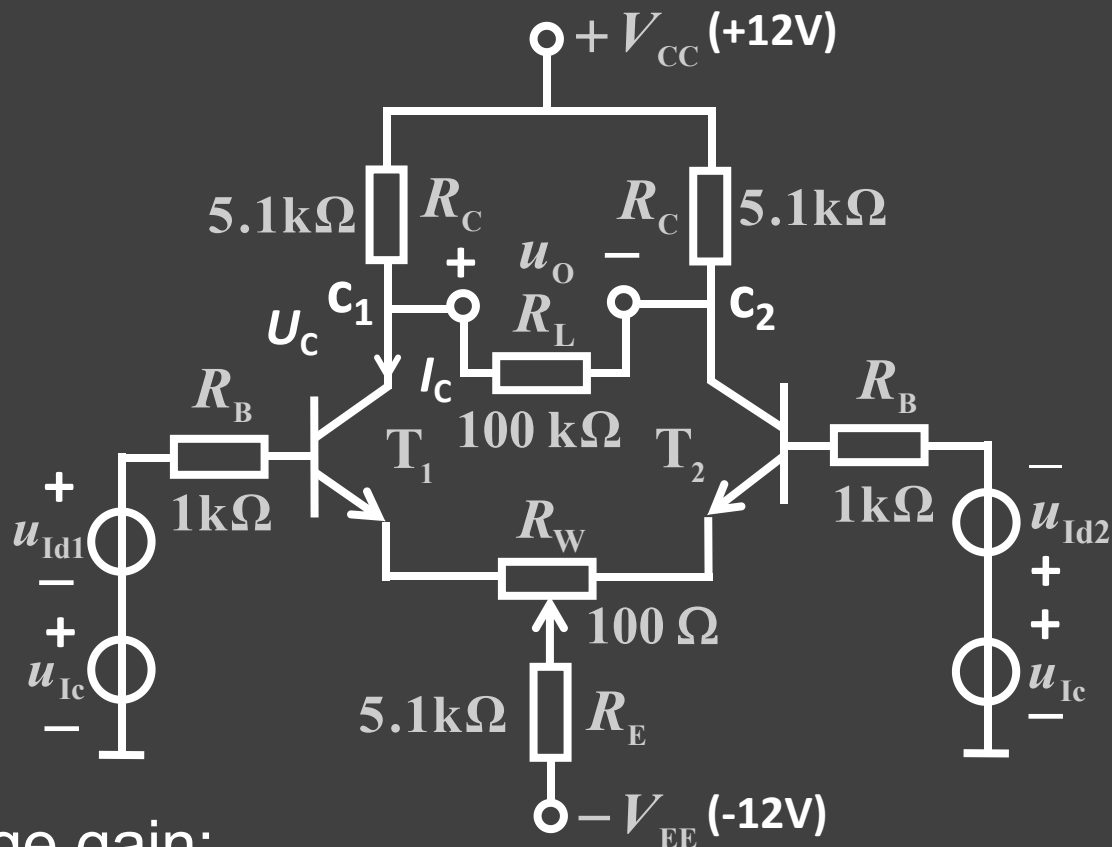
$$\begin{aligned}
 \text{(b)} \quad r_{be} &= r_{bb'} + \beta \frac{U_T}{|I_{CQ}|} \\
 &= 300 + 50 \frac{26}{1.1} \\
 &\approx 1.5 \text{ k}\Omega
 \end{aligned}$$

Common-mode
voltage gain:

$$\dot{A}_c = \frac{\dot{U}_{oc}}{\dot{U}_{ic}} = 0$$

Differential-mode voltage gain:

$$\dot{A}_d = \frac{\dot{U}_{od}}{\dot{U}_{id}} = - \frac{\beta R_C // \frac{R_L}{2}}{R_B + r_{be} + (1 + \beta) \frac{R_W}{2}} \approx -46$$



Calculate this result by yourself

Common-mode rejection ratio

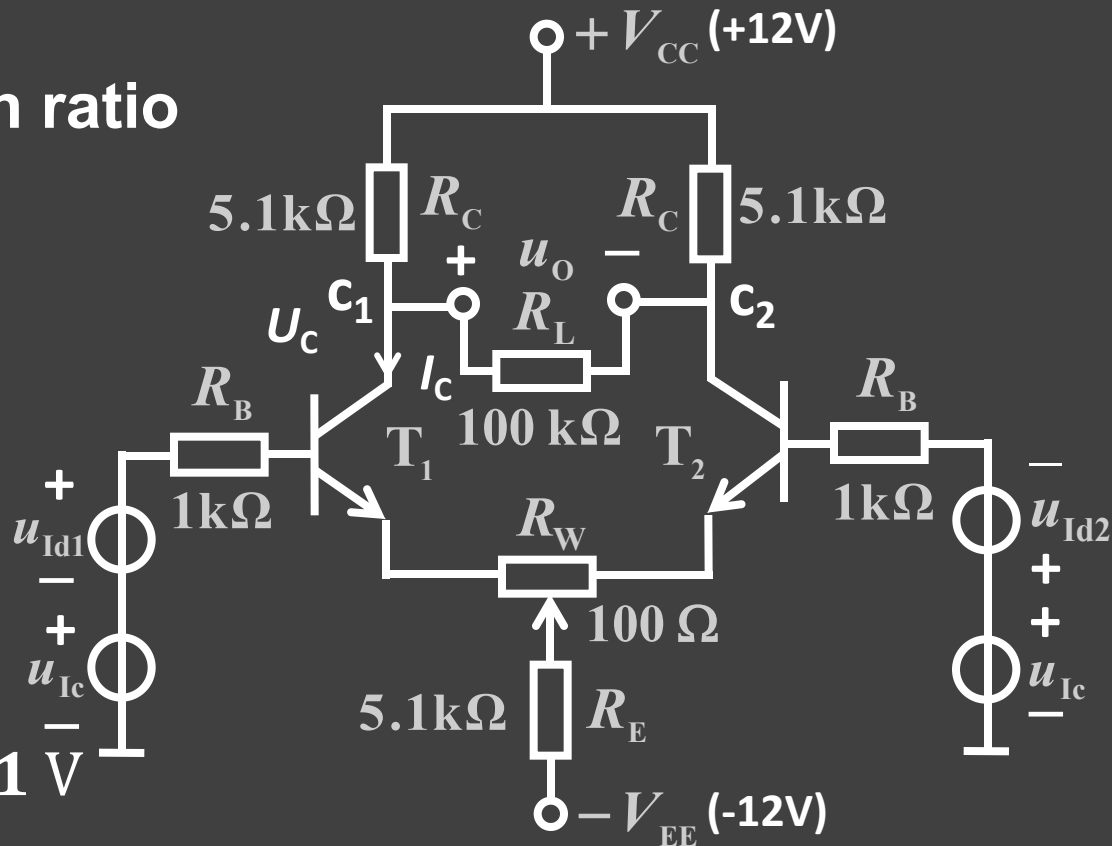
$$K_{\text{CMR}} = \left| \frac{\dot{A}_d}{\dot{A}_c} \right| \rightarrow \infty$$

Output AC voltage:

$$u_o = \dot{A}_d u_{id} + \dot{A}_c u_{ic}$$

$$= -46 \times 0.05 \text{ V} + 0 \times 1 \text{ V}$$

$$= -2.3 \text{ V}$$



(c) Single-ended output

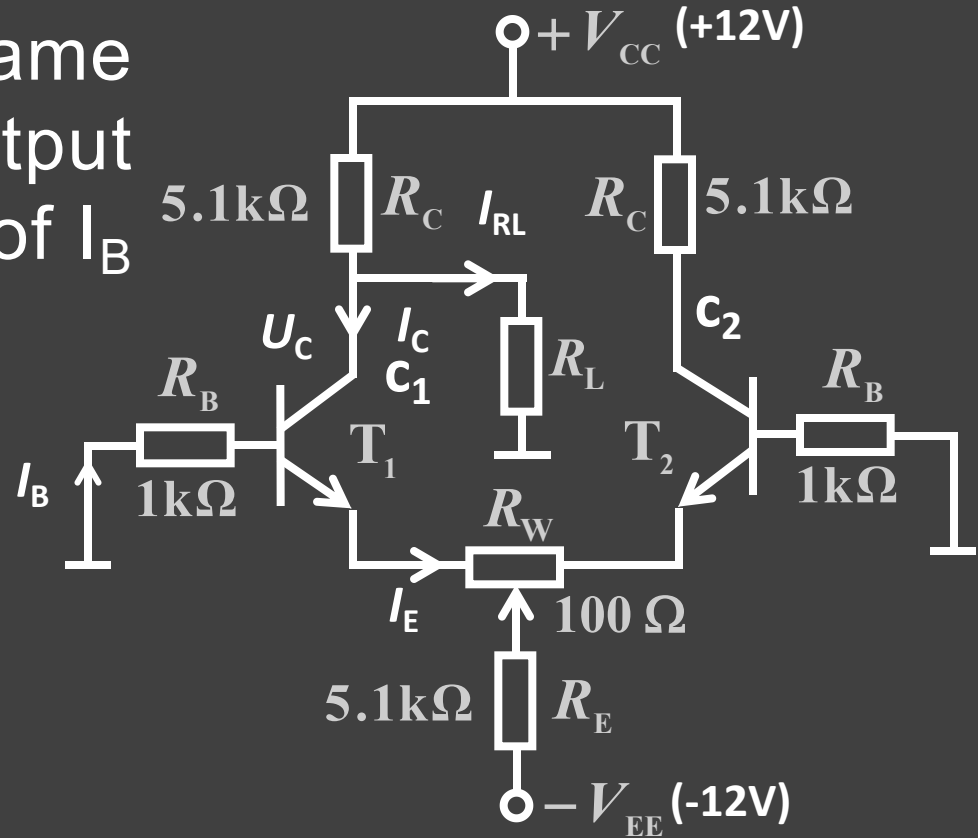
The input circuit is the same as that in dual-ended output mode. Hence the value of I_B and I_C are the same.

Current through left R_C :

$$I_{RC} = I_C + I_{RL} = I_C + \frac{U_C}{R_L}$$

$$U_C = V_{CC} - I_{RC}R_C$$

$$U_C = \frac{V_{CC} - I_C R_C}{1 + \frac{R_C}{R_L}} \approx 6V$$

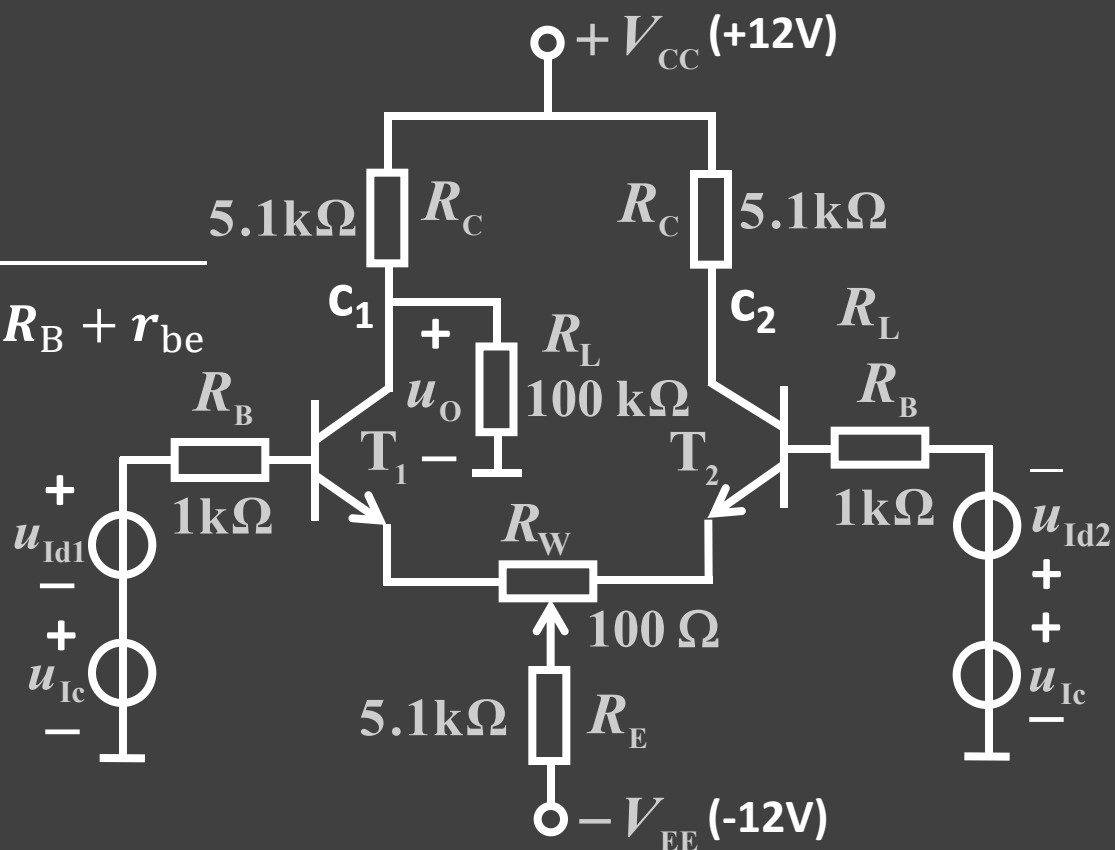


**Common-mode
voltage gain:**

$$\dot{A}_c = \frac{\dot{U}_{oc}}{\dot{U}_{ic}}$$

$$= - \frac{\beta(R_C // R_L)}{2(1 + \beta)R_E + \frac{(1 + \beta)}{2}R_W + R_B + r_{be}}$$

$$\approx -0.5$$



Differential-mode voltage gain: $\dot{A}_d = \frac{\dot{U}_{od}}{\dot{U}_{id}}$

$$= - \frac{\beta R_C // R_L}{2[R_B + r_{be} + (1 + \beta) \frac{R_W}{2}]}$$

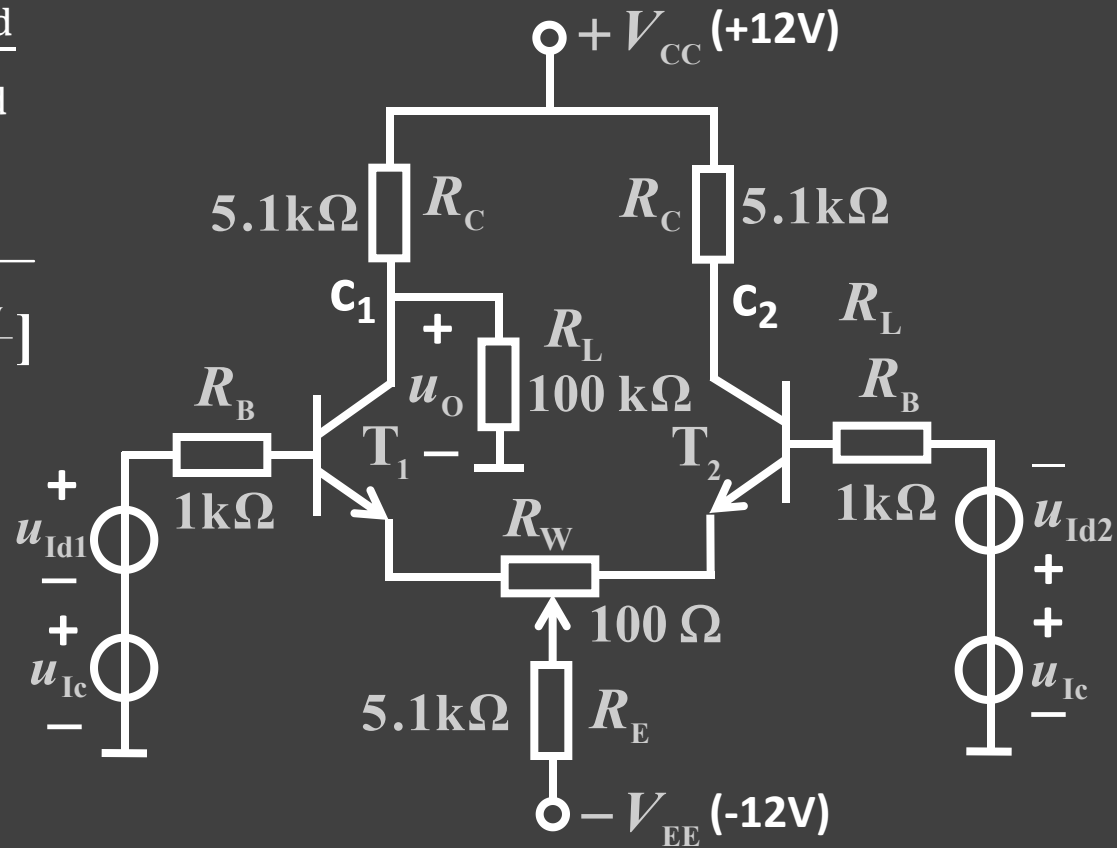
$$= -24$$

Common-mode rejection ratio

$$K_{CMR} = \left| \frac{\dot{A}_d}{\dot{A}_c} \right| = 48$$

Output AC voltage:

$$u_o = \dot{A}_d u_{id} + \dot{A}_c u_{ic} = -24 \times 0.05 - 0.5 \times 1 = -1.7V$$



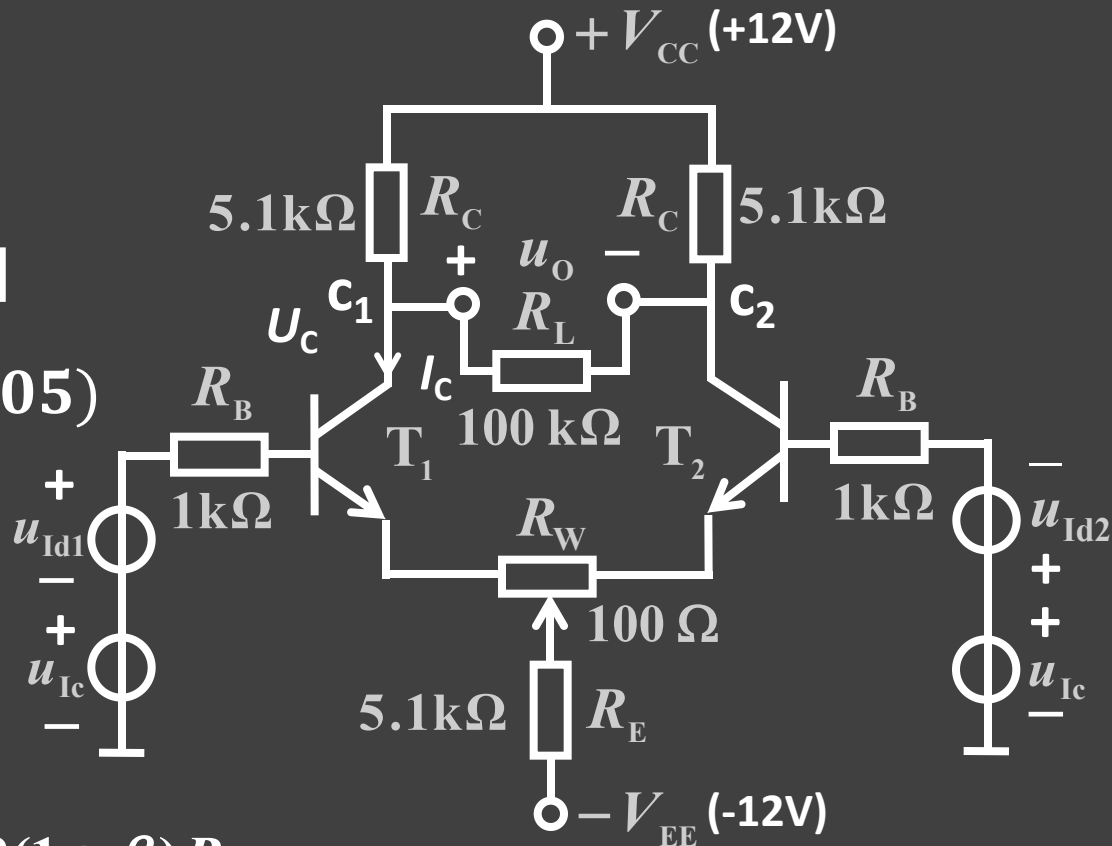
(d) Input resistance

Differential mode

$$\begin{aligned} R_{id} &= 2[R_B + r_{be} + (1 + \beta) \frac{R_W}{2}] \\ &= 2 \times (1 + 1.5 + 51 \times 0.05) \\ &= 10.1 \text{ k}\Omega \end{aligned}$$

Common mode

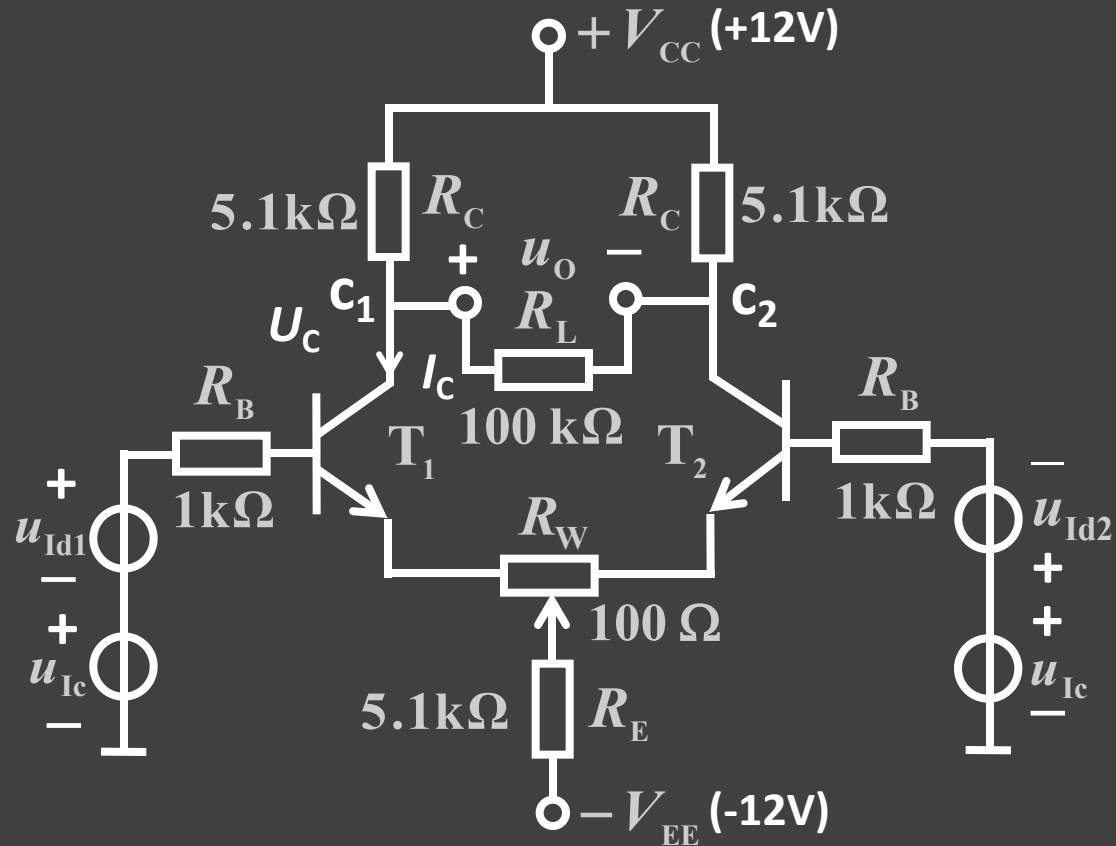
$$\begin{aligned} R_{ic} &= R_B + r_{be} + (1 + \beta) \frac{R_W}{2} + 2(1 + \beta) R_E \\ &\approx 520 \text{ k}\Omega \end{aligned}$$



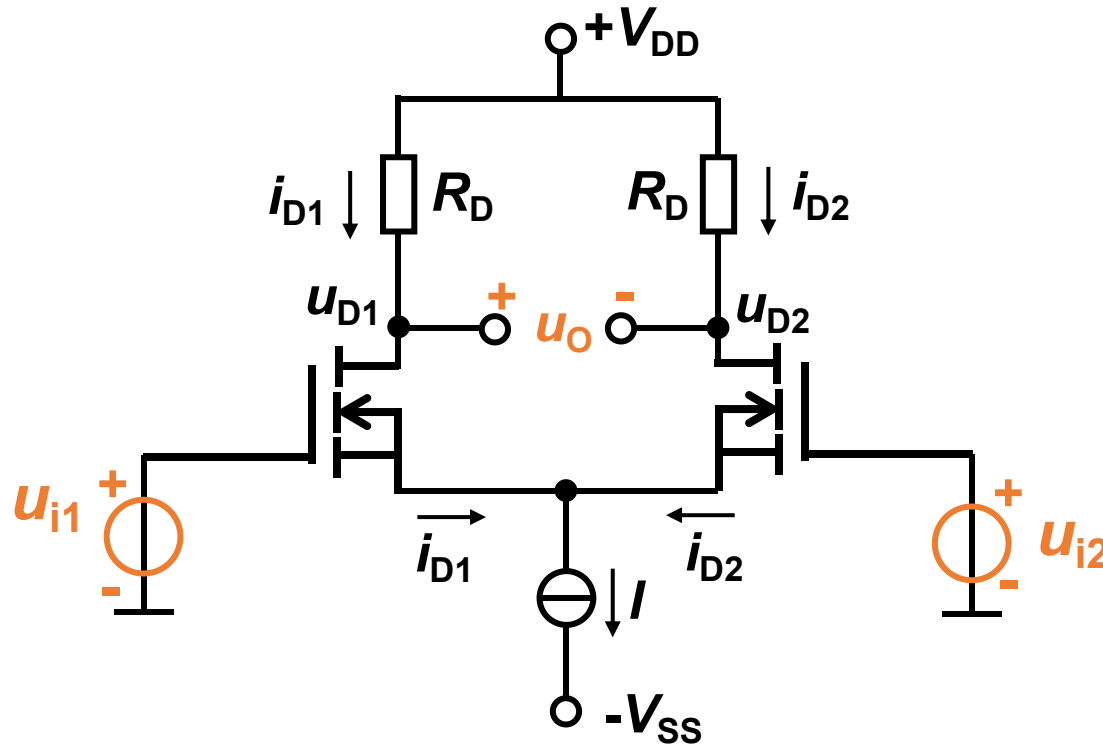
Output resistance

For dual-ended output mode

$$R_o = 2R_C = 2 \times 5.1 = 10.2\text{k}\Omega$$



4.2.3 Differential amplifier using FETs



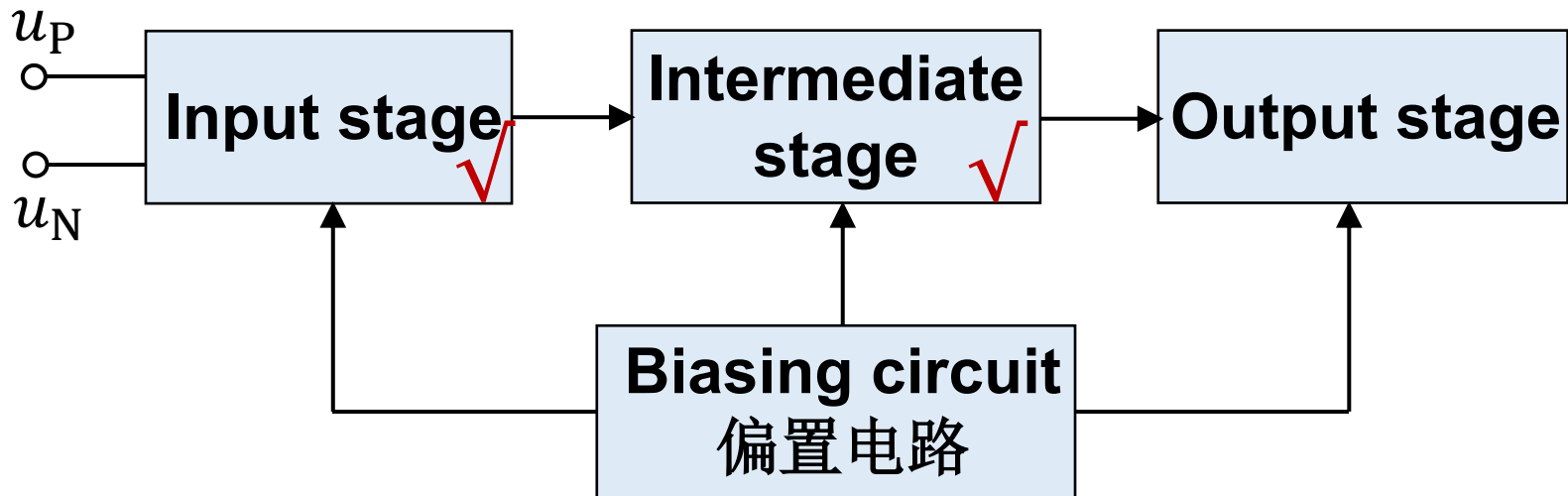
Calculate the voltage gain, input resistance and output resistance in differential mode ($u_{i1} = u_{i2}$) by yourself.

You have a homework regarding to this problem.

Homework 4-1: refer to blackboard

4.3 Complementary output stage 互补输出极

The structure of the integrated operational circuit (IOC)



4.3 Complementary output stage 互补输出极

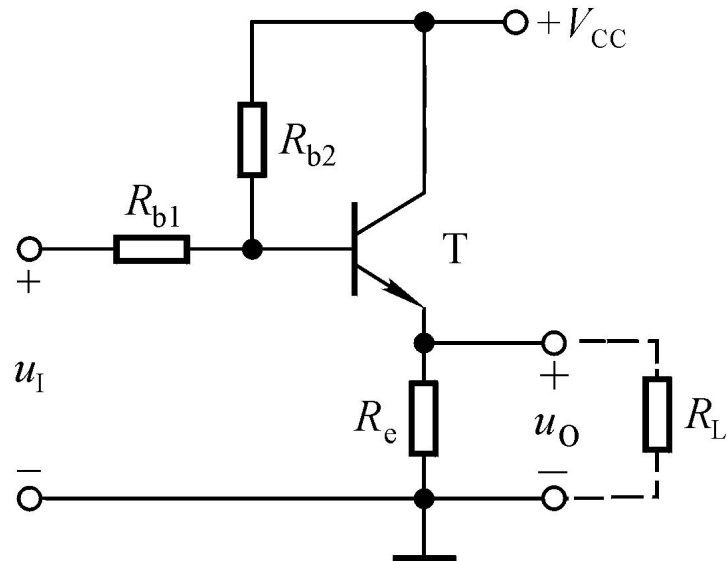
Requirement of the output stage:

- 1) **Low output resistance** 输出电阻低：负载能力强
- 2) **Low DC power consumption** 直流功耗小
- 3) **No DC power consumption on load resistor** 负载上无直流功耗：输入为零时输出为零
- 4) **High dynamic range** 动态范围尽量大：使最大不失真电压尽可能大

Compare of common -e, -c and -b amplifier

	Common-e	Common-c	Common-b
$\dot{A}_u$	High	<1	High
$\dot{A}_i$ No load resistor	β	$(1 + \beta)$	$\alpha = \beta / (1 + \beta)$
R_i	Medium	High	Low
R_o	High	Low	High
Bandwidth	Narrow	Medium	Wide

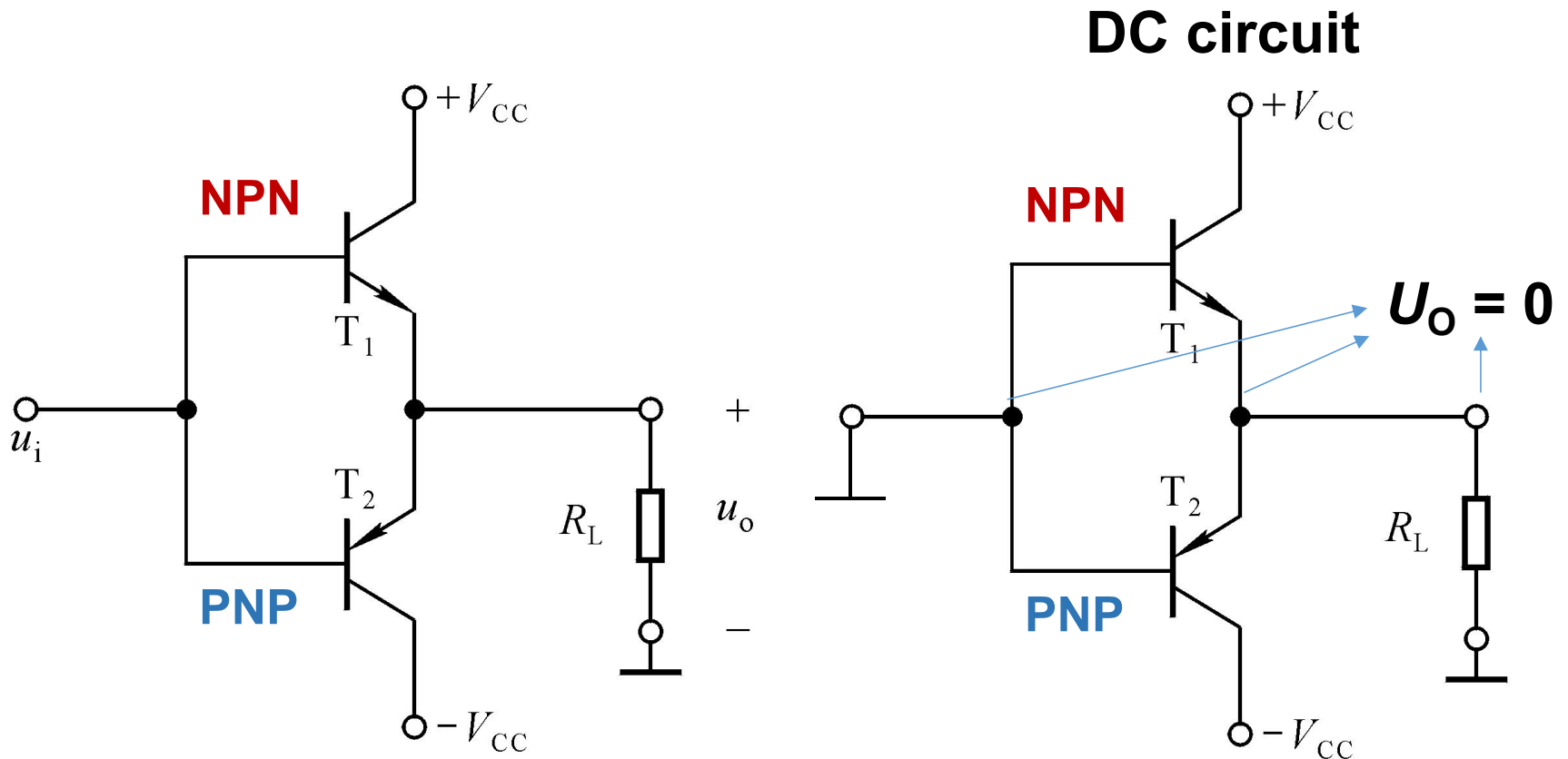
Common-collector amplifier has low output resistance



However,

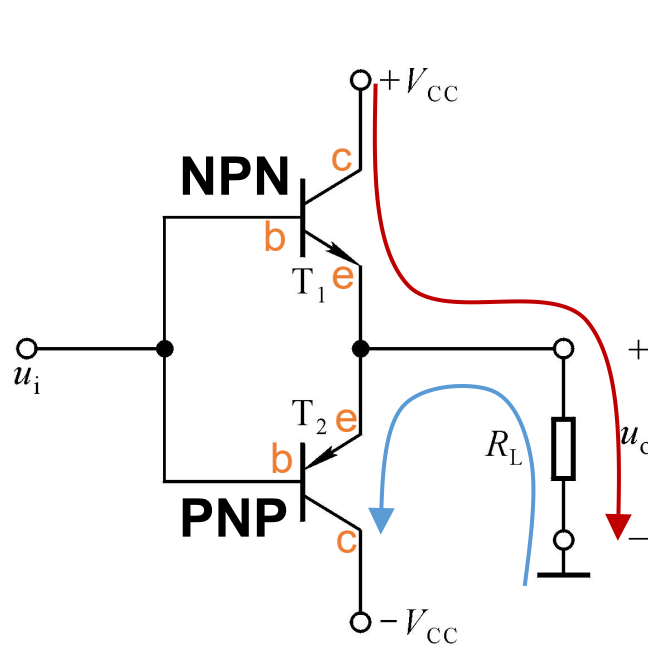
- 1) When $u_i=0$, there is DC current on load resistor R_L
- 2) When R_L changes, the quiescent working point Q changes
- 3) When there is R_L , the dynamic range decreases

The basic structure of complementary output stage



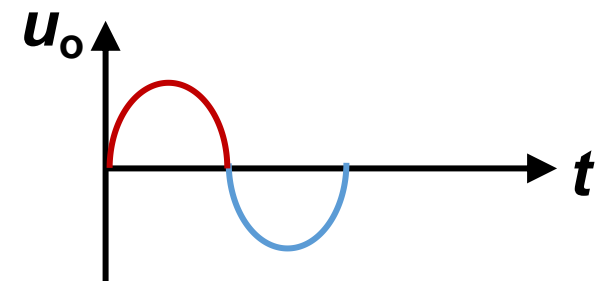
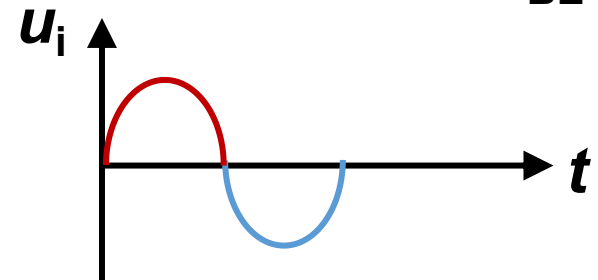
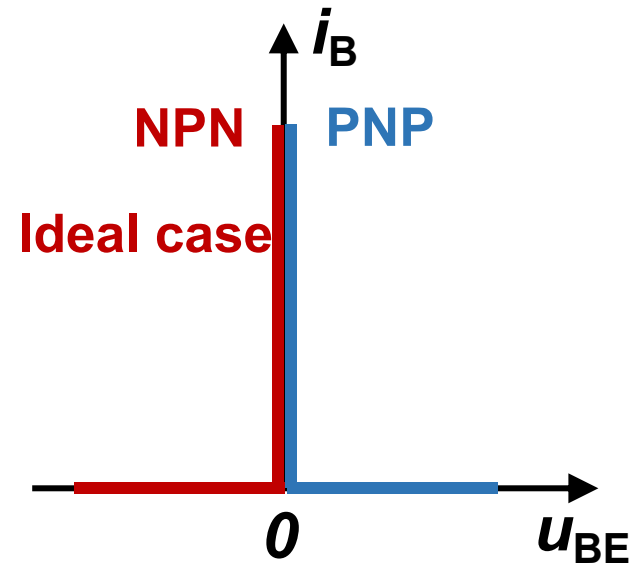
No DC current flow through R_L

1) Ideal case: Threshold voltage of PN junction $U_{ON}=0$

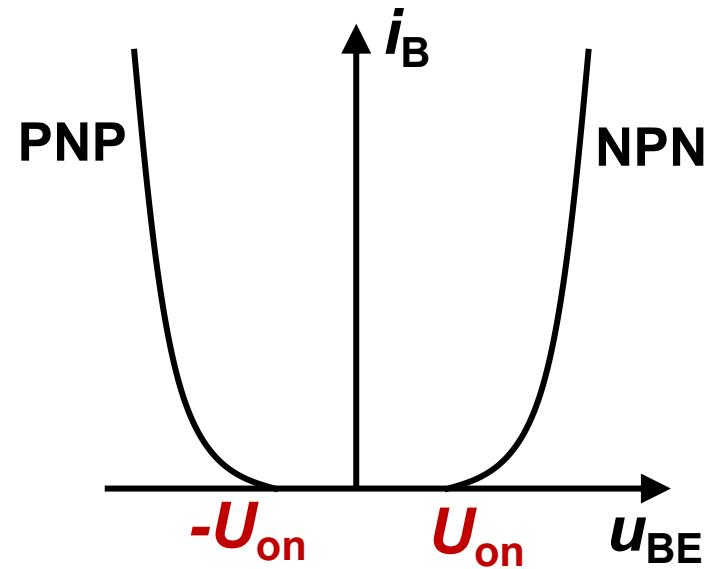
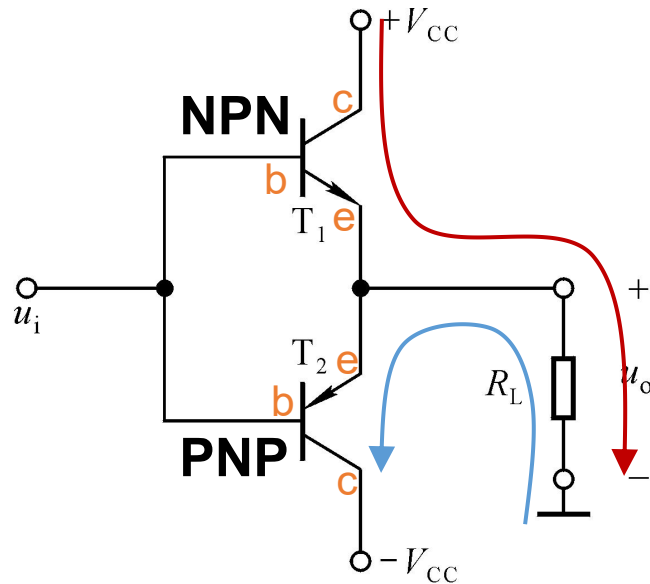


$u_i > 0$ T1 ✓ T2 ✗

$u_i < 0$ T1 ✗ T2 ✓



2) Real case: Crossover distortion



(1) For 1st half period

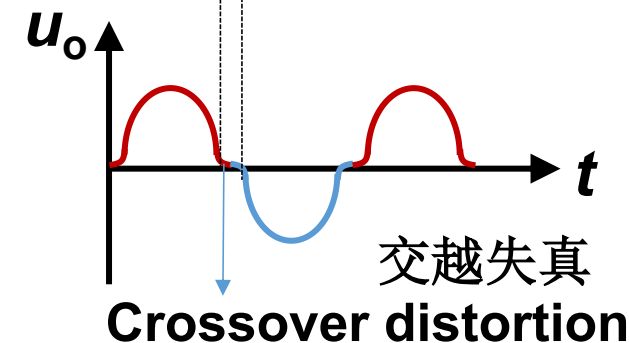
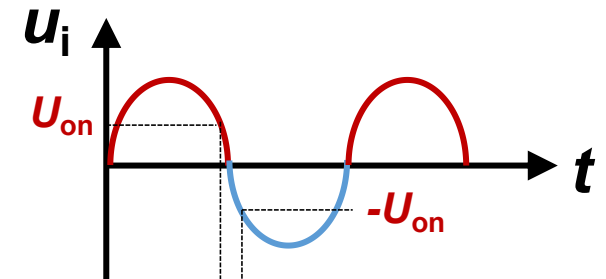
$u_i > U_{on}$ T1 (amplification) ✓ T2 (off) ✗

$U_{on} > u_i > 0$ T1 (off) ✗ T2 (off) ✗

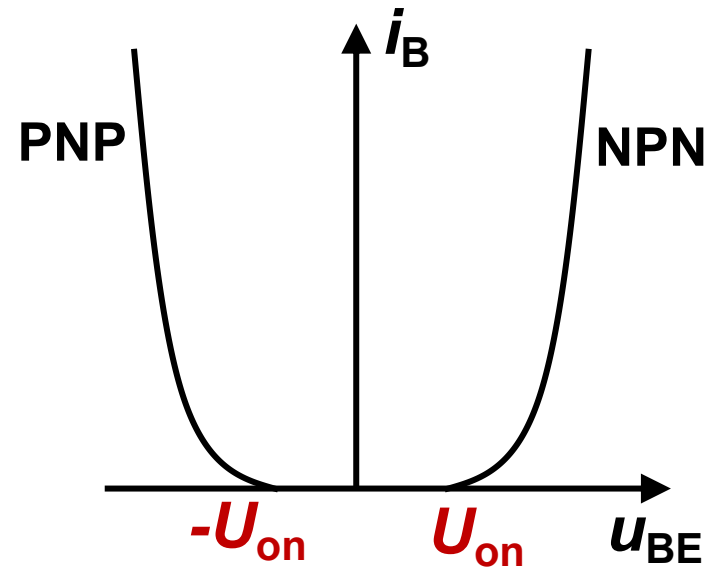
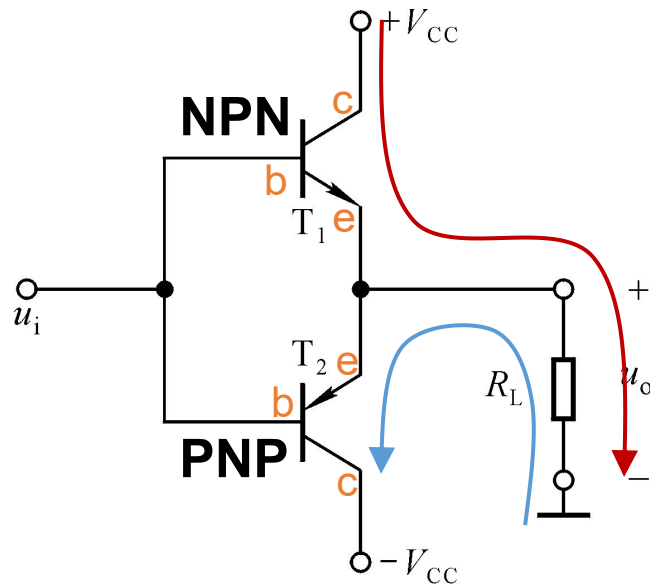
(2) For 2nd half period

$u_i < -U_{on}$ T1 (amplification) ✗ T2 (off) ✓

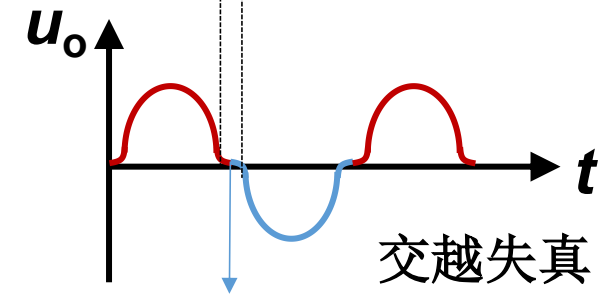
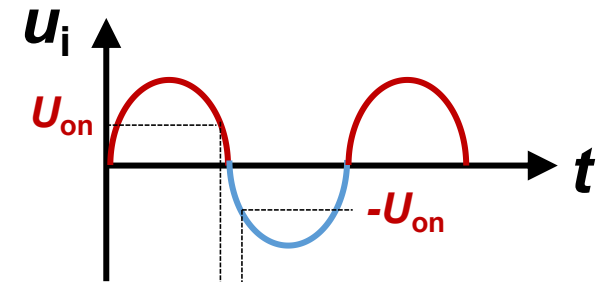
$0 > u_i > -U_{on}$ T1 (off) ✗ T2 (off) ✗



2) Real case: Crossover distortion

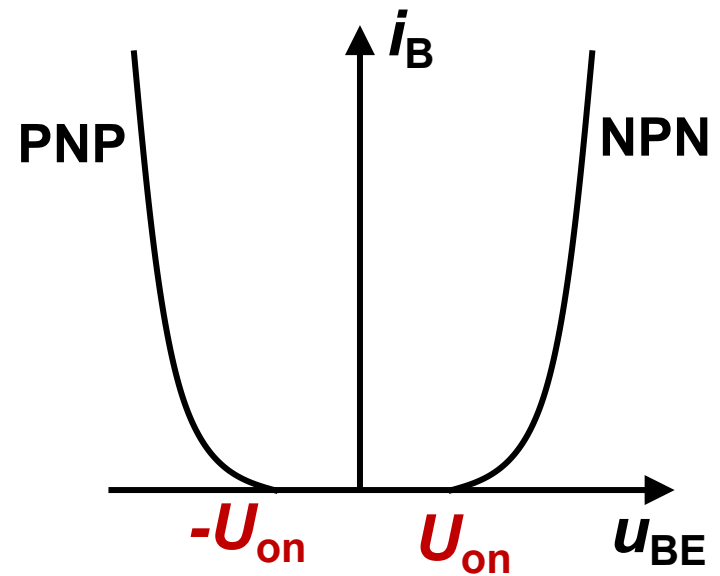
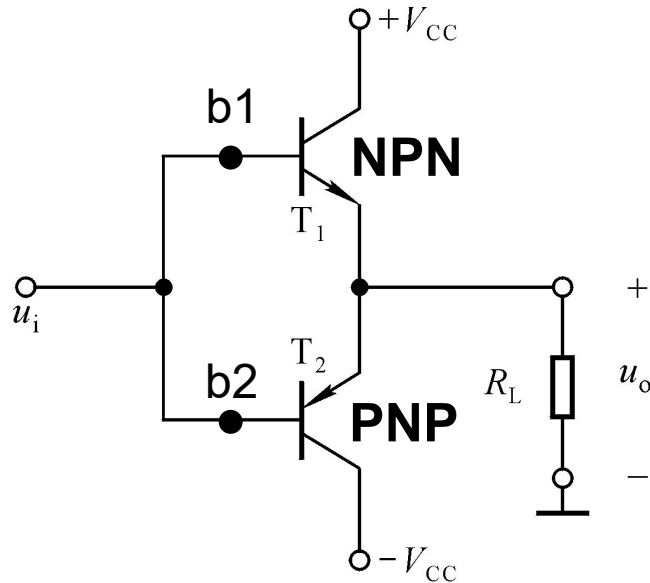


In conclusion, when $U_{on} > u_i > -U_{on}$, there will be crossover distortion in output signal.



交越失真
Crossover distortion

How to reduce crossover distortion

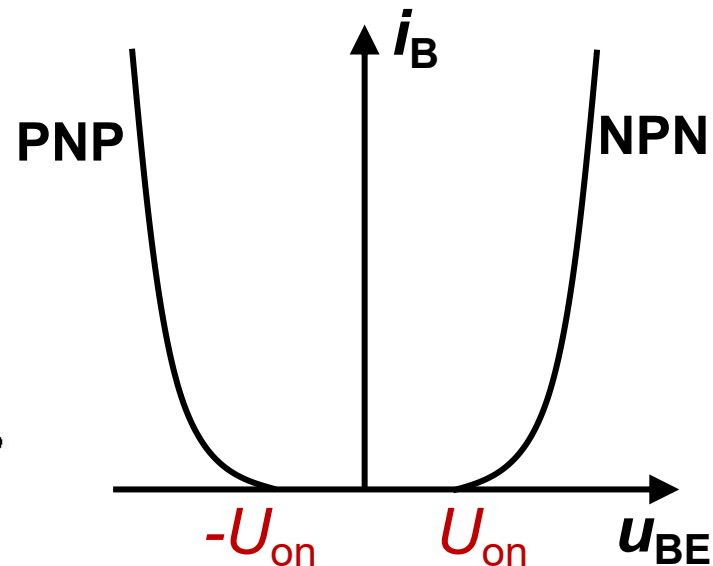
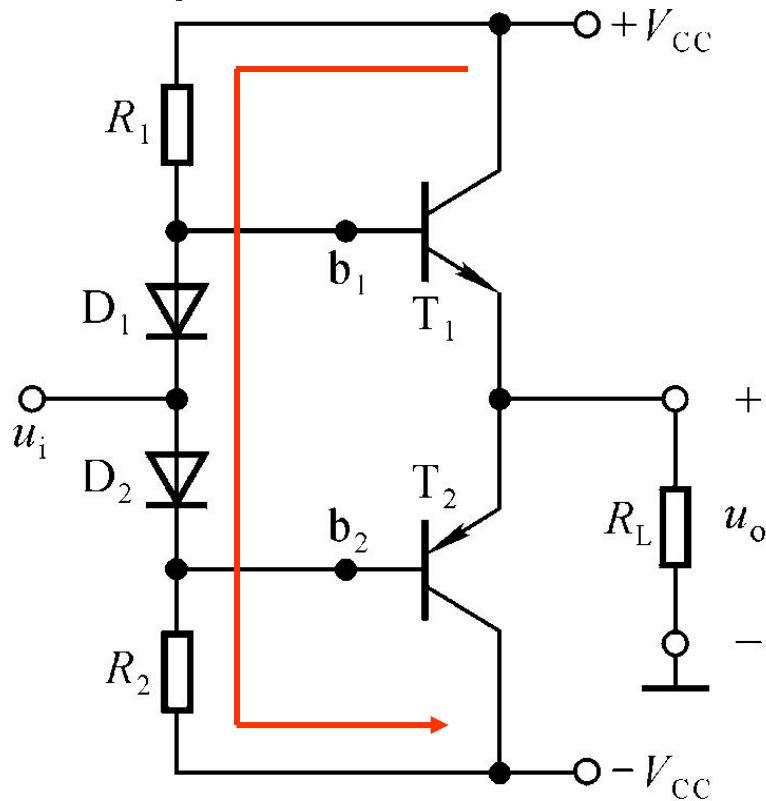


Quiescent voltages at $b1$ and $b2$: $U_{b1} = U_{b2} = 0$

Make quiescent U_{b1} slightly larger than U_{on} , and U_{b2} slightly smaller than $-U_{on}$!

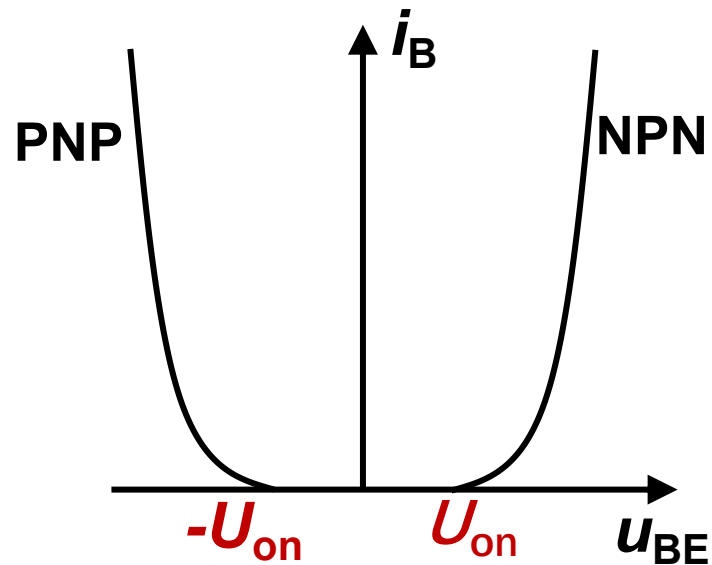
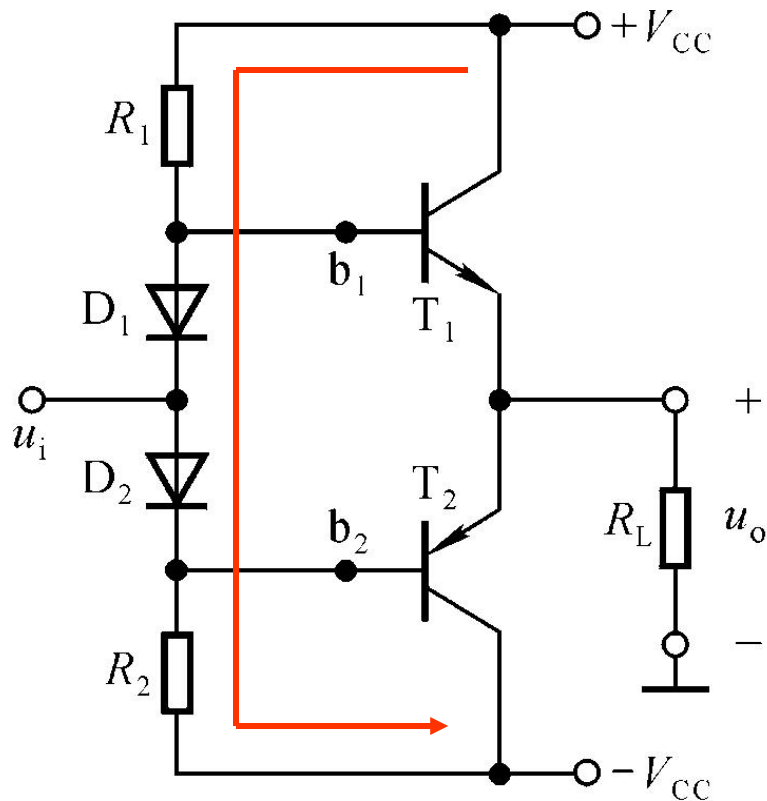
4.3.1 Eliminate crossover distortion

1) Use diodes



Use the same material for transistors and diodes. As a result, the working voltage of diodes D_1 and D_2 :

$$U_{D1} = U_{D2} \approx U_{on}$$

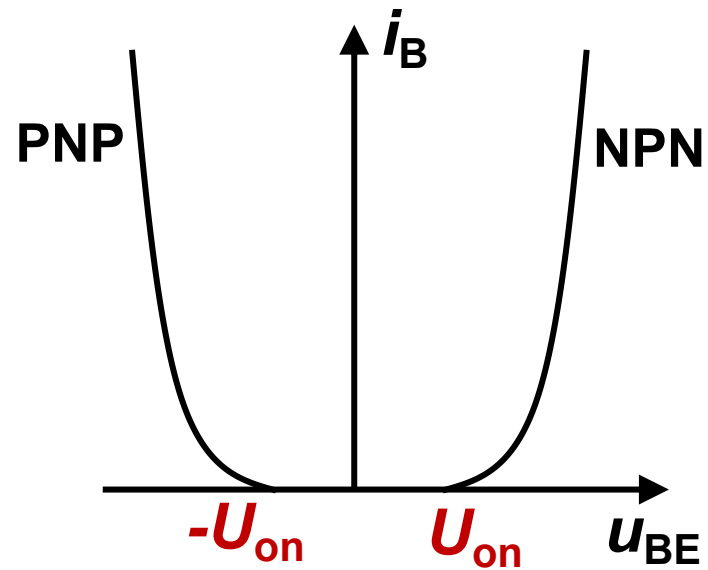
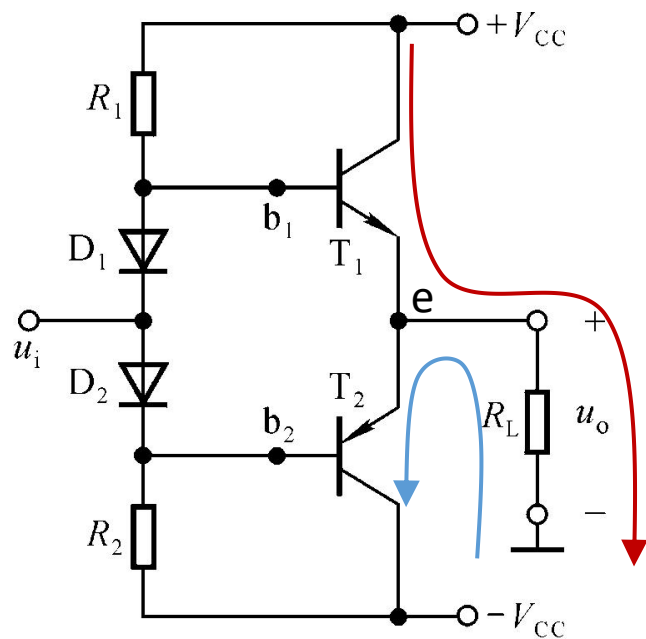


The quiescent voltages at b_1 and b_2 are nonzero:

$$U_{b1} = U_{D1} \quad U_{b2} = -U_{D2} \quad U_{D1} = U_{D2} \approx U_{on}$$

After we apply AC voltage u_i , the total voltage at b_1 and b_2 are:

$$U_{b1} \approx U_{on} + u_i \quad U_{b2} \approx -U_{on} + u_i$$



(1) For 1st half period

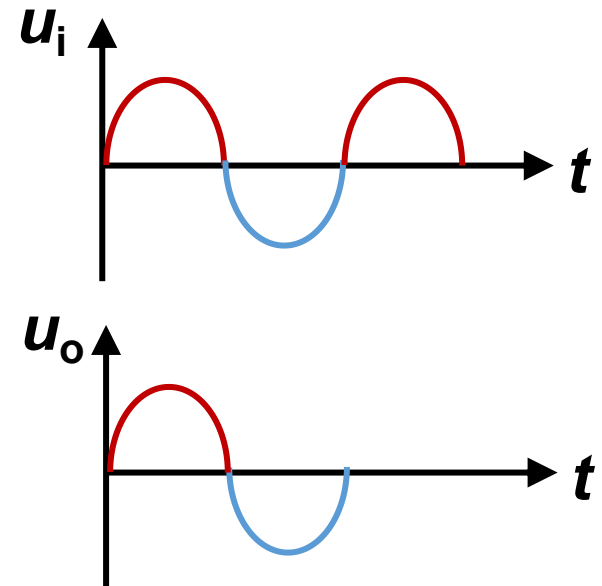
$$u_i > 0 \Rightarrow u_{be1} = U_{b1} > U_{on}$$

T1 (amplification) ✓ T2 (off) ✗

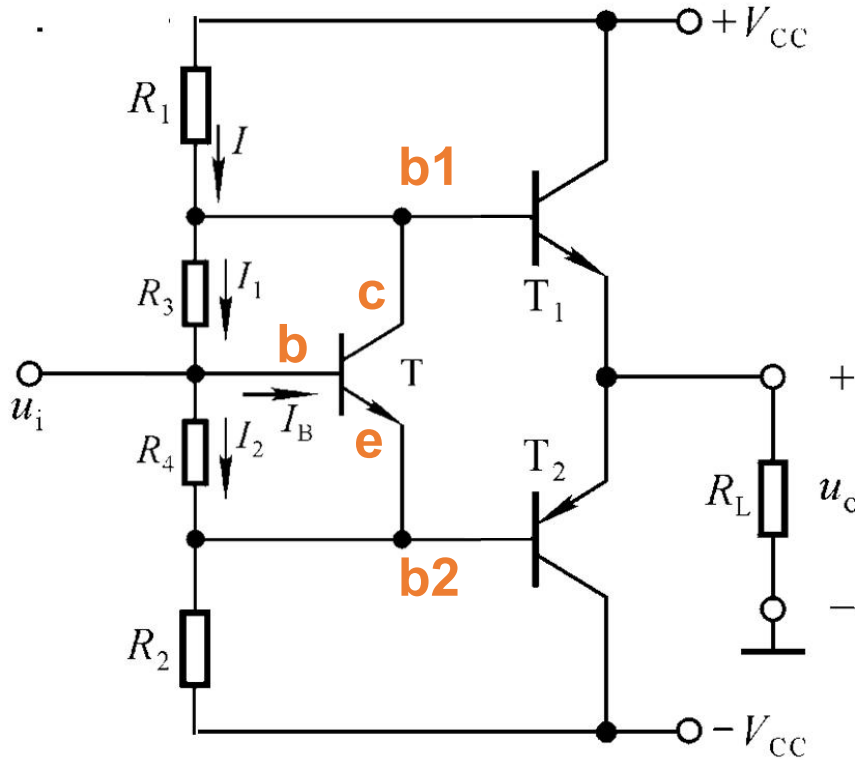
(2) For 2nd half period

$$u_i < 0 \Rightarrow u_{be2} = U_{b2} < -U_{on}$$

T1 (off) ✗ T2 (amplification) ✓



2) U_{BE} multiplying circuit U_{BE} 倍增电路

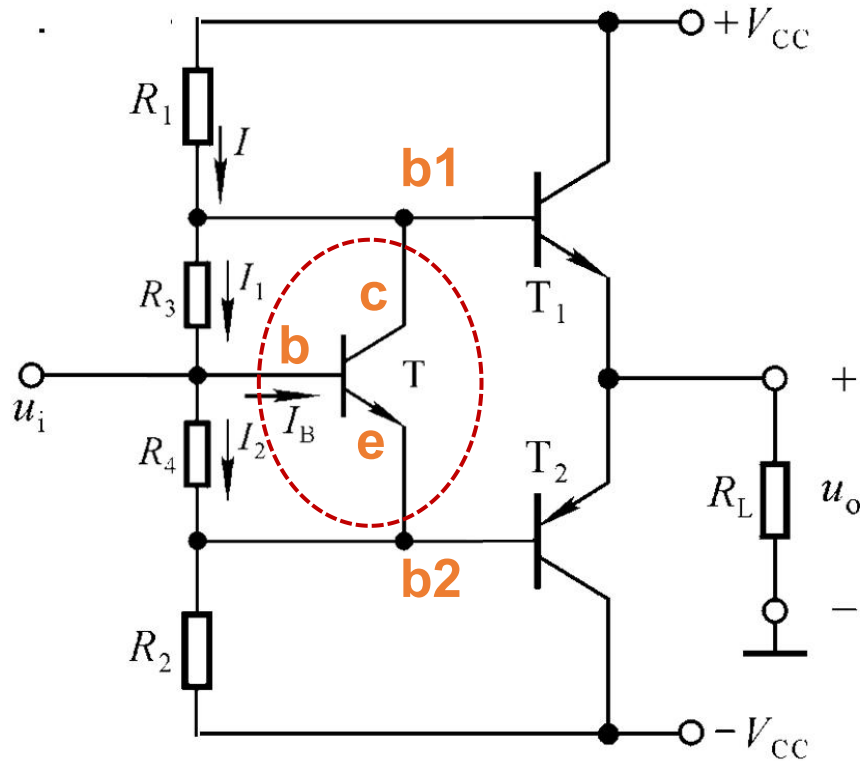


Choose R_3 and R_4 , let
 $I_2 \gg I_B$, $I_2 \approx I_1$

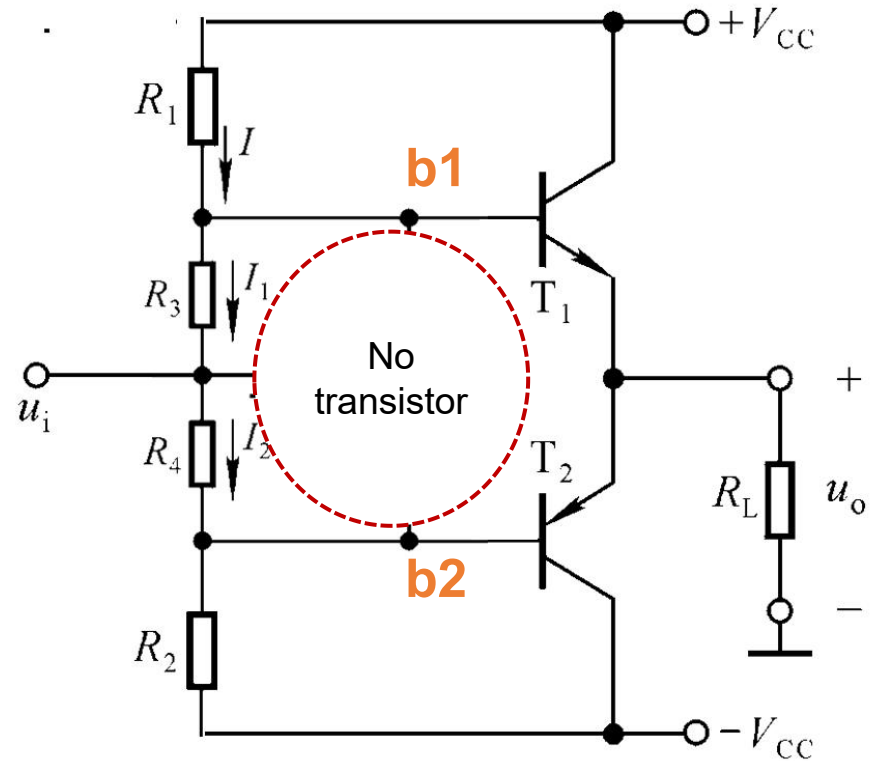
$$U_{b1b2} = U_{CE}$$

$$\approx \frac{R_3 + R_4}{R_4} U_{BE}$$

Why use T?



$$U_{b1b2} \approx \frac{R_3 + R_4}{R_4} U_{BE}$$



$$U_{b1b2} \approx \frac{R_3 + R_4}{R_1 + R_2 + R_3 + R_4} V_{CC}$$

Use composite transistor

